Class 8

Mathematics

Solutions

Table of Content

1.	Irrational Numbers, Surds and Logarithms	1 - 47
2.	Trigonometry	48 – 65
3.	Polynomials and Pair of Linear Equations	66 – 82
4.	Quadratic Equations	83 – 98
5.	Plane Geometry	99 – 121
6.	Progressions	122 – 159
7.	Statistics and Probability	160 – 183
8.	Coordinate Geometry	184 – 212

1. Irrational number, Surds and Logarithms

Irrational Numbers

LEVEL-1

1. Which of the following numbers is an irrational number?

A) $\sqrt{225} = 15$(Rational)

B) $\sqrt{23}$ is not a perfect square (Irrational)

C) 0.3796is a terminating decimal (Rational)

D) $\frac{7}{4}$is a fraction (Rational)

Correct Answer: B) $\sqrt{23}$

2. The decimal expansion of an irrational number is:

Irrational numbers have decimal expansions that never end and never repeat in a pattern.

Correct Answer: C) Non-terminating and non-recurring

3. Which of the following numbers is irrational?

A) $\sqrt{49} = 7$(Rational)

B) $\frac{22}{7}$is a fraction (Rational)

C) $\sqrt{18} = 3\sqrt{2}$(Irrational)

D) 0.625is a terminating decimal (Rational)

Correct Answer: C) $\sqrt{18}$

4. If $a = \sqrt{5}$ and $b = \sqrt{20}$, then $a + b =$

$\sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$

$a + b = \sqrt{5} + 2\sqrt{5} = 3\sqrt{5}$

Correct Answer: A) $3\sqrt{5}$

5. Which of the following is irrational?

A) 0.121212 ...is recurring (Rational)

B) $\frac{9}{11}$is a fraction (Rational)

C) $\sqrt{11}$is not a perfect square (Irrational)

D) 0.75is terminating (Rational)

Correct Answer: C) $\sqrt{11}$

6. If $x = \sqrt{2} + \sqrt{3}$, then $x^2 =$

$x^2 = (\sqrt{2} + \sqrt{3})^2$

$= 2 + 3 + 2\sqrt{6}$

$= 5 + 2\sqrt{6}$

Correct Answer: C) $5 + 2\sqrt{6}$

7. Which of the following numbers lies between 2 and 3?

$\sqrt{3} \approx 1.73$

$\sqrt{5} \approx 2.23$

$\sqrt{10} \approx 3.16$

$\sqrt{15} \approx 3.87$

Only $\sqrt{5}$ lies between 2 and 3.

Correct Answer: B) $\sqrt{5}$

8. Which of the following is NOT irrational?

$\sqrt{50} = 5\sqrt{2}$(Irrational)

$\sqrt{72} = 6\sqrt{2}$(Irrational)

$\sqrt{64} = 8$(Rational)

$\sqrt{45} = 3\sqrt{5}$(Irrational)

Correct Answer: C) $\sqrt{64}$

9. Simplify: $\sqrt{48}$

$\sqrt{48} = \sqrt{16 \times 3}$

$= 4\sqrt{3}$

Correct Answer: B) $4\sqrt{3}$

10. If $2 + \sqrt{7}$ is added to 5, the result is:

$$5 + (2 + \sqrt{7}) = 7 + \sqrt{7}$$

Since $\sqrt{7}$ is irrational, the sum is also irrational.

Correct Answer: B) Irrational

11. Which of the following numbers is irrational?

A) $\sqrt{48} - 4\sqrt{3} = 4\sqrt{3} - 4\sqrt{3} = 0$

(Rational)

B)$(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2}) = 5 - 2 = 3$

(Rational)

C)$2\sqrt{7} + \sqrt{28} = 2\sqrt{7} + 2\sqrt{7} = 4\sqrt{7}$

(Irrational)

D)$\frac{\sqrt{121}}{11} = \frac{11}{11} = 1$

(Rational)

Correct Answer: C) $2\sqrt{7} + \sqrt{28}$

12. Find the value of $\left(\sqrt{7} - \sqrt{3}\right)^2$

Using $(a - b)^2 = a^2 + b^2 - 2ab$,

$\left(\sqrt{7} - \sqrt{3}\right)^2 = 7 + 3 - 2\sqrt{21}$

$= 10 - 2\sqrt{21}$

Correct Answer: A) $10 - 2\sqrt{21}$

13. If $x = \sqrt{8} + \sqrt{2}$. find x^2.

$x^2 = (\sqrt{8} + \sqrt{2})^2$

$= 8 + 2 + 2\sqrt{16}$

$= 10 + 2(4)$

$= 18$

Correct Answer: B) $10 + 8$

14. Problem: Which of the following has a terminating decimal expansion?

A rational number has a terminating decimal if the denominator has only factors 2and/or 5.

$\frac{5}{16} = \frac{5}{2^4}$

So it is terminating.

Correct Answer: B) $\frac{5}{16}$

15. Simplify $\frac{5}{\sqrt{5}}$

$\frac{5}{\sqrt{5}} = \sqrt{5}$

because

$\frac{5}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{5\sqrt{5}}{5} = \sqrt{5}$

Correct Answer: B) $\sqrt{5}$

16. If $a = \sqrt{5} - \sqrt{2}$.find a^2.

$a^2 = (\sqrt{5} - \sqrt{2})^2$

$= 5 + 2 - 2\sqrt{10}$

$= 7 - 2\sqrt{10}$

Correct Answer: A) $7 - 2\sqrt{10}$

17. Which of the following lies between $\sqrt{10}$ and $\sqrt{11}$?

Approximate values:

$\sqrt{10} \approx 3.16, \sqrt{11} \approx 3.32$

Among the options, 3.2 lies between them.

Correct Answer: B) 3.2

18. Find the product $(3 + \sqrt{2})(3 - \sqrt{2})$

Using

$(a + b)(a - b) = a^2 - b^2$

$= 3^2 - (\sqrt{2})^2$

$= 9 - 2$

$= 7$

Correct Answer: A) 7

19. **Given:**

$\boldsymbol{x = \sqrt{3} + \sqrt{2}, y = \sqrt{3} - \sqrt{2}}$

We need to find xy.

Using the identity:

$\boldsymbol{(a + b)(a - b) = a^2 - b^2}$

$\boldsymbol{xy} = (\sqrt{\boldsymbol{3}} + \sqrt{\boldsymbol{2}})(\sqrt{\boldsymbol{3}} - \sqrt{\boldsymbol{2}})$

$= (\sqrt{\boldsymbol{3}})^{\boldsymbol{2}} - (\sqrt{\boldsymbol{2}})^{\boldsymbol{2}}$

$= \boldsymbol{3 - 2 = 1}$

Therefore, $\boxed{\mathbf{1}}$

Correct Answer: A) 1

20. If $x = \sqrt{5} + \sqrt{6}$.find x^2.

$x^2 = (\sqrt{5} + \sqrt{6})^2$

$= 5 + 6 + 2\sqrt{30}$

$= 11 + 2\sqrt{30}$

Correct Answer: B) $11 + 2\sqrt{30}$

LEVEL-2

1. **Which of the following is irrational?**

A) $\sqrt{32} \times \sqrt{2} = \sqrt{64} = 8$(rational)

B) $\frac{\sqrt{50}}{5} = \frac{5\sqrt{2}}{5} = \sqrt{2}$(irrational)

C) $\sqrt{49} + \sqrt{81} = 7 + 9 = 16$(rational)

D) $\sqrt{18} \times \sqrt{2} = \sqrt{36} = 6$(rational)

Correct Answer: **B)** $\frac{\sqrt{50}}{5}$

2. If $\sqrt{a} = 3\sqrt{2}$, then a =

Squaring both sides:

$a = (3\sqrt{2})^2 = 9 \times 2 = 18$

Correct Answer: **C) 18**

3. Which of the following is irrational?

A) $\sqrt{98} - 7\sqrt{2} = 7\sqrt{2} - 7\sqrt{2} = 0$

B) $\sqrt{45} + \sqrt{20} = 3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$(irrational)

C) $\sqrt{27} - 3\sqrt{3} = 3\sqrt{3} - 3\sqrt{3} = 0$

D) $\left(\sqrt{6}\right)^2 = 6$

Correct Answer: **B)** $\sqrt{45} + \sqrt{20}$

4. Which of the following is irrational?

A) $(\sqrt{5})(\sqrt{20}) = \sqrt{100} = 10$

B) $\sqrt{72} - 6\sqrt{2} = 6\sqrt{2} - 6\sqrt{2} = 0$

C) $\sqrt{12} + \sqrt{3} = 2\sqrt{3} + \sqrt{3} = 3\sqrt{3}$(irrational)

D) $(\sqrt{9} + \sqrt{16}) = 3 + 4 = 7$

Correct Answer: **C)** $\sqrt{12} + \sqrt{3}$

5. Simplify $3\sqrt{8} - 2\sqrt{18}$

$= 3(2\sqrt{2}) - 2(3\sqrt{2})$

$= 6\sqrt{2} - 6\sqrt{2} = 0$

Correct Answer: **A) 0**

6. If $a = \sqrt{3} + \sqrt{5}$, then a^2 equals:

$a^2 = (\sqrt{3} + \sqrt{5})^2$

$= 3 + 5 + 2\sqrt{15}$

$= 8 + 2\sqrt{15}$

Correct Answer: **C)** $8 + 2\sqrt{15}$

7. The number $\sqrt{45} + \sqrt{20}$ is equal to:

$\sqrt{45} = 3\sqrt{5}, \sqrt{20} = 2\sqrt{5}$

$3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$

Correct Answer: **A)** $5\sqrt{5}$

8. Which of the following is irrational?

A) $\frac{\sqrt{50}}{\sqrt{2}} = \sqrt{25} = 5$

B) $\frac{\sqrt{81}}{9} = \frac{9}{9} = 1$

C) $\frac{3}{\sqrt{5}}$is irrational

D) $\sqrt{144} = 12$

Correct Answer: **C)** $\frac{3}{\sqrt{5}}$

9. If $x = \sqrt{2} - 1$, then $x^2 + 2x$ equals:

$x^2 + 2x = (\sqrt{2} - 1)^2 + 2(\sqrt{2} - 1)$

$= 2 + 1 - 2\sqrt{2} + 2\sqrt{2} - 2$

$= 1$

Correct Answer: **A) 1**

10. Which of the following lies between $\sqrt{7}$and $\sqrt{8}$?

$\sqrt{7} \approx 2.64, \sqrt{8} \approx 2.83$

Among the options, 2.7 lies between them.

Correct Answer: **B) 2.7**

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

11. Which of the following expressions are irrational?

A) $\sqrt{2}(1+\sqrt{2}) = \sqrt{2}+2 \rightarrow$ irrational

B) $(\sqrt{3})^2 + \sqrt{5} = 3+\sqrt{5} \rightarrow$ irrational

C) $\sqrt{27} - 3\sqrt{3} = 3\sqrt{3} - 3\sqrt{3} = 0 \rightarrow$ rational

D) $\sqrt{6} + \sqrt{24} = \sqrt{6} + 2\sqrt{6} = 3\sqrt{6} \rightarrow$ irrational

Correct Answers: **A, B, D**

12. Which of the following irrational numbers lie between 3 and 4?

$\sqrt{10} \approx 3.16$

$\sqrt{12} \approx 3.46$

$\sqrt{18} \approx 4.24$

$\sqrt{15} \approx 3.87$

Numbers between 3 and 4 are:

Correct Answers: **A, B, D**

13. Which of the following are always irrational?

A) $\sqrt{2} + \sqrt{3} \rightarrow$ irrational

B) $5\sqrt{7} \rightarrow$ irrational

C) $\sqrt{8} - \sqrt{2} = 2\sqrt{2} - \sqrt{2} = \sqrt{2} \rightarrow$ irrational

D) $\sqrt{9} + \sqrt{16} = 3 + 4 = 7 \rightarrow$ rational

Correct Answers: **A, B, C**

14. Which expressions are irrational?

A) $(\sqrt{3}+1)^2 = 3 + 1 + 2\sqrt{3} = 4 + 2\sqrt{3} \rightarrow$ irrational

B) $(\sqrt{5} - \sqrt{5}) = 0 \rightarrow$ rational

C) $\sqrt{6} + \sqrt{24} = \sqrt{6} + 2\sqrt{6} = 3\sqrt{6} \rightarrow$ irrational

D) $\sqrt{20} - 2\sqrt{5} = 2\sqrt{5} - 2\sqrt{5} = 0 \rightarrow$ rational

Correct Answers: **A, C**

COMPREHENSION TYPE

PASSAGE

15. Which of the following numbers is irrational?

A) $\sqrt{64} + \sqrt{25} = 8 + 5 = 13 \rightarrow$ rational

B) $\sqrt{45} - 3\sqrt{5} = 3\sqrt{5} - 3\sqrt{5} = 0 \rightarrow$ rational

C) $2\sqrt{3} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3} \rightarrow$ irrational

D) $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4 \rightarrow$ rational

Correct Answer: **C)** $2\sqrt{3} + \sqrt{27}$

16. If $x = \sqrt{5} + \sqrt{3}$, then x^2 is:

$x^2 = (\sqrt{5} + \sqrt{3})^2$

$= 5 + 3 + 2\sqrt{15}$

$= 8 + 2\sqrt{15}$

Since it contains $\sqrt{15}$, it is irrational.

Correct Answer: **B) Irrational**

17. The number $0.101001000100001\ldots$ is:

The decimal expansion is non-terminating and non-repeating.

Hence, it is irrational.

Correct Answer: **B) Irrational**

MATRIX MATCHING TYPE

A → r

$\sqrt{3} \times \sqrt{27} = \sqrt{81} = 9$

B → p

$\sqrt{32} - \sqrt{8} = 4\sqrt{2} - 2\sqrt{2} = 2\sqrt{2}$

C → s

$\sqrt{121} = 11$

D → q

$\pi + \sqrt{2}$ is an irrational number.

INTEGER TYPE

19. If $2\sqrt{3} + \sqrt{27} = a\sqrt{3}$, find a.

$\sqrt{27} = \sqrt{9 \times 3} = 3\sqrt{3}$

So, $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$

Comparing with $a\sqrt{3}$,

$a = 5$

Correct Answer: $\boxed{5}$

20. Find the value of $(\sqrt{5} + \sqrt{4})^2 - (\sqrt{5} - \sqrt{4})^2$

Since $\sqrt{4} = 2$,

$(\sqrt{5} + 2)^2 - (\sqrt{5} - 2)^2$

Using identity:

$(a + b)^2 - (a - b)^2 = 4ab$

Here $a = \sqrt{5}$, $b = 2$

$4(\sqrt{5})(2) = 8\sqrt{5}$

Correct Answer: $\boxed{8\sqrt{5}}$

STUDENT PRACTISE CORNER

1. Which of the following is an irrational number?

A) $\frac{7}{8} \rightarrow$ Rational

B) $0.625 \rightarrow$ Rational

C) $\sqrt{11} \rightarrow$ Irrational

D) $-4 \rightarrow$ Rational

Correct Answer C: $\boxed{\sqrt{11}}$

2. Which of the following numbers lies between 4 and 5?

$4^2 = 16, 5^2 = 25$

A number between 16 and 25 will have square root between 4 and 5.

18 lies between 16 and 25

So, $\sqrt{18}$

lies between 4 and 5.

Correct Answer B: $\boxed{\sqrt{18}}$

3. The value of $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$ is:

Using identity:

$(a+b)(a-b) = a^2 - b^2$

$= (\sqrt{7})^2 - (\sqrt{5})^2$

$= 7 - 5 = 2$

Correct Answer B: $\boxed{2}$

4. Which of the following numbers is NOT irrational?

A) $\sqrt{72}\rightarrow$ Irrational

B) $\sqrt{0.81} = 0.9\rightarrow$ Rational

C) $\sqrt{17}\rightarrow$ Irrational

D) $\sqrt{19}\rightarrow$ Irrational

Correct Answer B: $\boxed{\sqrt{0.81}}$

5. If $x = 2 + \sqrt{3}$, then $\frac{1}{x}$ equals:

$$\frac{1}{2+\sqrt{3}}$$

Rationalize the denominator:

$$\frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$= \frac{2-\sqrt{3}}{4-3}$$

$$= 2 - \sqrt{3}$$

Correct Answer B: $\boxed{2-\sqrt{3}}$

6. If $a = \sqrt{3} + \sqrt{2}$ and $b = \sqrt{3} - \sqrt{2}$, then ab =

$ab = (\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})$

Using identity:

$(a+b)(a-b) = a^2 - b^2$

$= 3 - 2 = 1$

Correct Answer A: $\boxed{1}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Which of the following are irrational numbers between 1 and 2?

A) $\sqrt{2} \approx 1.414\rightarrow$ irrational and between 1 and 2

B) $\sqrt{3} \approx 1.732\rightarrow$ irrational and between 1 and 2

C) $\frac{3}{2} = 1.5\rightarrow$ rational

D) $\sqrt{\frac{5}{2}} \approx 1.58\rightarrow$ irrational and between 1 and 2

Correct Answers: A, B, D

8. Which of the following are irrational?

A) $3\sqrt{2}\rightarrow$ irrational

B) $5 - \sqrt{7}\rightarrow$ irrational

C) $\sqrt{81} = 9\rightarrow$ rational

D) $\frac{2}{\sqrt{5}}\rightarrow$ irrational

Correct Answers: A, B, D

9. Which of the following square roots are irrational?

A) $\sqrt{18} = 3\sqrt{2}\rightarrow$ irrational

B) $\sqrt{64} = 8\rightarrow$ rational

C) $\sqrt{27} = 3\sqrt{3}$→ irrational

D) $\sqrt{121} = 11$→ rational

Correct Answers: A, C

COMPREHENSION TYPE

PASSAGE

10. **Which of the following is irrational?**

A) $\sqrt{100} - 5 = 10 - 5 = 5$(Rational)

B) $\sqrt{45} - \sqrt{20} = 3\sqrt{5} - 2\sqrt{5} = \sqrt{5}$(Irrational)

C) $\sqrt{25} + \sqrt{49} = 5 + 7 = 12$(Rational)

D) $\sqrt{16} + \sqrt{9} = 4 + 3 = 7$(Rational)

Correct Answer: B) $\sqrt{45} - \sqrt{20}$

11. **Which of the following lies between 4 and 5?**

Since $4^2 = 16$ and $5^2 = 25$, the number under the root must lie between 16 and 25.

A) $14 < 16$

B) 19 lies between 16and 25

C) $28 > 25$

D) $35 > 25$

Correct Answer: B) $\sqrt{19}$

12. **If** $a = \sqrt{3}$**, then** $a^2 + a$ **is:**

$a^2 + a = (\sqrt{3})^2 + \sqrt{3} = 3 + \sqrt{3}$

Since $\sqrt{3}$is irrational, $3 + \sqrt{3}$is also irrational.

Correct Answer: C) Irrational

MATRIX MATCHING TYPE

P. $\sqrt{36}$

So, it is a **rational number**.

Q.$\sqrt{7}$

is an **irrational number**.

R. π

is a **non-terminating non-repeating decimal**.

S. 0.272727 ...

is a **repeating decimal**.

Matching: P → 2, Q → 1, R → 3, S → 4

INTEGER TYPE

14. Find the value of $(\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2})$

Using identity:

$(a - b)(a + b) = a^2 - b^2$

$= (\sqrt{5})^2 - (\sqrt{2})^2$

$= 5 - 2 = 3$

Answer: 3

15. Find the value of $(\sqrt{18} - \sqrt{8})^2$

$\sqrt{18} = 3\sqrt{2}, \sqrt{8} = 2\sqrt{2}$

$(3\sqrt{2} - 2\sqrt{2})^2$

$= (\sqrt{2})^2 = 2$

Answer: 2

Surds

LEVEL-I

1. Let $x = 0.12\bar{3}$

$100x = 12.\bar{3}$

$1000x = 123.\bar{3}$

Subtract:

$1000x - 100x = 123.\bar{3} - 12.\bar{3}$

$900x = 111$

$x = \frac{111}{900}$

Correct Answer: A) $\mathbf{\frac{111}{900}}$

2. Let $x = 0.2353535\ldots$

$10x = 2.353535\ldots$

$1000x = 235.353535\ldots$

Subtract:

$1000x - 10x = 235.353535\ldots - 2.353535\ldots$

$990x = 233$

$x = \frac{233}{990}$

Correct Answer: C) $\mathbf{\frac{233}{990}}$

3. The bar is only on 1, so the number is:

$0.62\bar{1} = 0.6211111\ldots$

Let

$x = 0.6211111\ldots$

Then

$100x = 62.111111\ldots$

$1000x = 621.111111\ldots$

Subtract:

$1000x - 100x = 621.111111\ldots - 62.111111\ldots$

$900x = 559$

$x = \frac{559}{900}$

Correct Answer B: $\boxed{\mathbf{\frac{559}{900}}}$

4. Check each option:

A) $\sqrt{\frac{16}{25}} = \frac{4}{5} \rightarrow$ Rational

B) $\sqrt{\frac{12}{75}} = \sqrt{\frac{4}{25}} = \frac{2}{5} \rightarrow$ Rational

C) $\sqrt{\frac{80}{16}} = \sqrt{5} \rightarrow$ Irrational

D) $\sqrt{121} = 11 \rightarrow$ Rational

Correct Answer: C) $\sqrt{\mathbf{\frac{80}{16}}}$

5. Let $x = 0.7\bar{1}2$

$1000x = 712.7\bar{1}2$

Subtract:

$1000x - x = 712$

$999x = 712$

$x = \frac{712}{999}$

Correct Answer: C) $\frac{712}{999}$

6. If $\sqrt{a+\sqrt{b}} = 2+\sqrt{3}$, then $\sqrt{a^2-b} =$?

Square both sides:

$a+\sqrt{b} = (2+\sqrt{3})^2$

$= 4+3+4\sqrt{3}$

$= 7+4\sqrt{3}$

Comparing,

$a = 7, \sqrt{b} = 4\sqrt{3}$

$b = 48$

Now, $\sqrt{a^2-b} = \sqrt{49-48} = \sqrt{1} = 1$

Correct Answer: D) 1

7. If $\sqrt{19-4\sqrt{x}} = \sqrt{12}-\sqrt{7}$, then $x =$?

Square RHS:

$\left(\sqrt{12}-\sqrt{7}\right)^2$

$= 12+7-2\sqrt{84}$

$= 19-2\sqrt{84}$

Compare with:

$19-4\sqrt{x}$

So,

$4\sqrt{x} = 2\sqrt{84}$

$2\sqrt{x} = \sqrt{84}$

Squaring:

$4x = 84$

$x = 21$

Correct Answer: A) 21

8. The value of

$$\sqrt{12+\sqrt{12+\sqrt{12+\cdots}}}$$

Let the value be x.

Then,

$x = \sqrt{12+x}$

Square both sides:

$x^2 = 12+x$

$x^2-x-12 = 0$

$(x-4)(x+3) = 0$

Since value is positive,

$x = 4$

Correct Answer: A) 4

9. There are 99 terms.

Use: $\frac{1}{\sqrt{n}+\sqrt{n+1}} = \frac{\sqrt{n+1}-\sqrt{n}}{(\sqrt{n+1}+\sqrt{n})(\sqrt{n+1}-\sqrt{n})}$

$= \sqrt{n+1}-\sqrt{n}$

So the series becomes:

$(\sqrt{2}-1)+(\sqrt{3}-\sqrt{2})+(\sqrt{4}-\sqrt{3})+\cdots+(\sqrt{100}-\sqrt{99})$

All middle terms cancel.

$=\sqrt{100}-1$

$=10-1$

$=9$

Correct Answer B: $\boxed{9}$

10. $\frac{5\sqrt{3}+\sqrt{7}}{4\sqrt{3}-2\sqrt{7}}$

Rationalize the denominator:

$$=\frac{(5\sqrt{3}+\sqrt{7})(4\sqrt{3}+2\sqrt{7})}{(4\sqrt{3})^2-(2\sqrt{7})^2}$$

Denominator:

$48-28=20$

Numerator:

$60+10\sqrt{21}+4\sqrt{21}+14=74+14\sqrt{21}$

So,

$$=\frac{74+14\sqrt{21}}{20}=\frac{37}{10}+\frac{7\sqrt{21}}{10}$$

Hence,

$a=\frac{37}{10}, b=\frac{7}{10}$

Correct Answer: A

11. $\sqrt[3]{81}=\sqrt[3]{3^4}$

To make it rational, multiply by:

$\sqrt[3]{3^2}=\sqrt[3]{9}$

Then,

$\sqrt[3]{81}\times\sqrt[3]{9}=\sqrt[3]{729}=9$

Correct Answer: D

12. Compare approximate values:

$\sqrt[4]{10}\approx 1.78$

$\sqrt[3]{6}\approx 1.82$

$\sqrt{3}\approx 1.73$

Ascending order:

$\sqrt{3}<\sqrt[4]{10}<\sqrt[3]{6}$

Correct Answer: B

13. Compare the values:

$\sqrt[6]{2}=2^{1/6}, \sqrt[4]{2}=2^{1/4}, \sqrt[3]{2}=2^{1/3}$

Since,

$\frac{1}{6}<\frac{1}{4}<\frac{1}{3}$

for base $2>1$, larger exponent gives larger value.

Thus,

$\sqrt[6]{2}<\sqrt[4]{2}<\sqrt[3]{2}$

Correct Answer: A

14. Given,

$x = \frac{1}{2}\left(\sqrt{a}+\frac{1}{\sqrt{a}}\right)$

Use identity:

$\left(\sqrt{a}+\frac{1}{\sqrt{a}}\right)^2 = a+\frac{1}{a}+2$

So,

$x^2 = \frac{1}{4}\left(a+\frac{1}{a}+2\right)$

Then,

$x^2-1 = \frac{1}{4}\left(a+\frac{1}{a}-2\right) = \frac{1}{4}\left(\sqrt{a}-\frac{1}{\sqrt{a}}\right)^2$

Hence,

$\sqrt{x^2-1} = \frac{1}{2}\left(\sqrt{a}-\frac{1}{\sqrt{a}}\right)$

Now, $x-\sqrt{x^2-1} = \frac{1}{2}\left(\sqrt{a}+\frac{1}{\sqrt{a}}\right)-\frac{1}{2}\left(\sqrt{a}-\frac{1}{\sqrt{a}}\right) = \frac{1}{\sqrt{a}}$

Therefore,

$$\frac{\sqrt{x^2-1}}{x-\sqrt{x^2-1}} = \frac{\frac{1}{2}\left(\sqrt{a}-\frac{1}{\sqrt{a}}\right)}{\frac{1}{\sqrt{a}}}$$

$$= \frac{a-1}{2}$$

Correct Answer: B

15. Given,

$x = \sqrt[3]{2}+\sqrt[3]{4}$

Let,

$a = \sqrt[3]{2}, b = \sqrt[3]{4}$

Then,

$ab = \sqrt[3]{8} = 2$

Using identity:

$(a+b)^3 = a^3+b^3+3ab(a+b)$

$x^3 = 2+4+3(2)x$

$x^3 = 6+6x$

$x^3-6x = 6$

Correct Answer: B

16. If $\sqrt{x} + \sqrt{(x - \sqrt{(1 - x)})} = 1$, then the value of x is:

Steps:Isolate the nested radical term:

$\sqrt{(x - \sqrt{(1 - x)})} = 1 - \sqrt{x}$

Square both sides:

$x - \sqrt{(1 - x)} = (1 - \sqrt{x})^2$

$x - \sqrt{(1 - x)} = 1 - 2\sqrt{x} + x$

Cancel x from both sides:

$-\sqrt{(1 - x)} = 1 - 2\sqrt{x}$

$2\sqrt{x} - 1 = \sqrt{(1 - x)}$

Square both sides again:

$(2\sqrt{x} - 1)^2 = (\sqrt{(1 - x)})^2$

$4x - 4\sqrt{x} + 1 = 1 - x$

Simplify:

$5x = 4\sqrt{x}$
Square both sides:
$25x^2 = 16x$
$25x^2 - 16x = 0$
$x(25x - 16) = 0$
Possible solutions:
$x = 0$ or $x = 16/25$
Check $x = 0$:
$\sqrt{0} + \sqrt{(0 - \sqrt{1})} = \sqrt{(-1)}$, which is not real.
Hence, $x = 16/25$
Answer: D) 16/25

17. $27^{\frac{1}{9}} = (3^3)^{\frac{1}{9}} = 3^{\frac{1}{3}}, 4^{\frac{1}{6}} = (2^2)^{\frac{1}{6}} = 2^{\frac{1}{3}}$
So the expression becomes:
$3^{\frac{1}{3}} - 2^{\frac{1}{3}}$
Using identity:
$(a-b)(a^2+ab+b^2) = a^3 - b^3$
Here, $a = 3^{\frac{1}{3}}, b = 2^{\frac{1}{3}}$
Rationalizing factor:
$a^2 + ab + b^2$
$= 3^{\frac{2}{3}} + 3^{\frac{1}{3}}2^{\frac{1}{3}} + 2^{\frac{2}{3}}$
Correct Answer: B

18.
$$\frac{5-\sqrt{3}}{2+\sqrt{3}}$$
Multiply numerator and denominator by conjugate $(2-\sqrt{3})$:
$$= \frac{(5-\sqrt{3})(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})}$$
Denominator:
$4 - 3 = 1$
Numerator:
$10 - 5\sqrt{3} - 2\sqrt{3} + 3 = 13 - 7\sqrt{3}$
Comparing with:
$x + y\sqrt{3}$
$x = 13, y = -7$
Correct Answer: D

19. $\sqrt{13 - a\sqrt{10}} = \sqrt{8} + \sqrt{5}$
Square both sides:
$13 - a\sqrt{10} = 8 + 5 + 2\sqrt{40}$
$= 13 + 2(2\sqrt{10})$
$= 13 + 4\sqrt{10}$
Compare both sides:
$-a\sqrt{10} = 4\sqrt{10}$
$a = -4$
Correct Answer: C

20. $x = 2\sqrt{2} + \sqrt{7}$

Find $\frac{1}{x}$:

$\frac{1}{x} = \frac{1}{2\sqrt{2} + \sqrt{7}}$

Rationalizing:

$= \frac{2\sqrt{2} - \sqrt{7}}{8 - 7}$

$= 2\sqrt{2} - \sqrt{7}$

Now,

$x + \frac{1}{x} = (2\sqrt{2} + \sqrt{7}) + (2\sqrt{2} - \sqrt{7})$

$= 4\sqrt{2}$

Therefore,

$\frac{1}{2}\left(x + \frac{1}{x}\right) = \frac{1}{2}(4\sqrt{2}) = 2\sqrt{2}$

Correct Answer: B

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. Evaluate: $\sqrt[4]{193 + 132\sqrt{2}}$

To solve this, we can find the square root of the expression twice.

First, let

$\sqrt{193 + 132\sqrt{2}} = a + b\sqrt{2}$

Squaring both sides:

$(a + b\sqrt{2})^2 = a^2 + 2b^2 + 2ab\sqrt{2}$

Comparing with $193 + 132\sqrt{2}$:

$a^2 + 2b^2 = 193$

$2ab = 132 \Rightarrow ab = 66$

Taking $a = 11$, $b = 6$:

$11^2 + 2(6)^2 = 121 + 72 = 193$

So, $\sqrt{193 + 132\sqrt{2}} = 11 + 6\sqrt{2}$

Now find:

$\sqrt{11 + 6\sqrt{2}}$

Assume:

$= c + d\sqrt{2}$

Then, $(c + d\sqrt{2})^2 = c^2 + 2d^2 + 2cd\sqrt{2}$

Comparing:

$c^2 + 2d^2 = 11$

$2cd = 6 \Rightarrow cd = 3$

Taking $c = 3$, $d = 1$:

$3^2 + 2(1)^2 = 9 + 2 = 11$

Hence, $\sqrt[4]{193 + 132\sqrt{2}} = 3 + \sqrt{2}$

Correct Option: C) $3 + \sqrt{2}$

2. Evaluate: $\sqrt[4]{137-36\sqrt{14}}$

First, find:

$\sqrt{137-36\sqrt{14}}$

Assume:

$=a-b\sqrt{14}$

Squaring:

$(a-b\sqrt{14})^2=a^2+14b^2-2ab\sqrt{14}$

Comparing:

$a^2+14b^2=137$

$2ab=36\Rightarrow ab=18$

Taking $a=9,\ b=2$:

$9^2+14(2)^2=81+56=137$

So,$\sqrt{137-36\sqrt{14}}=9-2\sqrt{14}$

Now find:

$\sqrt{9-2\sqrt{14}}$

Use:

$(\sqrt{x}-\sqrt{y})^2=x+y-2\sqrt{xy}$

Comparing:

$x+y=9$

$xy=14$

The numbers are 7and 2.

Therefore, $\sqrt{9-2\sqrt{14}}=\sqrt{7}-\sqrt{2}$

Hence, $\sqrt[4]{137-36\sqrt{14}}=\sqrt{7}-\sqrt{2}$

Correct Option: B) $\sqrt{7}-\sqrt{2}$

3. Evaluate:

$$\sqrt{\frac{15}{2}+2\sqrt{3}+2\sqrt{6}+4\sqrt{2}}$$

Assume the expression inside the root is:

$(x+y+z)^2$

Using:

$(x+y+z)^2=x^2+y^2+z^2+2xy+2yz+2zx$

Compare the cross terms:

$2xy=2\sqrt{3}\Rightarrow xy=\sqrt{3}$

$2yz=2\sqrt{6}\Rightarrow yz=\sqrt{6}$

$2zx=4\sqrt{2}\Rightarrow zx=2\sqrt{2}$

Multiplying:

$(xyz)^2=(\sqrt{3})(\sqrt{6})(2\sqrt{2})=12$

$xyz=2\sqrt{3}$

Now,

$$z=\frac{xyz}{xy}=\frac{2\sqrt{3}}{\sqrt{3}}=2$$

$$x=\frac{xyz}{yz}=\frac{2\sqrt{3}}{\sqrt{6}}=\sqrt{2}$$

$y = \frac{xyz}{zx} = \frac{2\sqrt{3}}{2\sqrt{2}} = \sqrt{\frac{3}{2}}$

Check constant term:

$\left(\sqrt{2})^2 + \left(\sqrt{\frac{3}{2}}\right)^2 + 2^2 = 2 + \frac{3}{2} + 4 = \frac{15}{2}\right.$

Hence,

$\sqrt{\frac{15}{2} + 2\sqrt{3} + 2\sqrt{6} + 4\sqrt{2}} = 2 + \sqrt{2} + \sqrt{\frac{3}{2}}$

Correct Option: A) $2 + \sqrt{2} + \sqrt{\frac{3}{2}}$

4. Evaluate:

$\sqrt{a + 3b + 4 + 4\sqrt{a} - 4\sqrt{3b} - 2\sqrt{3ab}}$

Assume the expression inside the root is:

$\left(\sqrt{a} - \sqrt{3b} + 2\right)^2$

Expanding:

$\left(\sqrt{a})^2 + (-\sqrt{3b})^2 + 2^2 + 2(\sqrt{a})(-\sqrt{3b}) + 2(\sqrt{a})(2) + 2(-\sqrt{3b})(2\right)$

$= a + 3b + 4 - 2\sqrt{3ab} + 4\sqrt{a} - 4\sqrt{3b}$

This matches the given expression.

Therefore,

$\sqrt{a + 3b + 4 + 4\sqrt{a} - 4\sqrt{3b} - 2\sqrt{3ab}} = 2 + \sqrt{a} - \sqrt{3b}$

Correct Option: A) $2 + \sqrt{a} - \sqrt{3b}$

5. Simplify $\sqrt[3]{72 - 32\sqrt{5}}$

The easiest way is to test the options by cubing them using the identity:

$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Test Option A: $\left(3 - \sqrt{5}\right)^3$

Here,

$a = 3, b = \sqrt{5}$

Expand:

$\left(\sqrt{5})^3 = 3^3 - 3(3)^2(\sqrt{5}) + 3(3)(\sqrt{5})^2 - (\sqrt{5}\right)^3$

Simplify:

$= 27 - 27\sqrt{5} + 9(5) - 5\sqrt{5}$

$= 27 - 27\sqrt{5} + 45 - 5\sqrt{5}$

Combine like terms:

$= (27 + 45) - (27 + 5)\sqrt{5}$

$= 72 - 32\sqrt{5}$

This exactly matches the expression inside the cube root.

Therefore, $\sqrt[3]{72 - 32\sqrt{5}} = 3 - \sqrt{5}$

Correct Answer: A) $3 - \sqrt{5}$

6. Problem: Simplify $\sqrt[3]{38+17\sqrt{5}}$

Use the identity:

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Test Option C: $\left(2+\sqrt{5}\right)^3$

Here, $a = 2, b = \sqrt{5}$

Expand:

$\left(\sqrt{5}\right)^3 = 2^3 + 3(2)^2(\sqrt{5}) + 3(2)(\sqrt{5})^2 + (\sqrt{5})^3$

Simplify:

$= 8 + 12\sqrt{5} + 6(5) + 5\sqrt{5}$

$= 8 + 12\sqrt{5} + 30 + 5\sqrt{5}$

Combine like terms:

$= (8+30) + (12+5)\sqrt{5}$

$= 38 + 17\sqrt{5}$

This matches the expression inside the cube root.

Therefore,

$\sqrt[3]{38+17\sqrt{5}} = 2+\sqrt{5}$

Correct Answer: C) $2+\sqrt{5}$

7. If $a = x + \sqrt{x^2+1}$

find x.

Step 1: Isolate the radical.

$a - x = \sqrt{x^2+1}$

Step 2: Square both sides.

$(a-x)^2 = x^2 + 1$

Expand:

$a^2 - 2ax + x^2 = x^2 + 1$

Subtract x^2 from both sides:

$a^2 - 2ax = 1$

Step 3: Solve for x.

$a^2 - 1 = 2ax$

$x = \dfrac{a^2-1}{2a}$

Split the fraction:

$x = \dfrac{a^2}{2a} - \dfrac{1}{2a}$

$x = \dfrac{a}{2} - \dfrac{1}{2a}$

$x = \dfrac{1}{2}\left(a - \dfrac{1}{a}\right)$

Correct Answer: A) $\frac{1}{2}\left(a-\frac{1}{a}\right)$

8. Simplify $\sqrt[3]{54\sqrt{3}+41\sqrt{5}}$

We test the options using:

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Test Option C: $\left(\sqrt{3}+2\sqrt{5}\right)^3$

Let $a = \sqrt{3}, b = 2\sqrt{5}$

Now expand:

$(\sqrt{3}+2\sqrt{5})^3 = (\sqrt{3})^3 + 3(\sqrt{3})^2(2\sqrt{5}) + 3(\sqrt{3})(2\sqrt{5})^2 + (2\sqrt{5})^3$

Simplify each term:

$(\sqrt{3})^3 = 3\sqrt{3}$

$3(\sqrt{3})^2(2\sqrt{5}) = 3(3)(2\sqrt{5}) = 18\sqrt{5}$

$3(\sqrt{3})(2\sqrt{5})^2 = 3(\sqrt{3})(20) = 60\sqrt{3}$

$(2\sqrt{5})^3 = 8(5)\sqrt{5} = 40\sqrt{5}$

Combine like terms:

$(3\sqrt{3} + 60\sqrt{3}) + (18\sqrt{5} + 40\sqrt{5})$

$= 63\sqrt{3} + 58\sqrt{5}$

Not correct.

Now test Option A:

$(\sqrt{3}+\sqrt{5})^3$

Expand:

$= 3\sqrt{3} + 9\sqrt{5} + 15\sqrt{3} + 5\sqrt{5}$

$= 18\sqrt{3} + 14\sqrt{5}$

Not correct.

Test Option B:

$(2\sqrt{3}+\sqrt{5})^3$

Expand:

$= (2\sqrt{3})^3 + 3(2\sqrt{3})^2(\sqrt{5}) + 3(2\sqrt{3})(\sqrt{5})^2 + (\sqrt{5})^3$

Simplify:

$= 24\sqrt{3} + 36\sqrt{5} + 30\sqrt{3} + 5\sqrt{5}$

$= (24+30)\sqrt{3} + (36+5)\sqrt{5}$

$= 54\sqrt{3} + 41\sqrt{5}$

This matches exactly.

Therefore,

$\sqrt[3]{54\sqrt{3} + 41\sqrt{5}} = 2\sqrt{3} + \sqrt{5}$

Correct Answer: B) $2\sqrt{3} + \sqrt{5}$

9. Evaluate $\sqrt[3]{(10\sqrt{7} - 22)}$

Assume $\sqrt[3]{(10\sqrt{7} - 22)} = \sqrt{m} - \sqrt{n}$

Now cube both sides using:

$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

Take option D:

$(\sqrt{6} - \sqrt{2})^3$

$= (\sqrt{6})^3 - (\sqrt{2})^3 - 3(\sqrt{6})(\sqrt{2})(\sqrt{6} - \sqrt{2})$

$= 6\sqrt{6} - 2\sqrt{2} - 3\sqrt{12}(\sqrt{6} - \sqrt{2})$

Since $\sqrt{12} = 2\sqrt{3}$,

$= 6\sqrt{6} - 2\sqrt{2} - 6\sqrt{3}(\sqrt{6} - \sqrt{2})$

$= 6\sqrt{6} - 2\sqrt{2} - 6\sqrt{18} + 6\sqrt{6}$

Since $\sqrt{18} = 3\sqrt{2}$,

$= 6\sqrt{6} - 2\sqrt{2} - 18\sqrt{2} + 6\sqrt{6}$

$= 12\sqrt{6} - 20\sqrt{2}$

Factor $\sqrt{2}$:

$= \sqrt{2}(12\sqrt{3} - 20)$

$= 2(6\sqrt{3} - 10)\sqrt{2}$

$= 12\sqrt{6} - 20\sqrt{2}$
This does not match.
Now test option C:
$(\sqrt{5} - \sqrt{3})^3$
$= 5\sqrt{5} - 3\sqrt{3} - 3\sqrt{15}(\sqrt{5} - \sqrt{3})$
$= 5\sqrt{5} - 3\sqrt{3} - 15\sqrt{3} + 9\sqrt{5}$
$= 14\sqrt{5} - 18\sqrt{3}$
Not matching.
Now test option A:
$(\sqrt{2} - \sqrt{6})^3$
$= 2\sqrt{2} - 6\sqrt{6} - 3\sqrt{12}(\sqrt{2} - \sqrt{6})$
$= 2\sqrt{2} - 6\sqrt{6} - 6\sqrt{6} + 18\sqrt{2}$
$= 20\sqrt{2} - 12\sqrt{6}$
Not matching.
Now test option B:
$\sqrt{7} - \sqrt{1} = \sqrt{7} - 1$
$(\sqrt{7} - 1)^3$
$= 7\sqrt{7} - 1 - 21 + 3\sqrt{7}$
$= 10\sqrt{7} - 22$
Hence,
$\sqrt[3]{(10\sqrt{7} - 22)} = \sqrt{7} - 1$
Correct Answer: B) $\sqrt{7} - \sqrt{1}$

10. $x = \sqrt{17} - \sqrt{16}, y = \sqrt{15} - \sqrt{14}$
Find the relation between $x$and y.
Simplify:$x = \sqrt{17} - 4$
Rationalize both expressions:
$x = \frac{17 - 16}{\sqrt{17} + 4} = \frac{1}{\sqrt{17} + 4}$
Similarly,$y = \frac{15-14}{\sqrt{15}+\sqrt{14}} = \frac{1}{\sqrt{15}+\sqrt{14}}$
Compare denominators: $\sqrt{17} + 4 > \sqrt{15} + \sqrt{14}$
A larger denominator gives a smaller value.
Therefore,$x < y$
Correct Answer: C) $x < y$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. If $x > 0$, then
$\sqrt{x\sqrt{x\sqrt{x \cdots \infty \text{ times}}}}$
equals:
Let $y = \sqrt{x\sqrt{x\sqrt{x \cdots}}}$
Then, $y = \sqrt{xy}$
Squaring both sides:
$y^2 = xy$
$y = x(\because y > 0)$
So the value is x.
Also, writing the infinite power form:

$y = x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots}$

$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots = 1$

Hence, $y = x^1 = x$

Therefore:

A) x

B) $x^{0.\bar{9}} = x^1 = x$

Correct Options: A, B

12. If $\sqrt{18 + x\sqrt{2}} = \sqrt{12} + \sqrt{6}$,

then the value of x is:

Square both sides:

$18 + x\sqrt{2} = (\sqrt{12} + \sqrt{6})^2$

$= 12 + 6 + 2\sqrt{72}$

$= 18 + 2(6\sqrt{2})$

$= 18 + 12\sqrt{2}$

Comparing:

$x\sqrt{2} = 12\sqrt{2}$

$x = 12$

Now check options:

A) $\sqrt{144} = 12$

B) 12

C) $\left(2\sqrt{3}\right)^2 = 12$

D) $6 \times 2 = 12$

Correct Options: A, B, C, D

13. If $x = 5 + 2\sqrt{6}$,

then the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$

is:

Notice: $5 + 2\sqrt{6} = (\sqrt{3} + \sqrt{2})^2$

So, $\sqrt{x} = \sqrt{3} + \sqrt{2}$

Now, $\frac{1}{\sqrt{x}} = \frac{1}{\sqrt{3}+\sqrt{2}}$

Rationalizing:

$$= \frac{\sqrt{3} - \sqrt{2}}{\left(\sqrt{3} + \sqrt{2}\right)(\sqrt{3} - \sqrt{2})}$$

$= \sqrt{3} - \sqrt{2}$

Therefore,

$$\sqrt{x} + \frac{1}{\sqrt{x}} = (\sqrt{3} + \sqrt{2}) + (\sqrt{3} - \sqrt{2})$$

$= 2\sqrt{3}$

Hence:

B) $\sqrt{12}$ D) $2\sqrt{3}$

Correct Options: B, D

14. The value of the continued fraction

$$1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cdots}}}$$

is:

Let

$$x=1+\cfrac{1}{1+\cfrac{1}{1+\cdots}}$$

Then,

$$x=1+\frac{1}{x}$$

Multiplying by x:

$x^2=x+1$

$x^2-x-1=0$

Using quadratic formula:

$$x=\frac{1\pm\sqrt{5}}{2}$$

Since the value is positive,

$$x=\frac{1+\sqrt{5}}{2}$$

Also,

$$\frac{2}{\sqrt{5}-1}=\frac{2(\sqrt{5}+1)}{5-1}=\frac{\sqrt{5}+1}{2}$$

So:

A) $\frac{1+\sqrt{5}}{2}$

C) $\frac{2}{\sqrt{5}-1}$

Correct Options: A, C

COMPREHENSION TYPE

Passage

15. If

$$x=\sqrt{30+\sqrt{30+\sqrt{30+\cdots}}}$$

then $x=$

Let $x=\sqrt{30+x}$

Squaring:

$x^2=30+x$

$x^2-x-30=0$

$(x-6)(x+5)=0$

$x=6$ or -5

Since square root values are non-negative,

$x=6$

Hence:

A) 6

Correct Option: A

16. The value of

$$\sqrt{1+\sqrt{7+\sqrt{1+\sqrt{7}+\cdots}}}$$

is:

Let $x = \sqrt{1+\sqrt{7+x}}$

Square both sides:

$x^2 = 1+\sqrt{7+x}$

$x^2 - 1 = \sqrt{7+x}$

Squaring again:

$(x^2-1)^2 = 7+x$

Checking options:

For $x = 2$:

$\sqrt{1+\sqrt{7+2}} = \sqrt{1+\sqrt{9}} = \sqrt{1+3} = \sqrt{4} = 2$

So $x = 2$ satisfies.

Hence:

C) 2

Correct Option: C

17. The value of

$$2+\sqrt{2+\sqrt{2+\cdots}}$$

is:

Let

$$x = \sqrt{2+\sqrt{2+\cdots}}$$

Then,

$x = \sqrt{2+x}$

Squaring:

$x^2 = 2+x$

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

Since the expression is positive,

$x = 2$

Therefore,

$2 + x = 2 + 2 = 4$

Hence:

B) 4

Correct Option: B

MATRIX MATCHING TYPE

MATCH Column-I and Column-II

(a) If $x = 4 - \sqrt{15}$, then the value of $x^2 - \frac{1}{x^2} =$

Find $\frac{1}{x}$:

$$\frac{1}{x} = \frac{1}{4-\sqrt{15}}$$

Rationalize the denominator:

$\frac{1}{x} = \frac{1}{4-\sqrt{15}} \times \frac{4+\sqrt{15}}{4+\sqrt{15}} = \frac{4+\sqrt{15}}{16-15} = 4+\sqrt{15}$

Calculate $\left(x-\frac{1}{x}\right)$and $\left(x+\frac{1}{x}\right)$:

$x - \frac{1}{x} = (4-\sqrt{15}) - (4+\sqrt{15}) = -2\sqrt{15}$

$x + \frac{1}{x} = (4-\sqrt{15}) + (4+\sqrt{15}) = 8$

Use the identity $a^2 - b^2 = (a-b)(a+b)$:

$x^2 - \frac{1}{x^2} = \left(x-\frac{1}{x}\right)\left(x+\frac{1}{x}\right)$

$= (-2\sqrt{15})(8) = -16\sqrt{15}$

Match: (a) →(R)

(b). Problem: $x = 2+\sqrt{3}$, $xy = 1$, then the value of

$\frac{x}{x-2} + \frac{y}{y-2} =$

Find y:

$xy = 1 \implies y = \frac{1}{x} = \frac{1}{2+\sqrt{3}}$

Rationalize the denominator:

$y = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}$

Find $(x-2)$and $(y-2)$:

$x - 2 = (2+\sqrt{3}) - 2 = \sqrt{3}$

$y - 2 = (2-\sqrt{3}) - 2 = -\sqrt{3}$

Substitute into the expression:

$\frac{x}{x-2} + \frac{y}{y-2} = \frac{2+\sqrt{3}}{\sqrt{3}} + \frac{2-\sqrt{3}}{-\sqrt{3}}$

$= \frac{2+\sqrt{3}}{\sqrt{3}} - \frac{2-\sqrt{3}}{\sqrt{3}}$

$= \frac{(2+\sqrt{3}) - (2-\sqrt{3})}{\sqrt{3}}$

$= \frac{2\sqrt{3}}{\sqrt{3}} = 2$

Match: (b) →(P)

(c). $\frac{5-\sqrt{11}}{7+2\sqrt{11}} = a\sqrt{11} - b$

Then the value of b is

Rationalize the denominator:

$\frac{5-\sqrt{11}}{7+2\sqrt{11}} \times \frac{7-2\sqrt{11}}{7-2\sqrt{11}}$

Expand the numerator:

$(5-\sqrt{11})(7-2\sqrt{11})$

$= 35 - 10\sqrt{11} - 7\sqrt{11} + 2(11)$

$= 35 - 17\sqrt{11} + 22$

$= 57 - 17\sqrt{11}$

Expand the denominator:

$(7)^2 - (2\sqrt{11})^2 = 49 - 4(11) = 49 - 44 = 5$

So,

$$\frac{57-17\sqrt{11}}{5}=\frac{57}{5}-\frac{17}{5}\sqrt{11}$$

Compare with:

$$a\sqrt{11}-b$$

$$-\frac{17}{5}\sqrt{11}+\frac{57}{5}=a\sqrt{11}-b$$

Comparing rational parts:

$$-b=\frac{57}{5}\implies b=-\frac{57}{5}$$

Match: (c) →(Q)

(d). Problem: If $x=7+4\sqrt{3}$, then the value of

$$x+\frac{1}{x}=$$

Find $\frac{1}{x}$:

$$\frac{1}{x}=\frac{1}{7+4\sqrt{3}}$$

Rationalize the denominator:

$$\frac{1}{x}=\frac{1}{7+4\sqrt{3}}\times\frac{7-4\sqrt{3}}{7-4\sqrt{3}}$$

$$=\frac{7-4\sqrt{3}}{49-16(3)}$$

$$=\frac{7-4\sqrt{3}}{49-48}$$

$$=7-4\sqrt{3}$$

Add x and $\frac{1}{x}$:

$$x+\frac{1}{x}=(7+4\sqrt{3})+(7-4\sqrt{3})=14$$

Match: (d) →(S)

INTEGER TYPE

19. If $x=(49-20\sqrt{6})^{-\frac{1}{2}}$

then find the value of

$$x^3+\frac{1}{x^3}-9\left(x^2+\frac{1}{x^2}\right)-7\left(x+\frac{1}{x}\right)-18$$

Steps:

First simplify x.

$$49-20\sqrt{6}=(5-2\sqrt{6})^2$$

So,

$$x=\left((5-2\sqrt{6})^2\right)^{-\frac{1}{2}}=\frac{1}{5-2\sqrt{6}}$$

Rationalize:

$$x=\frac{5+2\sqrt{6}}{25-24}=5+2\sqrt{6}$$

Hence,

$$\frac{1}{x}=5-2\sqrt{6}$$

Therefore,

$x + \frac{1}{x} = (5 + 2\sqrt{6}) + (5 - 2\sqrt{6}) = 10$

Now use identities.

$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$

$= 10^2 - 2 = 98$

Also,

$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$

$= 10^3 - 3(10) = 1000 - 30 = 970$

Substitute:

$970 - 9(98) - 7(10) - 18$

$= 970 - 882 - 70 - 18$

$= 0$

Correct Answer: 0

20. If

$\frac{\sqrt{13} - \sqrt{11}}{\sqrt{13} + \sqrt{11}} = a - b\sqrt{143}$

then find the value of

$a^2 - 143b$

Steps:

Rationalize the denominator:

$\frac{\sqrt{13} - \sqrt{11}}{\sqrt{13} + \sqrt{11}} \times \frac{\sqrt{13} - \sqrt{11}}{\sqrt{13} - \sqrt{11}}$

$= \frac{\left(\sqrt{13} - \sqrt{11}\right)^2}{13 - 11}$

Expand numerator:

$= \frac{13 + 11 - 2\sqrt{143}}{2}$

$= \frac{24 - 2\sqrt{143}}{2}$

$= 12 - \sqrt{143}$

Comparing with

$a - b\sqrt{143}$

we get $a = 12, b = 1$

Now,

$a^2 - 143b = 12^2 - 143(1)$

$= 144 - 143$

$= 1$

Correct Answer: 1

Student Practice Corner

1. Express 1.0001 in $\frac{p}{q}$form

Given:

$1.0001 = \frac{10001}{10000}$

Correct Answer: A) $\frac{10001}{10000}$

2. The value of $1.5\bar{6}$ in rational form is

Let

$x = 1.5\bar{6}$

$100x = 156.5\bar{6}$

Subtract:

$100x - x = 156.5\bar{6} - 1.5\bar{6}$

$99x = 155$

$x = \frac{155}{99}$

Correct Answer: A) $\frac{155}{99}$

3. Conjugate of $a + \sqrt{b}$ is

The conjugate is obtained by changing the sign between the terms.

$a + \sqrt{b} \rightarrow a - \sqrt{b}$

Correct Answer: A) $a - \sqrt{b}$

4. If $\sqrt{18 - x\sqrt{2}} = \sqrt{12} - \sqrt{6}$, find x

Square RHS:

$(\sqrt{12} - \sqrt{6})^2$

$= 12 + 6 - 2\sqrt{72}$

$= 18 - 12\sqrt{2}$

Compare with:

$18 - x\sqrt{2}$

So,

$x = 12$

Correct Answer: B) 12

5. The square root of $7 - \sqrt{48}$ is

$\sqrt{48} = 4\sqrt{3}$

So,

$7 - \sqrt{48} = 7 - 4\sqrt{3}$

Assume

$\sqrt{7 - 4\sqrt{3}} = a - b$

Check option $2 - \sqrt{3}$:

$(2 - \sqrt{3})^2$

$= 4 + 3 - 4\sqrt{3}$

$= 7 - 4\sqrt{3}$

Hence, $\sqrt{7 - \sqrt{48}} = 2 - \sqrt{3}$

Correct Answer: B) $2 - \sqrt{3}$

6. The positive square root of $8 + 2\sqrt{15}$ is

Assume

$\sqrt{8 + 2\sqrt{15}} = \sqrt{m} + \sqrt{n}$

Then,

$(\sqrt{m} + \sqrt{n})^2 = m + n + 2\sqrt{mn}$

Compare:

$m + n = 8, mn = 15$

Numbers satisfying this are:

$m = 5,\ n = 3$

Therefore,

$$\sqrt{8 + 2\sqrt{15}} = \sqrt{5} + \sqrt{3}$$

Correct Answer: C) $\sqrt{5} + \sqrt{3}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. Which expressions are equal to $4\sqrt{2}$?

A)$\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$

Correct

B)$2\sqrt{8} = 2(2\sqrt{2}) = 4\sqrt{2}$

Correct

C)$\sqrt{18} = 3\sqrt{2}$

Not equal

D)$\frac{\sqrt{128}}{2} = \frac{8\sqrt{2}}{2} = 4\sqrt{2}$

Correct

Correct Answers: A, B and D

8. Which of the following simplify to rational numbers?

A)$(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2}) = 5 - 2 = 3$

Rational

B)$(\sqrt{7})^2 = 7$

Rational

C)$\sqrt{12} \times \sqrt{3} = \sqrt{36} = 6$

Rational

D)$\sqrt{2} + \sqrt{8} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

Irrational

Correct Answers: A, B and C

9. Which of the following products are rational?

A)$(\sqrt{7} + \sqrt{2})(\sqrt{7} - \sqrt{2}) = 7 - 2 = 5$

Rational

B)$(\sqrt{3} + \sqrt{5})^2 = 3 + 5 + 2\sqrt{15} = 8 + 2\sqrt{15}$

Irrational

C)$(\sqrt{6})(\sqrt{24}) = \sqrt{144} = 12$

Rational

D)$(\sqrt{2} + \sqrt{3})(\sqrt{2} - \sqrt{3}) = 2 - 3 = -1$

Rational

Correct Answers: A, C and D

COMPREHENSION PASSAGE

Passage:

10. R.F of $7^{\frac{1}{3}} + 7^{-\frac{1}{3}}$

$$7^{\frac{1}{3}} = \sqrt[3]{7}, 7^{-\frac{1}{3}} = \sqrt[3]{\frac{1}{7}}$$

Expression:

$$\sqrt[3]{7} + \sqrt[3]{\frac{1}{7}}$$

Using,

$(\sqrt[3]{a}+\sqrt[3]{b})(\sqrt[3]{a^2}-\sqrt[3]{ab}+\sqrt[3]{b^2})=a+b$

Here,

$a=7, b=\frac{1}{7}$

So the rationalising factor is:

$\sqrt[3]{49}-1+\sqrt[3]{\frac{1}{49}}$

Correct Answer: B) $\boldsymbol{\sqrt[3]{49}-1+\sqrt[3]{\frac{1}{49}}}$

11. R.F of $\sqrt[3]{25}+1+\sqrt[3]{\frac{1}{25}}$

Observe:

$\sqrt[3]{25}+1+\sqrt[3]{\frac{1}{25}}$

matches the identity:

a^2+ab+b^2

where

$a=\sqrt[3]{5}, b=\sqrt[3]{\frac{1}{5}}$

Its rationalising factor is:

$a-b=\sqrt[3]{5}-\sqrt[3]{\frac{1}{5}}$

Answer: B) $\boldsymbol{\sqrt[3]{5}-\sqrt[3]{\frac{1}{5}}}$

12. Problem: Find the R.F. of $\sqrt[4]{3}+1$

First multiply by the conjugate:

$(\sqrt[4]{3}+1)(\sqrt[4]{3}-1)=\sqrt{3}-1$

Now rationalize $\sqrt{3}-1$ using its conjugate:

$(\sqrt{3}-1)(\sqrt{3}+1)=3-1=2$

Hence, the complete rationalising factor is:

$\left(\sqrt[4]{3}-1\right)(\sqrt{3}+1)$

Answer: Option A.

MATRIX MATCHING TYPE

13. MATCH Column-I and Column-II

Given:

$x=2+\sqrt{3}$

Since

$\frac{1}{x}=2-\sqrt{3}$

a) $x+\frac{1}{x}=(2+\sqrt{3})+(2-\sqrt{3})=4$

So, $a \rightarrow S$

b) $x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$

$=4^2-2=16-2=14$

So,

$b \to Q$

c)

$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$

$= 4^3 - 3(4) = 64 - 12 = 52$

So,

$c \to R$

d)

$x^2 - \frac{1}{x^2} = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)$

Now, $x - \frac{1}{x} = (2 + \sqrt{3}) - (2 - \sqrt{3}) = 2\sqrt{3}$

Hence, $x^2 - \frac{1}{x^2} = 4(2\sqrt{3}) = 8\sqrt{3}$

Then,

$x^4 - \frac{1}{x^4} = (8\sqrt{3})(14) = 112\sqrt{3}$

Therefore,

$\left(x^4 - \frac{1}{x^4}\right)^2 = (112\sqrt{3})^2 = 112^2 \times 3$

So, $d \to P$

INTEGER TYPE

14. Find the value of: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$

Use identity:

$(a + b)(a - b) = a^2 - b^2$

$= (\sqrt{5})^2 - (\sqrt{3})^2$

$= 5 - 3 = 2$

Answer: 2

15. If $(\sqrt{x})^2 = 169$, find x.

Since, $(\sqrt{x})^2 = x$

Therefore,

$x = 169$

Answer: 169

Logarithms

LEVEL-I

1. If $\log_e(x - 1) + \log_e x + \log_e(x + 1) = 0$, then

Using log property:

$\log_e[(x - 1)x(x + 1)] = 0$

$(x - 1)x(x + 1) = 1$

$x(x^2 - 1) = 1$

$x^3 - x - 1 = 0$

Answer: B) $x^3 - x - 1 = 0$

2. If $P = \log_3 5$ and $Q = \log_{17} 25$

$Q = \log_{17} 25 = \log_{17}(5^2) = 2\log_{17} 5$

Using change of base:

$P = \frac{\log 5}{\log 3}, Q = \frac{2\log 5}{\log 17}$

Since $\log 17 > 2\log 3$,
$Q < P$
Therefore,
$P > Q$
Answer: C) $P > Q$

3. The value of x for which
$\log_9 x - \log_9\left(\frac{x}{10}+\frac{1}{9}\right) = 1$
Using log property:
$$\log_9\left(\frac{x}{\frac{x}{10}+\frac{1}{9}}\right) = 1$$
Since $\log_9 9 = 1$,
$$\frac{x}{\frac{x}{10}+\frac{1}{9}} = 9$$
$$x = 9\left(\frac{x}{10}+\frac{1}{9}\right)$$
$$x = \frac{9x}{10} + 1$$
$$10x = 9x + 10$$
$$x = 10$$
Answer: D) 10

4. The value of
$$\frac{\log\sqrt{27} + \log 8 + \log\sqrt{1000}}{\log 120}$$
Combine logs:
$$= \frac{\log(\sqrt{27} \times 8 \times \sqrt{1000})}{\log 120}$$
$\sqrt{27} = 3\sqrt{3}, \sqrt{1000} = 10\sqrt{10}$
$3\sqrt{3} \times 8 \times 10\sqrt{10} = 240\sqrt{30}$
$240\sqrt{30} = 120\sqrt{120}$
So,
$$= \frac{\log(120\sqrt{120})}{\log 120}$$
$$= \frac{\log 120 + \log\sqrt{120}}{\log 120}$$
$$= \frac{\log 120 + \frac{1}{2}\log 120}{\log 120} = \frac{3}{2}$$
Answer: B) $\frac{3}{2}$

5. The value of
$\left(2^{\log_3 7} - 7^{\log_3 2}\right)$
Using identity:
$a^{\log_c b} = b^{\log_c a}$
$2^{\log_3 7} = 7^{\log_3 2}$
Therefore,
$7^{\log_3 2} - 7^{\log_3 2} = 0$
Answer: A) 0

6. The value of

$\log_{y^2} x^3 \cdot \log_{z^2} y^3 \cdot \log_{x^2} z^3$

Using change of base:

$= \frac{\log x^3}{\log y^2} \times \frac{\log y^3}{\log z^2} \times \frac{\log z^3}{\log x^2}$

$= \frac{3\log x}{2\log y} \times \frac{3\log y}{2\log z} \times \frac{3\log z}{2\log x}$

Cancel common terms:

$= \frac{3 \times 3 \times 3}{2 \times 2 \times 2} = \frac{27}{8}$

Answer: B) $\frac{27}{8}$

7. The value of

$$\frac{1}{\log_2(\log_2(\log_3 81))}$$

First, $\log_3 81 = 4$

Then, $\log_2 4 = 2$

Then, $\log_2 2 = 1$

So,

$\frac{1}{1} = 1$

Answer: B) 1

8. The value of

$$\log_{\sqrt{a}} \sqrt{a\sqrt{a\sqrt{a\sqrt{a}}}}$$

Write the nested root as powers of a:

$\sqrt{a} = a^{1/2}$

Inside exponent:

$$\sqrt{a\sqrt{a\sqrt{a\sqrt{a}}}} = a^{31/32}$$

Now,

$\log_{a^{1/2}} a^{31/32} = \frac{31/32}{1/2} = \frac{31}{16}$

Answer: D) $\frac{31}{16}$

9. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, find

$\log_{10}\left(\frac{2^3 \times 3^2}{5^2}\right)$

Using log laws:

$= 3\log 2 + 2\log 3 - 2\log 5$

Since,

$\log 5 = \log 10 - \log 2 = 1 - 0.3010 = 0.6990$

So,

$= 3(0.3010) + 2(0.4771) - 2(0.6990)$

$= 0.9030 + 0.9542 - 1.3980$

$= 0.4592$

Answer: A) 0.4592

10. The value of

$\log_3 \sqrt[4]{729 \cdot \sqrt[3]{9^{-1}27^{-4/3}}}$

Convert everything to powers of 3:

$729 = 3^6, 9^{-1} = 3^{-2}, 27^{-4/3} = 3^{-4}$

Inside cube root:

$3^{-2} \times 3^{-4} = 3^{-6}$

$\sqrt[3]{3^{-6}} = 3^{-2}$

Then, $729 \times 3^{-2} = 3^6 \times 3^{-2} = 3^4$

Now fourth root:

$\sqrt[4]{3^4} = 3$

Therefore,

$\log_3 3 = 1$

Answer: B) 1

11. If

$3^{\log_3 5} + 5^{\log_x 3} = 8$

Using,

$a^{\log_a b} = b$

$3^{\log_3 5} = 5$

So,

$5 + 5^{\log_x 3} = 8$

$5^{\log_x 3} = 3$

Check options:

For $x = 5$,

$\log_5 3$

Hence,

$5^{\log_5 3} = 3$

Therefore,

$x = 5$

Answer: B) 5

12. If $\log 3 = 0.4771$, then the number of digits in $3^{100}$is

Number of digits in N:

$= [\log N] + 1$

$\log(3^{100}) = 100\log 3$

$= 100(0.4771) = 47.71$

Digits:

$= 47 + 1 = 48$

Answer: B) 48

13. The value of

$\log[1-\{1-(1-x^2)^{-1}\}^{-1}]^{-1/2}$

Simplify inside:

$1-(1-x^2)^{-1} = 1 - \frac{1}{1-x^2}$

$= \frac{1-x^2-1}{1-x^2} = \frac{-x^2}{1-x^2}$

Taking inverse:

$\left(\frac{-x^2}{1-x^2}\right)^{-1} = \frac{1-x^2}{-x^2}$

Now,

$1 - \frac{1-x^2}{-x^2} = \frac{-x^2-(1-x^2)}{-x^2} = \frac{-1}{-x^2} = \frac{1}{x^2}$

Then, $\left(\frac{1}{x^2}\right)^{-1/2} = x$

Hence, $\log x$

Answer: B) $\log x$

14. Find the value of

$\log_{10} 2 + 16\log_{10}\frac{16}{15} + 12\log_{10}\frac{25}{24} + 7\log_{10}\frac{81}{80}$

Using

$n\log a = \log a^n$

$= \log\left[2 \cdot \left(\frac{16}{15}\right)^{16} \cdot \left(\frac{25}{24}\right)^{12} \cdot \left(\frac{81}{80}\right)^{7}\right]$

Write in prime factors:

$16 = 2^4, 25 = 5^2, 81 = 3^4$

$15 = 3 \cdot 5, 24 = 2^3 \cdot 3, 80 = 2^4 \cdot 5$

So,

$= \log\left[2 \cdot \frac{2^{64}}{3^{16}5^{16}} \cdot \frac{5^{24}}{2^{36}3^{12}} \cdot \frac{3^{28}}{2^{28}5^{7}}\right]$

Combine powers:

For 2:

$2^{1+64-36-28} = 2^1$

For 3:

$3^{-16-12+28} = 3^0$

For 5:

$5^{-16+24-7} = 5^1$

Thus,

$= \log(2 \times 5) = \log 10$

$= 1$

Correct Answer: D) 1

15. The value of $\left(\frac{1}{\log_5 210} + \frac{1}{\log_6 210} + \frac{1}{\log_7 210}\right)$

Using,

$\frac{1}{\log_a b} = \log_b a$

$= \log_{210} 5 + \log_{210} 6 + \log_{210} 7$

$= \log_{210}(5 \times 6 \times 7)$

$= \log_{210} 210 = 1$

Answer: B) 1

16. The value of

$\log\frac{1}{2} + \log\frac{2}{3} + \log\frac{3}{4} + \cdots + \log\frac{99}{100}$

Combine logs:

$= \log\left(\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \cdots \times \frac{99}{100}\right)$

All middle terms cancel:

$= \log\frac{1}{100}$

$= \log 10^{-2} = -2$

Answer: C) -2

16. Find the value of

$\log\left(\frac{1}{2}\right) + \log\left(\frac{2}{3}\right) + \log\left(\frac{3}{4}\right) + \cdots + \log\left(\frac{99}{100}\right)$

Using

$\log a + \log b = \log(ab)$

$= \log\left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{99}{100}\right)$

All middle terms cancel:

$= \log\left(\frac{1}{100}\right)$

$= \log(10^{-2}) = -2$

Correct Answer: C) -2

17. If $\log_3 2 = x$, find

$\frac{\log_{10} 72}{\log_{10} 24}$

Using change of base,

$x = \frac{\log 2}{\log 3}$

So,

$\log 2 = x\log 3$

Now,

$72 = 2^3 \cdot 3^2$

$\log 72 = 3\log 2 + 2\log 3$

Substitute $\log 2 = x \log 3$:

$= 3x\log 3 + 2\log 3 = (3x + 2)\log 3$

Similarly,

$24 = 2^3 \cdot 3$

$\log 24 = 3\log 2 + \log 3$

$= 3x\log 3 + \log 3 = (3x + 1)\log 3$

Therefore,

$\frac{\log 72}{\log 24} = \frac{(3x + 2)\log 3}{(3x + 1)\log 3} = \frac{3x + 2}{3x + 1}$

Correct Answer C:

$\boxed{\frac{3x + 2}{3x + 1}}$

18. If $p = \log_6 216$ and $q = \log_3 25$, find p^q

Since

$216 = 6^3$

$p = \log_6 216 = 3$

Also,

$q = \log_3 25$

Therefore,

$p^q = 3^{\log_3 25} = 25$

Correct Answer: B) 25

19. If $x^4 + y^4 = 83x^2y^2$, find the value of

$\log\left(\frac{x^2 - y^2}{9}\right)$

Given,

$x^4 + y^4 = 83x^2y^2$

Add $2x^2y^2$ on both sides:

$x^4 + 2x^2y^2 + y^4 = 85x^2y^2$

$(x^2 + y^2)^2 = 85x^2y^2$

Now,

$(x^2 - y^2)^2 = (x^2 + y^2)^2 - 4x^2y^2$

$= 85x^2y^2 - 4x^2y^2$

$= 81x^2y^2$

$x^2 - y^2 = 9xy$

So,

$\frac{x^2 - y^2}{9} = xy$

Therefore,

$\log\left(\frac{x^2 - y^2}{9}\right) = \log(xy)$

$= \log x + \log y$

Correct Answer: D) $\log x + \log y$

20. Find the value of

$\frac{1}{\log_{\frac{a}{b}} x} + \frac{1}{\log_{\frac{b}{c}} x} + \frac{1}{\log_{\frac{c}{a}} x}$

Using $\frac{1}{\log_a x} = \log_x a$

$= \log_x \frac{a}{b} + \log_x \frac{b}{c} + \log_x \frac{c}{a}$

Combine logs:

$= \log_x \left(\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{a}\right)$

$= \log_x 1$

$= 0$

Correct Answer: A) 0

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. If $\log_{10} x + \log_{10} \sqrt{x} + \log_{10} \sqrt[4]{x} + \log_{10} \sqrt[8]{x} + \cdots = 2$

Write roots as powers:

$= \log x + \log x^{1/2} + \log x^{1/4} + \log x^{1/8} + \cdots$

$= \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots\right) \log x$

The GP sum is:

$\frac{1}{1 - \frac{1}{2}} = 2$

So,

$2\log x = 2$

$\log x = 1$

$x = 10$

Correct Answer: D) 10

2. If $\log_r 6 = s$ and $\log_r 3 = t$, find $\log_r \left(\frac{r}{2}\right)$

$\log_r 6 = \log_r 2 + \log_r 3$

$s = \log_r 2 + t$

$\log_r 2 = s - t$

Now,

$\log_r \left(\frac{r}{2}\right) = \log_r r - \log_r 2$

$= 1 - (s - t)$

$= 1 - s + t$

Correct Answer: B) $1 - s + t$

3. If $\log\left(\frac{13^2}{\sqrt{2^3}\times 5}\right) = 2\log_a 13 - \log_a 5 - X$

Using log laws:

$\log_a \left(\frac{13^2}{5 \cdot 2^{3/2}}\right) = \log_a 13^2 - \log_a 5 - \log_a 2^{3/2}$

$= 2\log_a 13 - \log_a 5 - \frac{3}{2}\log_a 2$

Comparing,

$X = \frac{3}{2}\log_a 2$

$a^X = 2^{3/2}$

Correct Answer: A) $a^X = 2^{3/2}$

4. If $\frac{\log x}{a^2+ab+b^2} = \frac{\log y}{b^2+bc+c^2} = \frac{\log z}{c^2+ca+a^2}$

let the common value be k.

Then,

$\log x = k(a^2 + ab + b^2)$

$\log y = k(b^2 + bc + c^2)$

$\log z = k(c^2 + ca + a^2)$

Now,

$\log\left(x^{a-b}y^{b-c}z^{c-a}\right)$

$= (a - b)\log x + (b - c)\log y + (c - a)\log z$

Substitute values:

$= k[(a - b)(a^2 + ab + b^2) + (b - c)(b^2 + bc + c^2) + (c - a)(c^2 + ca + a^2)]$

Using

$(a - b)(a^2 + ab + b^2) = a^3 - b^3$

Similarly, $= b^3 - c^3 + c^3 - a^3 = 0$

Hence, $\log\left(x^{a-b}y^{b-c}z^{c-a}\right) = 0$

Therefore, $x^{a-b}y^{b-c}z^{c-a} = 1$

Correct Answer: B) 1

5. If $0.0001 < n < 0.001$

Write in powers of 10:

$10^{-4} < n < 10^{-3}$

Taking $\log_{10}$throughout,

$-4 < \log_{10} n < -3$

Correct Answer: A) $-4 < \log_{10} n < -3$

6. If $x = \log_5 7, y = \log_5 3$

Since base $5 > 1$, logarithmic function is increasing.

Because $7 > 3$

therefore,

$\log_5 7 > \log_5 3$

$x > y$

Correct Answer: A) $x > y$

7. If $5^x = (0.5)^y = 1000$

Then, $5^x = 1000 \Rightarrow x = \log_5 1000$

and

$\left(0.5\right)^y = 1000 \Rightarrow \left(\frac{1}{2}\right)^y = 1000$

$2^{-y} = 1000 \Rightarrow -y = \log_2 1000$

$y = -\log_2 1000$

Now,

$\frac{1}{x} - \frac{1}{y} = \frac{1}{\log_5 1000} + \frac{1}{\log_2 1000}$

Using

$\frac{1}{\log_a b} = \log_b a$

$= \log_{1000} 5 + \log_{1000} 2$

$= \log_{1000}(10)$

Since $1000 = 10^3$,

$\log_{1000} 10 = \frac{1}{3}$

Correct Answer: C) $\frac{1}{3}$

8. Solve $\log_2(x-7) < \log_4(x-7)$

Domain:

$x - 7 > 0 \Rightarrow x > 7$

Using change of base,

$\log_4(x-7) = \frac{1}{2}\log_2(x-7)$

So,

$\log_2(x-7) < \frac{1}{2}\log_2(x-7)$

$\frac{1}{2}\log_2(x-7) < 0$

$\log_2(x-7) < 0$

Since base $2 > 1$,

$0 < x - 7 < 1$

$7 < x < 8$

Correct Answer: B) $(7, 8)$

9. Number of digits in 16^{50}

We know:

$16^{50} = (2^4)^{50} = 2^{200}$

Number of digits in $N$is:

$\lfloor \log_{10} N \rfloor + 1$

So,$\log_{10}(2^{200}) = 200\log_{10} 2$

Given:

$\log_{10} 2 = 0.3010$

$200 \times 0.3010 = 60.2$

Hence number of digits:

$\lfloor 60.2 \rfloor + 1 = 60 + 1 = 61$

Correct Answer: B) 61

10. Number of digits in $(214)^{15}$

Number of digits:
$\lfloor \log_{10}(214^{15}) \rfloor + 1$
$= \lfloor 15\log_{10} 214 \rfloor + 1$
Given:
$\log_{10} 21.4 = 1.3304$
Since:
$214 = 21.4 \times 10$
$\log_{10} 214 = \log_{10} 21.4 + \log_{10} 10$
$= 1.3304 + 1 = 2.3304$
Now,
$15 \times 2.3304 = 34.956$
Number of digits:
$\lfloor 34.956 \rfloor + 1 = 34 + 1 = 35$
Correct Answer: C) 35

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Given
$\frac{\log a}{b-c} = \frac{\log b}{c-a} = \frac{\log c}{a-b} = k$
Then
$\log a = k(b-c), \log b = k(c-a), \log c = k(a-b)$
Adding all three:
$\log a + \log b + \log c = 0$
$\log(abc) = 0$
$abc = 1$
So, option C is correct.
Now check others:

A) $a^a b^b c^c$
Taking log:
$a\log a + b\log b + c\log c$
$= ak(b-c) + bk(c-a) + ck(a-b) = 0$
Hence value = 1.
So A is correct.

B) Exponent sum:
$(b+c)\log a + (c+a)\log b + (a+b)\log c$
Substitute:
$k[(b+c)(b-c) + (c+a)(c-a) + (a+b)(a-b)]$
$= k[(b^2 - c^2) + (c^2 - a^2) + (a^2 - b^2)] = 0$
Hence value = 1.
So B is correct.

D) Exponent expression:
$(b^2 - bc + c^2)\log a + (a^2 - ac + b^2)\log b + (b^2 - ab + a^2)\log c$
After substitution and simplification, total becomes 0.
Hence value = 1.
So D is also correct.
Correct answers: A, B, C, D

12. Let
$\log_2 x = t$
Then
$x^{(\log_2 x)^2 - 6\log_2 x + 11} = 64$
becomes
$(2^t)^{t^2-6t+11} = 2^6$
$2^{t(t^2-6t+11)} = 2^6$
$t^3 - 6t^2 + 11t = 6$
$t^3 - 6t^2 + 11t - 6 = 0$
$(t-1)(t-2)(t-3) = 0$
So $t = 1,2,3$
Hence $x = 2,4,8$
Correct answers: A, B, C

13. $x = \log_{0.1}(0.001) = 3$
because
$(0.1)^3 = 0.001$
Also,
$y = \log_9 81 = 2$
Now, $\sqrt{x - 2\sqrt{y}} = \sqrt{3 - 2\sqrt{2}}$
Using $3 - 2\sqrt{2} = (\sqrt{2} - 1)^2$
Therefore,
$\sqrt{3 - 2\sqrt{2}} = \sqrt{2} - 1$
Correct answer: C

14. Given $|\log_{10} x| < 1$
This means
$-1 < \log_{10} x < 1$
Taking antilog:
$10^{-1} < x < 10^1$
$0.1 < x < 10$
Correct answer: A

COMPREHENSION TYPE

Passage - I

15. Let $t = \log_x 5$
Also, $\log_{\sqrt{x}} 5 = \frac{\log_x 5}{\log_x \sqrt{x}}$
Since
$\log_x \sqrt{x} = \frac{1}{2}$
we get
$\log_{\sqrt{x}} 5 = 2t$
Equation becomes
$t^3 - t^2 - 3(2t) = 0$
$t^3 - t^2 - 6t = 0$
$t(t^2 - t - 6) = 0$
$t(t-3)(t+2) = 0$

So,$t = 0,3,-2$
Now,$\log_x 5 = 0$
is impossible because $5 \neq 1$.
For $t = 3, x^3 = 5 \Rightarrow x = 5^{1/3}$
For
$t = -2, x^{-2} = 5 \Rightarrow x = \frac{1}{\sqrt{5}}$
Hence number of solutions = 2.
Correct answer: B

16. Convert all logs to base 2.
$\log_{\sqrt{2}} x = \frac{\log_2 x}{\log_2 \sqrt{2}} = \frac{\log_2 x}{1/2} = 2\log_2 x$
Also,
$\log_{1/2} x = \frac{\log_2 x}{\log_2 (1/2)} = -\log_2 x$
Let $t = \log_2 x$
Then
$2t + 2t - t = 9$
$3t = 9$
$t = 3$
Thus, $x = 2^3 = 8$
Solution set: {8}
Correct answer: B

17. Let $t = \log_5 x$
Then $x = 5^t$
Now, $\log_{5x}\left(\frac{5}{x}\right) = \log_{5^{t+1}}(5^{1-t}) = \frac{1-t}{1+t}$
Equation becomes
$t^2 + \frac{1-t}{1+t} = 1$
Multiply by (1+t):
$t^2(1+t) + 1 - t = 1 + t$
$t^3 + t^2 - 2t = 0$
$t(t^2 + t - 2) = 0$
$t(t+2)(t-1) = 0$
So,$t = 0, -2, 1$
Hence,$x = 1, \frac{1}{25}, 5$
Solution set:$\left\{1, 5, \frac{1}{25}\right\}$
Correct answer: C

MATRIX MATCHING TYPE

a) $\frac{\log_2 32}{\log_3 \sqrt[5]{243}}$
$\log_2 32 = 5$
Also,$\sqrt[5]{243} = \sqrt[5]{3^5} = 3$
So,$\log_3 3 = 1$
Hence value
$= \frac{5}{1} = 5$
5 is a prime number.

Matches: S

b) $\frac{2\log 6}{\log 12+\log 3}$

$= \frac{2\log 6}{\log 36} = \frac{\log 36}{\log 36} = 1$

1 is a positive integer.

Matches: P

c) $\log_{1/4}\left(\left(\frac{1}{16}\right)^{-2}\right)$

$\left(\frac{1}{16}\right)^{-2} = 256$

$\log_{1/4} 256 = -4$

-4 is a negative integer.

Matches: Q

d) $\frac{\log_5 16-\log_5 4}{\log_5 128}$

$= \frac{\log_5(16/4)}{\log_5 128} = \frac{\log_5 4}{\log_5 128}$

$= \log_{128} 4 = \log_{2^7} 2^2 = \frac{2}{7}$

$\frac{2}{7}$ is rational but not integer.

Matches: R

INTEGER TYPE

19. If $x = \log_3 \log_2 \log_2 256$, then $2^{\log_4 2^x} = ?$

Steps:

$\log_2 256 = 8$

$\log_2 8 = 3$

$x = \log_3 3 = 1$

Now, $2^{\log_4 2^x} = 2^{\log_4 2}$

Since $x = 1$,

$\log_4 2 = \frac{1}{2}$

So,

$2^{1/2} = \sqrt{2}$

Answer: $\sqrt{2}$

20. The value of $5^{\sqrt{\log_5 7}} - 7^{\sqrt{\log_7 5}}$ is _____.

Let $a = \sqrt{\log_5 7}$

Then, $a^2 = \log_5 7$

Using log property: $5^{a^2} = 7$

Similarly, if $b = \sqrt{\log_7 5}$

then $7^{b^2} = 5$

Also, $\log_5 7 \cdot \log_7 5 = 1$

So, $ab = 1$

Hence,

$a = \frac{1}{b}$

Therefore, $5^a = 7^b$

So, $5^{\sqrt{\log_5 7}} - 7^{\sqrt{\log_7 5}} = 0$

Answer: 0

Student Practise Corner

1. The expanding form of $\log\left(\frac{xy^2}{z^3}\right)$

Using logarithm laws:

$\log\left(\frac{a}{b}\right) = \log a - \log b$

$\log(ab) = \log a + \log b$

$\log(a^n) = n\log a$

So,$\log\left(\frac{xy^2}{z^3}\right) = \log x + \log y^2 - \log z^3$

$= \log x + 2\log y - 3\log z$

Correct Answer: C) $\log x + 2\log y - 3\log z$

2. The value of $\log_2 \log_{25} 5$

First find:$\log_{25} 5 = \frac{1}{2}$

because $25^{1/2} = 5$

Now,$\log_2\left(\frac{1}{2}\right) = -1$

Correct Answer: C) -1

3. The value of $\log_{5\sqrt{5}} 125$

Write in powers of 5:

$125 = 5^3$

$5\sqrt{5} = 5 \cdot 5^{1/2} = 5^{3/2}$

So, $\log_{5^{3/2}} 5^3 = \frac{3}{3/2} = 2$

Correct Answer: A) 2

4. If $\log[4 - 5\log_{32}(x+5)] = 0$

Then $4 - 5\log_{32}(x+5) = 1$

because $\log 1 = 0$.

So,$5\log_{32}(x+5) = 3$

$\log_{32}(x+5) = \frac{3}{5}$

Since $32 = 2^5$,

$x + 5 = 32^{3/5} = (2^5)^{3/5} = 2^3 = 8$

$x = 8 - 5 = 3$

Correct Answer: C) 3

5. The value of $\log_{\sqrt{6}} 216$

Write 216 as a power of 6:

$216 = 6^3$

Also,$\sqrt{6} = 6^{1/2}$

So, $\log_{6^{1/2}} 6^3 = \frac{3}{1/2} = 6$

Correct Answer: D) 6

6. If $\log_{10}\left(98+\sqrt{x^2 - 12x + 36}\right) = 2$

then $98 + \sqrt{x^2 - 12x + 36} = 100$

because $\log_{10} 100 = 2$

So, $\sqrt{x^2 - 12x + 36} = 2$

Notice, $x^2 - 12x + 36 = (x-6)^2$
Thus,
$\sqrt{(x-6)^2} = 2$
$|\, x-6 \,| = 2$

So, $x - 6 = 2$ or $x - 6 = -2$
$x = 8 \text{or} x = 4$
Correct Answer: C) 8,4

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. The solutions of $x^{\log_4 x} = 2^{3(\log_4 x+3)}$
Let $\log_4 x = t$
Then, $x = 4^t = (2^2)^t = 2^{2t}$
Now, $x^{\log_4 x} = (2^{2t})^t = 2^{2t^2}$
and $2^{3(\log_4 x+3)} = 2^{3(t+3)} = 2^{3t+9}$
Equating powers:
$2t^2 = 3t + 9$
$2t^2 - 3t - 9 = 0$
$(2t+3)(t-3) = 0$
So, $t = 3$ or $t = -\frac{3}{2}$
Case 1: $\log_4 x = 3 \Rightarrow x = 4^3 = 64$
Case 2: $\log_4 x = -\frac{3}{2} \Rightarrow x = 4^{-3/2} = \frac{1}{8}$
Correct Answers: B) 64 and C) $\frac{1}{8}$

8. Which of the following use the change of base rule correctly?
Change of base rule:
$\log_a b = \frac{\log b}{\log a}$

A) $\log_2 16 = \frac{\log 16}{\log 2}$
Correct
B) $\log_5 25 = \frac{\ln 25}{\ln 5}$
Correct
C) $\log_3 81 = \frac{\log 3}{\log 81}$
Incorrect
(It should be $\frac{\log 81}{\log 3}$)
D) $\log_7 49 = \frac{\log 49}{\log 7}$
Correct
Correct Answers: A, B and D

9. Which of the following are correct?
A) $\sqrt{a} \times \sqrt{a} = a$
Correct
B) $\sqrt[3]{a} \times \sqrt[3]{a^2} = \sqrt[3]{a^3} = a$
Correct
C) $\log_2 4 + \log_2 8 = \log_2(4 \times 8) = \log_2 32$
Correct

D) $\log_3 9 - \log_3 3 = 2 - 1 = 1$

Incorrect

Correct Answers: A, B and C

COMPREHENSION TYPE

PASSAGE

10. The number of solutions of

$\log_5(x^4 + 5) - \log_{\frac{1}{5}}(x^2 + 25) = \frac{3}{2}$

Use: $\log_{\frac{1}{5}} a = -\log_5 a$

So the equation becomes

$\log_5(x^4 + 5) + \log_5(x^2 + 25) = \frac{3}{2}$

$\log_5[(x^4 + 5)(x^2 + 25)] = \frac{3}{2}$

Convert to exponential form:

$(x^4 + 5)(x^2 + 25) = 5^{3/2} = 5\sqrt{5}$

Now, $x^4 + 5 \geq 5$

and $x^2 + 25 \geq 25$

Thus, $(x^4 + 5)(x^2 + 25) \geq 5 \times 25 = 125$

But $125 > 5\sqrt{5}$

So the equation is impossible.

Correct Answer: A) 0

11. The solution set of

$\log_3(3 + \sqrt{x}) + \frac{1}{2}\log_{\sqrt{3}}(1 + x^2) = 0$

Use change of base:

$\log_{\sqrt{3}}(1 + x^2) = \frac{\log_3(1 + x^2)}{\log_3 \sqrt{3}}$

Since $\log_3 \sqrt{3} = \frac{1}{2}$

we get $\log_{\sqrt{3}}(1 + x^2) = 2\log_3(1 + x^2)$

Therefore,

$\frac{1}{2}\log_{\sqrt{3}}(1 + x^2) = \log_3(1 + x^2)$

So,

$\log_3(3 + \sqrt{x}) + \log_3(1 + x^2) = 0$

$\log_3[(3 + \sqrt{x})(1 + x^2)] = 0$

Hence, $(3 + \sqrt{x})(1 + x^2) = 1$

But for $x \geq 0$, $3 + \sqrt{x} \geq 3$

and $1 + x^2 \geq 1$

Thus, $(3 + \sqrt{x})(1 + x^2) \geq 3$

which can never equal 1.

So there is no solution.

Correct Answer: D) ϕ

Matrix Matching Type

12. Match Column–I and Column–II

a) If $b = \frac{a^x + a^{-x}}{2}$

Then, $2b = a^x + a^{-x}$

Let $y = a^x$. Then $a^{-x} = \frac{1}{y}$.

$2b = y + \frac{1}{y}$

$2by = y^2 + 1$

$y^2 - 2by + 1 = 0$

$y = b \pm \sqrt{b^2 - 1}$

Since $y = a^x$, $x = \log_a\left(b + \sqrt{b^2 - 1}\right)$

Matches with T

b) If $b = \frac{a^x - a^{-x}}{2}$

Let $y = a^x$.

$2b = y - \frac{1}{y}$

$2by = y^2 - 1$

$y^2 - 2by - 1 = 0$

$y = b + \sqrt{b^2 + 1}$

Thus, $x = \log_a\left(b + \sqrt{b^2 + 1}\right)$

Matches with P

c) If $b = \frac{a^x - a^{-x}}{a^x + a^{-x}}$

Let $y = a^x$.

$$b = \frac{y - \frac{1}{y}}{y + \frac{1}{y}} = \frac{y^2 - 1}{y^2 + 1}$$

$b(y^2 + 1) = y^2 - 1$

$y^2(1 - b) = 1 + b$

$$y^2 = \frac{1 + b}{1 - b}$$

$$y = \sqrt{\frac{1 + b}{1 - b}}$$

Hence, $x = \frac{1}{2}\log_a\left(\frac{1+b}{1-b}\right)$

Matches with S

d) If $b = \frac{a^x + a^{-x}}{a^x - a^{-x}}$

Let $y = a^x$.

$$b = \frac{y + \frac{1}{y}}{y - \frac{1}{y}} = \frac{y^2 + 1}{y^2 - 1}$$

$b(y^2 - 1) = y^2 + 1$

$y^2(b - 1) = b + 1$

$$y^2 = \frac{b + 1}{b - 1}$$

$y = \sqrt{\frac{b+1}{b-1}}$

Therefore, $x = \frac{1}{2}\log_a\left(\frac{b+1}{b-1}\right)$

Matches with R

INTEGER TYPE

13. If $2^{\log_3 9} + 25^{\log_9 3} = 8^{\log_x 9}$

First, $\log_3 9 = 2$

So, $2^{\log_3 9} = 2^2 = 4$

Also, $\log_9 3 = \frac{1}{2}$

Thus, $25^{\log_9 3} = 25^{1/2} = 5$

Hence,

$4 + 5 = 8^{\log_x 9}$

$9 = 8^{\log_x 9}$

Write $8 = 2^3$:

$9 = 2^{3\log_x 9}$

Since $9 = 3^2$,

$3^2 = 2^{3\log_x 9}$

Taking log base 2:

$\log_2 9 = 3\log_x 9$

$\log_x 9 = \frac{1}{3}\log_2 9$

Using change of base:

$\frac{\log 9}{\log x} = \frac{1}{3} \cdot \frac{\log 9}{\log 2}$

$\frac{1}{\log x} = \frac{1}{3\log 2}$

$\log x = 3\log 2$

$x = 2^3 = 8$

Answer: 8

14. If $\log_{x+1}(2x-1) + \log_{2x-1}(x+1) = 2$

Let $a = \log_{x+1}(2x-1)$

Then,

$\log_{2x-1}(x+1) = \frac{1}{a}$

So,

$a + \frac{1}{a} = 2$

$a^2 - 2a + 1 = 0$

$(a-1)^2 = 0$

$a = 1$

Thus, $\log_{x+1}(2x-1) = 1$

$2x - 1 = x + 1$

$x = 2$

Answer: 2

15. The number of solutions of

$\log_2\left(\frac{8}{2^x}-1\right)=x-2$

Simplify:

$\frac{8}{2^x}=2^{3-x}$

So, $\log_2(2^{3-x}-1)=x-2$

Convert to exponential form:

$2^{3-x}-1=2^{x-2}$

Let $t=2^{x-2}$

Then, $2^{3-x}=\frac{2}{t}$

So,

$\frac{2}{t}-1=t$

$2-t=t^2$

$t^2+t-2=0$

$(t-1)(t+2)=0$

Since $t>0$, $t=1$

Thus,

$2^{x-2}=1$

$x-2=0$

$x=2$

Hence, the number of solutions is 1.

2.Trigonometry

LEVEL – I

1. If $\sin\theta = \frac{3}{5}$, find

$$\frac{\tan\theta - 2\cos\theta}{3\sin\theta + \sec\theta}$$

Given: $\sin\theta = \frac{3}{5}$

So, $\cos\theta = \frac{4}{5}, \tan\theta = \frac{3}{4}, \sec\theta = \frac{5}{4}$

Substitute:

$$\frac{\frac{3}{4} - 2\left(\frac{4}{5}\right)}{3\left(\frac{3}{5}\right) + \frac{5}{4}} = \frac{\frac{3}{4} - \frac{8}{5}}{\frac{9}{5} + \frac{5}{4}}$$

$$= \frac{\frac{15-32}{20}}{\frac{36+25}{20}} = \frac{-17/20}{61/20} = -\frac{17}{61}$$

Correct Answer: C) $-\frac{17}{61}$

2. If $7\tan\theta = 24$, find $\sin\theta + \cos\theta$

$$\tan\theta = \frac{24}{7}$$

Take opposite = 24, adjacent = 7

Hypotenuse:

$$\sqrt{24^2 + 7^2} = \sqrt{576 + 49} = 25$$

So,

$$\sin\theta = \frac{24}{25}, \cos\theta = \frac{7}{25}$$

$$\sin\theta + \cos\theta = \frac{24}{25} + \frac{7}{25} = \frac{31}{25}$$

Correct Answer: B) $\frac{31}{25}$

3. If $\sin\theta = \frac{1}{3}$, find $2\cot^2\theta + 2$

$$\sin\theta = \frac{1}{3}$$

$$\cos\theta = \sqrt{1 - \frac{1}{9}} = \sqrt{\frac{8}{9}} = \frac{2\sqrt{2}}{3}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{2\sqrt{2}/3}{1/3} = 2\sqrt{2}$$

$\cot^2\theta = (2\sqrt{2})^2 = 8$

$2\cot^2\theta + 2 = 2(8) + 2 = 18$

Correct Answer: C) 18

4. If $\operatorname{cosec}\theta = \frac{13}{12}$, find

$$\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$$

Given:

$$\operatorname{cosec}\theta = \frac{13}{12} \Rightarrow \sin\theta = \frac{12}{13}$$

$\cos\theta = \frac{5}{13}$

Substitute:

$$\frac{2\left(\frac{12}{13}\right) - 3\left(\frac{5}{13}\right)}{4\left(\frac{12}{13}\right) - 9\left(\frac{5}{13}\right)}$$

$$= \frac{\frac{24-15}{13}}{\frac{48-45}{13}} = \frac{9/13}{3/13} = 3$$

Correct Answer: B) 3

5. Find

$$\frac{\tan 45^\circ}{\sin 30^\circ + \cos 60^\circ}$$

Known values:

$\tan 45^\circ = 1, \sin 30^\circ = \frac{1}{2}, \cos 60^\circ = \frac{1}{2}$

$$\frac{1}{\frac{1}{2}+\frac{1}{2}} = \frac{1}{1} = 1$$

Correct Answer: B) 1

6. Find $(\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

Using identity:

$(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$

$= 2(\sin^2\theta + \cos^2\theta)$

$= 2(1) = 2$

Correct Answer: A) 2

7. Find $\frac{\cos A}{1+\sin A} + \frac{1+\sin A}{\cos A}$

Take LCM:

$$= \frac{\cos^2 A + (1+\sin A)^2}{\cos A(1+\sin A)}$$

Expand numerator:

$$= \frac{\cos^2 A + 1 + 2\sin A + \sin^2 A}{\cos A(1+\sin A)}$$

Using:

$\sin^2 A + \cos^2 A = 1$

$$= \frac{1+1+2\sin A}{\cos A(1+\sin A)}$$

$$= \frac{2(1+\sin A)}{\cos A(1+\sin A)} = \frac{2}{\cos A} = 2\sec A$$

Correct Answer: A) $2\sec A$

8. If $7\sin^2\theta + 3\cos^2\theta = 4$

find $\tan\theta$

Using:

$\cos^2\theta = 1 - \sin^2\theta$

$7\sin^2\theta + 3(1-\sin^2\theta) = 4$

$7\sin^2\theta + 3 - 3\sin^2\theta = 4$

$4\sin^2\theta = 1 \Rightarrow \sin^2\theta = \frac{1}{4}$

$\cos^2\theta = \frac{3}{4}$

$\tan^2\theta = \frac{\sin^2\theta}{\cos^2\theta} = \frac{1/4}{3/4} = \frac{1}{3}$

$\tan\theta = \frac{1}{\sqrt{3}}$

Correct Answer: A) $\mathbf{\frac{1}{\sqrt{3}}}$

9. If $\sec\theta + \tan\theta = k$

find $\sin\theta$

Using identity:

$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$

$\sec\theta - \tan\theta = \frac{1}{k}$

Add both equations:

$2\sec\theta = k + \frac{1}{k}$

$\sec\theta = \frac{k^2+1}{2k}$

Subtract:

$2\tan\theta = k - \frac{1}{k}$

$\tan\theta = \frac{k^2-1}{2k}$

Now,

$\sin\theta = \frac{\tan\theta}{\sec\theta}$

$= \frac{(k^2-1)/2k}{(k^2+1)/2k} = \frac{k^2-1}{k^2+1}$

Correct Answer: C) $\mathbf{\frac{k^2-1}{k^2+1}}$

10. If $x = a(\text{cosec}\,\theta + \cot\theta)^2$

and

$y = b(\text{cosec}\,\theta - \cot\theta)^2$

find $x^{1/2}y^{1/2}$

$\sqrt{x} = \sqrt{a}(\text{cosec}\,\theta + \cot\theta)$

$\sqrt{y} = \sqrt{b}(\text{cosec}\,\theta - \cot\theta)$

Multiply:

$\sqrt{xy} = \sqrt{ab}(\text{cosec}\,\theta + \cot\theta)(\text{cosec}\,\theta - \cot\theta)$

Using identity:

$\text{cosec}^2\theta - \cot^2\theta = 1$

$\sqrt{xy} = \sqrt{ab}$

Squaring both sides:

$xy = ab$

Hence, $x^2y^2 = (ab)^2 = a^2b^2$

Correct Answer: A) $\mathbf{a^2b^2}$

11. Find $\frac{1+\tan^2\theta}{1+\cot^2\theta}$

Using identities:

$1+\tan^2\theta=\sec^2\theta$

and

$1+\cot^2\theta=\operatorname{cosec}^2\theta$

So, $\frac{\sec^2\theta}{\operatorname{cosec}^2\theta}=\frac{1/\cos^2\theta}{1/\sin^2\theta}=\frac{\sin^2\theta}{\cos^2\theta}=\tan^2\theta$

Now check option B:

$\frac{1-\cot\theta}{1-\tan\theta}$

Put $\cot\theta=\frac{1}{\tan\theta}$:

$\frac{1-\frac{1}{\tan\theta}}{1-\tan\theta}=\frac{\tan\theta-1}{\tan\theta(1-\tan\theta)}=-\frac{1}{\tan\theta}$

Not equal to $\tan^2\theta$.

Check option C:

$\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2$

Let $t=\tan\theta$, then $\cot\theta=\frac{1}{t}$

$\left(\frac{1-t}{1-\frac{1}{t}}\right)^2=\left(\frac{1-t}{\frac{t-1}{t}}\right)^2=(-t)^2=t^2=\tan^2\theta$

Correct Answer: C) $\left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2$

12. Find $(1+\cos A)(1-\cos A)(1+\cot^2 A)$

Using: $(1+\cos A)(1-\cos A)=1-\cos^2 A=\sin^2 A$

Also, $1+\cot^2 A=\operatorname{cosec}^2 A$

So, $\sin^2 A\cdot\operatorname{cosec}^2 A=\sin^2 A\cdot\frac{1}{\sin^2 A}=1$

Correct Answer: B) 1

13. If $x=r\cos A\sin B, y=r\cos A\cos B, z=r\sin A$

find $x^2+y^2+z^2$

$x^2+y^2+z^2=r^2\cos^2 A\sin^2 B+r^2\cos^2 A\cos^2 B+r^2\sin^2 A$

Take $r^2\cos^2 A$common:

$=r^2\cos^2 A(\sin^2 B+\cos^2 B)+r^2\sin^2 A$

$=r^2\cos^2 A+r^2\sin^2 A$

$=r^2(\cos^2 A+\sin^2 A)=r^2$

Correct Answer: B) r^2

14. If $m=\tan\theta+\sin\theta, n=\tan\theta-\sin\theta$

find m^2-n^2

Using:

$m^2-n^2=(m+n)(m-n)$

$m+n=2\tan\theta$

$m-n=2\sin\theta$

$m^2-n^2=(2\tan\theta)(2\sin\theta)=4\sin\theta\tan\theta$

Now,

$mn=(\tan\theta+\sin\theta)(\tan\theta-\sin\theta)$

$=\tan^2\theta-\sin^2\theta$

$= \frac{\sin^2\theta}{\cos^2\theta} - \sin^2\theta$

$= \sin^2\theta\left(\frac{1-\cos^2\theta}{\cos^2\theta}\right) = \frac{\sin^4\theta}{\cos^2\theta}$

$\sqrt{mn} = \frac{\sin^2\theta}{\cos\theta} = \sin\theta\tan\theta$

Hence,

$m^2 - n^2 = 4\sqrt{mn}$

Correct Answer: D) $4\sqrt{mn}$

15. If $2\sin\theta = 2 - \cos\theta$

find $\sin\theta$

$2\sin\theta + \cos\theta = 2$

Let $s = \sin\theta$

$\cos\theta = 2 - 2s$

Using identity:

$\sin^2\theta + \cos^2\theta = 1$

$s^2 + (2-2s)^2 = 1$

$s^2 + 4 - 8s + 4s^2 = 1$

$5s^2 - 8s + 3 = 0$

$(5s-3)(s-1) = 0$

$s = \frac{3}{5}\text{or}1$

From options,

$\sin\theta = \frac{3}{5}$

Correct Answer: A) $\frac{3}{5}$

16. If $\tan^2\theta + \sec\theta = 5$

find $\cos\theta$

Using:

$\tan^2\theta = \sec^2\theta - 1$

$\sec^2\theta - 1 + \sec\theta = 5$

$\sec^2\theta + \sec\theta - 6 = 0$

$(\sec\theta + 3)(\sec\theta - 2) = 0$

$\sec\theta = -3\text{or}2$

Therefore,

$\cos\theta = -\frac{1}{3}\text{or}\frac{1}{2}$

Correct Answer: A) $-\frac{1}{3}$or $\frac{1}{2}$

17. If $\sec^2\theta = 2 + 2\tan\theta$

find $\tan\theta$

Using:

$\sec^2\theta = 1 + \tan^2\theta$

$1 + \tan^2\theta = 2 + 2\tan\theta$

$\tan^2\theta - 2\tan\theta - 1 = 0$

Using quadratic formula:

$\tan\theta = \frac{2 \pm \sqrt{4+4}}{2}$

$= \frac{2 \pm 2\sqrt{2}}{2}$

$= 1 \pm \sqrt{2}$

Correct Answer: C) $1 \pm \sqrt{2}$

18. First, convert all the trigonometric ratios into sine and cosine.

$\cot A = \frac{\cos A}{\sin A}$

$\csc A = \frac{1}{\sin A}$

$\tan A = \frac{\sin A}{\cos A}$

$\sec A = \frac{1}{\cos A}$

Substitute these into the expression:

$\left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right)\left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$

Find common denominators:

$\left(\frac{\sin A + \cos A - 1}{\sin A}\right)\left(\frac{\cos A + \sin A + 1}{\cos A}\right)$

Multiply the numerators and denominators:

$\frac{(\sin A + \cos A - 1)(\sin A + \cos A + 1)}{\sin A \cos A}$

Using

$(a - b)(a + b) = a^2 - b^2$

$= \frac{(\sin A + \cos A)^2 - 1}{\sin A \cos A}$

Expand:

$= \frac{\sin^2 A + \cos^2 A + 2\sin A \cos A - 1}{\sin A \cos A}$

Using identity:

$\sin^2 A + \cos^2 A = 1$

$= \frac{1 + 2\sin A \cos A - 1}{\sin A \cos A}$

$= \frac{2\sin A \cos A}{\sin A \cos A}$

$= 2$

Answer: $\boxed{A)\ 2}$

19. If $x = a(\text{cosec}\,\theta + \cot\theta), y = b(\text{cosec}\,\theta - \cot\theta)$

find the correct relation.

$xy = ab(\text{cosec}\,\theta + \cot\theta)(\text{cosec}\,\theta - \cot\theta)$

Using:

$\text{cosec}^2\,\theta - \cot^2\theta = 1$

$xy = ab$

Correct Answer: A) $xy = ab$

20. If $x = a\cos^2\theta, y = b\sin^2\theta$

find the relation.

$\frac{x}{a} = \cos^2\theta$

$\frac{y}{b} = \sin^2\theta$

Add: $\frac{x}{a}+\frac{y}{b}=\cos^2\theta+\sin^2\theta=1$

Correct Answer: C) $\frac{x}{a}+\frac{y}{b}=1$

LEVEL – II

Single Answer Questions

1. If $\sqrt{3}\sin\theta-\cos\theta=0$.find $\frac{\tan^2\theta+3\tan\theta}{1+\tan^2\theta}$

Solution: $\sqrt{3}\sin\theta=\cos\theta$

Divide by $\cos\theta$:

$\tan\theta=\frac{1}{\sqrt{3}}$

Substitute into the expression:

$$\frac{\left(\frac{1}{\sqrt{3}}\right)^2+3\left(\frac{1}{\sqrt{3}}\right)}{1+\left(\frac{1}{\sqrt{3}}\right)^2}$$

$$=\frac{\frac{1}{3}+\sqrt{3}}{1+\frac{1}{3}}$$

$$=\frac{\frac{1+3\sqrt{3}}{3}}{\frac{4}{3}}$$

$$=\frac{1+3\sqrt{3}}{4}$$

Answer: $\boxed{C)\ \frac{1+3\sqrt{3}}{4}}$

2. If $9\sin\theta+40\cos\theta=41$

find $41\cos\theta$.

Solution:

Compare with the Pythagorean triple:

$9^2+40^2=41^2$

Hence, $\sin\theta=\frac{9}{41},\cos\theta=\frac{40}{41}$

Therefore,

$41\cos\theta=41\left(\frac{40}{41}\right)=40$

Answer: $\boxed{A)\ 40}$

3. If $\cot\theta=\frac{q}{p}$.find $\frac{p\sin\theta-q\cos\theta}{p\sin\theta+q\cos\theta}$

Solution: $\cot\theta=\frac{\cos\theta}{\sin\theta}=\frac{q}{p}$

So, $p\cos\theta=q\sin\theta$

Using

$\tan\theta=\frac{p}{q}$

Take a right triangle:

Opposite $=p$, Adjacent $=q$

Hypotenuse: $\sqrt{p^2+q^2}$

Thus,

$\sin\theta = \frac{p}{\sqrt{p^2+q^2}}$

$\cos\theta = \frac{q}{\sqrt{p^2+q^2}}$

Substitute:

$$\frac{p\left(\frac{p}{\sqrt{p^2+q^2}}\right) - q\left(\frac{q}{\sqrt{p^2+q^2}}\right)}{p\left(\frac{p}{\sqrt{p^2+q^2}}\right) + q\left(\frac{q}{\sqrt{p^2+q^2}}\right)}$$

$$= \frac{p^2-q^2}{p^2+q^2}$$

Answer: $\boxed{D)\ \frac{p^2-q^2}{p^2+q^2}}$

4. If $15\cos A - 8\sin A = 0$.find $\frac{\sin A+\cos A}{2\cos A-\sin A}$

Solution:$15\cos A = 8\sin A$

Divide by $\cos A$: $\tan A = \frac{15}{8}$

Take a right triangle:

Opposite = 15, Adjacent = 8

Hypotenuse:$\sqrt{15^2+8^2} = 17$

So,$\sin A = \frac{15}{17}, \cos A = \frac{8}{17}$

Substitute:

$$\frac{\frac{15}{17}+\frac{8}{17}}{2\left(\frac{8}{17}\right)-\frac{15}{17}}$$

$$= \frac{23/17}{1/17} = 23$$

Answer: $\boxed{A)\ 23}$

5. If $\tan\theta + \cot\theta = 2$.find θ, where θ is acute.

Solution: $\tan\theta + \frac{1}{\tan\theta} = 2$

Let $\tan\theta = t$

Then,$t + \frac{1}{t} = 2$

Multiply by t:

$t^2 + 1 = 2t$

$t^2 - 2t + 1 = 0$

$(t-1)^2 = 0$

$t = 1$

So,$\tan\theta = 1$

$\theta = 45°$

Answer: $\boxed{A)\ 45°}$

6. If $x = a\sin\theta, y = b\tan\theta$.find $\frac{a^2}{x^2} - \frac{b^2}{y^2}$

Solution:

$$x = a\sin\theta \Rightarrow \frac{a^2}{x^2} = \frac{1}{\sin^2\theta} = \csc^2\theta$$

$$y = b\tan\theta \Rightarrow \frac{b^2}{y^2} = \frac{1}{\tan^2\theta} = \cot^2\theta$$

Therefore,

$\frac{a^2}{x^2} - \frac{b^2}{y^2} = \csc^2\theta - \cot^2\theta$

Using identity:

$\csc^2\theta - \cot^2\theta = 1$

Answer: $\boxed{A)\ 1}$

7. If $a\cos\theta + b\sin\theta = 1$ and $a\sin\theta - b\cos\theta = 1$.find $a^2 + b^2$.

Solution:

Square and add both equations:

$(a\cos\theta + b\sin\theta)^2 + (a\sin\theta - b\cos\theta)^2 = 1^2 + 1^2$

Expand:

$a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta$

$+a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta = 2$

Cross terms cancel:

$a^2(\sin^2\theta + \cos^2\theta) + b^2(\sin^2\theta + \cos^2\theta) = 2$

Using

$\sin^2\theta + \cos^2\theta = 1$

$a^2 + b^2 = 2$

Answer: $\boxed{B)\ 2}$

8. If $\tan\theta = \frac{a}{b}$.find $\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}$

Solution:

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{a}{b}$

Take $\sin\theta = \frac{a}{\sqrt{a^2+b^2}}, \cos\theta = \frac{b}{\sqrt{a^2+b^2}}$

Substitute:

$$\frac{\frac{b}{\sqrt{a^2+b^2}} + \frac{a}{\sqrt{a^2+b^2}}}{\frac{b}{\sqrt{a^2+b^2}} - \frac{a}{\sqrt{a^2+b^2}}}$$

$$= \frac{a+b}{b-a}$$

Answer: $\boxed{A)\ \frac{b+a}{b-a}}$

9. If $5\tan\alpha = 4$.find $\frac{5\sin\alpha-3\cos\alpha}{5\sin\alpha+2\cos\alpha}$

Solution:$\tan\alpha = \frac{4}{5}$

Take opposite = 4, adjacent = 5

Hypotenuse:

$\sqrt{4^2 + 5^2} = \sqrt{41}$

So,

$\sin\alpha = \frac{4}{\sqrt{41}}, \cos\alpha = \frac{5}{\sqrt{41}}$

Substitute:

$$\frac{5\left(\frac{4}{\sqrt{41}}\right) - 3\left(\frac{5}{\sqrt{41}}\right)}{5\left(\frac{4}{\sqrt{41}}\right) + 2\left(\frac{5}{\sqrt{41}}\right)}$$

$$= \frac{20-15}{20+10} = \frac{5}{30} = \frac{1}{6}$$

Answer: $\boxed{B)\ \frac{1}{6}}$

10. If $\tan A = \frac{4}{3}$.find $\frac{2\sin A - 3\cos A}{2\sin A + 3\cos A}$

Solution:

Take opposite = 4, adjacent = 3

Hypotenuse: $\sqrt{4^2 + 3^2} = 5$

Thus,$\sin A = \frac{4}{5}, \cos A = \frac{3}{5}$

Substitute:

$$\frac{2\left(\frac{4}{5}\right) - 3\left(\frac{3}{5}\right)}{2\left(\frac{4}{5}\right) + 3\left(\frac{3}{5}\right)}$$

$$= \frac{\frac{8}{5} - \frac{9}{5}}{\frac{8}{5} + \frac{9}{5}}$$

$$= \frac{-1/5}{17/5} = -\frac{1}{17}$$

Answer: $\boxed{B)\ -\frac{1}{17}}$

One or More Than One Correct Answer Questions

11. $\sin 0° \cdot \sin 1° \cdot \ldots\ldots \cdot \sin 90°$

Since $\sin 0° = 0$,

Product = 0

Also, $\tan 0° = 0$ and $\cos 90° = 0$.

Hence all are equal to the product.

Correct Answers: A, B, D

12. Given:

$\sin A + \cos A = \sqrt{2}$

Squaring both sides:

$(\sin A + \cos A)^2 = 2$

$\sin^2 A + \cos^2 A + 2\sin A \cos A = 2$

$1 + 2\sin A \cos A = 2$

$\sin A \cos A = 1/2$

Now,

$\sin A + \cos A = \sqrt{2}$ occurs when

$\sin A = \cos A = 1/\sqrt{2}$

Therefore $\tan A = 1$.

Correct Answers: A, B, C, D

13. Given:

$(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2$

$= k + \tan^2\theta + \cot^2\theta$

Expand:

$= \sin^2\theta + \operatorname{cosec}^2\theta + 2$

$\cos^2\theta + \sec^2\theta + 2$

Using $\sin^2\theta + \cos^2\theta = 1$,

$= 1 + \operatorname{cosec}^2\theta + \sec^2\theta + 4$

Now,
$\text{cosec}^2\theta = 1 + \cot^2\theta$
$\sec^2\theta = 1 + \tan^2\theta$
So,
$= 1 + (1 + \cot^2\theta) + (1 + \tan^2\theta) + 4$
$= 7 + \tan^2\theta + \cot^2\theta$
Hence, k = 7
Correct Answer: B

14. Given Equation: $\tan\theta = p - \frac{1}{4p}$

We need to find the value of $\sec\theta - \tan\theta$.Let's first find $\sec\theta$ using the identity $\sec^2\theta = 1 + \tan^2\theta$:

$$\sec^2\theta = 1 + \left(p - \frac{1}{4p}\right)^2$$

$$\sec^2\theta = 1 + \left(p^2 - 2(p)\left(\frac{1}{4p}\right) + \frac{1}{16p^2}\right)$$

$$\sec^2\theta = 1 + p^2 - \frac{1}{2} + \frac{1}{16p^2}$$

$$\sec^2\theta = p^2 + \frac{1}{2} + \frac{1}{16p^2}$$

$$\sec^2\theta = p^2 + 2(p)\left(\frac{1}{4p}\right) + \left(\frac{1}{4p}\right)^2$$

This is a perfect square: $\sec^2\theta = \left(p + \frac{1}{4p}\right)^2$

Taking the square root gives two possible values for $\sec\theta$:

$$\sec\theta = \pm\left(p + \frac{1}{4p}\right)$$

Now we evaluate $\sec\theta - \tan\theta$ for both cases:

Case 1: Let $\sec\theta = \left(p + \frac{1}{4p}\right)$

$$\sec\theta - \tan\theta = \left(p + \frac{1}{4p}\right) - \left(p - \frac{1}{4p}\right)$$

$$\sec\theta - \tan\theta = p + \frac{1}{4p} - p + \frac{1}{4p} = \frac{2}{4p} = \frac{1}{2\text{p}}$$

This matches Option B.

Case 2: Let $\sec\theta = -\left(p + \frac{1}{4p}\right) = -p - \frac{1}{4p}$

$$\sec\theta - \tan\theta = \left(p - \frac{1}{4p}\right) - \left(p - \frac{1}{4p}\right)$$

$$\sec\theta - \tan\theta = -p - \frac{1}{4p} - p + \frac{1}{4p} = -2\text{p}$$

This matches Option C.

Correct Options: B and C

Comprehension Type Questions

Passage I

First, let's establish the sides of the given right-angled triangleΔABC:

Angle B =90° (This means AC is the hypotenuse)

Side AB = x

Side BC = y

Using the Pythagorean theorem: $AC = \sqrt{x^2 + y^2}$

15. Find $\sin A$

The sine of an angle is the ratio of the opposite side to the hypotenuse.

Opposite side to angle A = BC = y

Hypotenuse = AC = $\sqrt{x^2+y^2}$

$\sin A = \frac{\text{Opposite}}{\text{Hypotenuse}}$

$\sin A = \frac{y}{\sqrt{x^2+y^2}}$

Correct Option: C) $\frac{y}{\sqrt{x^2+y^2}}$

16. Find sin C

Using the sine ratio for angle C:

Opposite side to angle C = AB = x

Hypotenuse = AC = $\sqrt{x^2+y^2}$

$\sin C = \frac{\text{Opposite}}{\text{Hypotenuse}}$

$\sin C = \frac{x}{\sqrt{x^2+y^2}}$

Correct Option: B) $\frac{x}{\sqrt{x^2+y^2}}$

17. Find $\sec^2 C - \tan^2 C$

Using the trigonometric identity:

$\sec^2\theta - \tan^2\theta = 1$

Therefore, $\sec^2 C - \tan^2 C = 1$

We can also verify using the triangle:

$\sec C = \frac{\text{Hypotenuse}}{\text{Adjacent}} = \frac{\sqrt{x^2+y^2}}{y}$

$\tan C = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{x}{y}$

Substituting:

$$\left(\frac{\sqrt{x^2+y^2}}{y}\right)^2 - \left(\frac{x}{y}\right)^2$$

$$= \frac{x^2+y^2}{y^2} - \frac{x^2}{y^2}$$

$$= \frac{x^2+y^2-x^2}{y^2}$$

$$= \frac{y^2}{y^2} = 1$$

Correct Option: D) 1

Matching Type Questions

Given:

$m = \cot\theta + \cos\theta, n = \cot\theta - \cos\theta$

$m+n = 2\cot\theta, m-n = 2\cos\theta$

a) $\frac{m+n}{m-n} = \frac{2\cot\theta}{2\cos\theta} = \frac{\cot\theta}{\cos\theta} = \text{cosec}\,\theta$

$a \to R$

b) $m^2 - n^2 = (m+n)(m-n)$

$= (2\cot\theta)(2\cos\theta) = 4\cot\theta\cos\theta$

$b \to S$

c) $2(m^2+n^2)$

Using identity:

$(m+n)^2+(m-n)^2=2(m^2+n^2)$

$=(2\cot\theta)^2+(2\cos\theta)^2$

$=4\cot^2\theta+4\cos^2\theta$

$=4(\cot^2\theta+\cos^2\theta)$

$c\to P$

d) $m^2+n^2+2mn=(m+n)^2$

$=(2\cot\theta)^2=4\cot^2\theta$

$d\to Q$

Final Answer: $a-R, b-S, c-P, d-Q$

Integer Type Questions

19.

$$\frac{\cos A}{1+\sin A}+\frac{\cos A}{1-\sin A}=k\sec A$$

$$=\frac{\cos A[(1-\sin A)+(1+\sin A)]}{1-\sin^2 A}$$

$$=\frac{2\cos A}{\cos^2 A}=\frac{2}{\cos A}=2\sec A$$

$k=2$

20. Given: $\cos\alpha+\cos^2\alpha=1$

Let $x=\cos\alpha$

Then

$x+x^2=1$

$x^2+x=1$

Also,

$\sin^2\alpha=1-\cos^2\alpha=1-x^2=x$

Required value:

$2\sin^4\alpha+2\sin^2\alpha+3$

$=2x^2+2x+3$

$=2(x^2+x)+3$

$=2(1)+3=5$

Answer: 5

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. Given:

$$\tan\theta=\frac{1}{\sqrt{7}}$$

Using identities:

$\sec^2\theta=1+\tan^2\theta$

$=1+\frac{1}{7}=\frac{8}{7}$

Also,

$\operatorname{cosec}^2\theta=1+\cot^2\theta$

Since

$\cot\theta=\sqrt{7}$

$\operatorname{cosec}^2\theta=1+7=8$

Now,

$$\frac{\text{cosec}^2\,\theta - \sec^2\theta}{\text{cosec}^2\,\theta + \sec^2\theta} = \frac{8 - \frac{8}{7}}{8 + \frac{8}{7}}$$

$$= \frac{\frac{56-8}{7}}{\frac{56+8}{7}} = \frac{48}{64} = \frac{3}{4}$$

Answer: A) $\frac{3}{4}$

2. Given:

$\sin\theta = \frac{1}{2}$

Expression:

$\cos\theta \cdot \tan\theta \cdot \sec\theta \cdot \text{cosec}\,\theta \cdot \cot\theta$

Using identities:

$\tan\theta = \frac{\sin\theta}{\cos\theta}, \sec\theta = \frac{1}{\cos\theta}, \text{cosec}\,\theta = \frac{1}{\sin\theta}, \cot\theta = \frac{\cos\theta}{\sin\theta}$

Substitute:

$\cos\theta \times \frac{\sin\theta}{\cos\theta} \times \frac{1}{\cos\theta} \times \frac{1}{\sin\theta} \times \frac{\cos\theta}{\sin\theta}$

After cancellation:

$= \frac{1}{\sin\theta}$

Since

$\sin\theta = \frac{1}{2}$

$\frac{1}{\sin\theta} = 2$

Answer: C) 2

3. Evaluate:

$\frac{1}{\cos\theta} - \cos\theta$

$= \sec\theta - \cos\theta$

Taking LCM:

$= \frac{1 - \cos^2\theta}{\cos\theta}$

Using

$1 - \cos^2\theta = \sin^2\theta$

$= \frac{\sin^2\theta}{\cos\theta}$

$= \sin\theta \cdot \frac{\sin\theta}{\cos\theta}$

$= \sin\theta \cdot \tan\theta$

Answer: C) $\sin\theta \cdot \tan\theta$

4. If $8\sin\theta = 4 + \cos\theta$, then $\sin\theta$ =?

Using the identity:

$\sin^2\theta + \cos^2\theta = 1$

Given:

$\cos\theta = 8\sin\theta - 4$

Substitute into the identity:

$\sin^2\theta + (8\sin\theta - 4)^2 = 1$

Let $\sin\theta = x$:

$x^2 + 64x^2 - 64x + 16 = 1$

$65x^2 - 64x + 15 = 0$

Solving:

$(5x - 3)(13x - 5) = 0$

$x = \frac{3}{5}$ or $\frac{5}{13}$

From the options, $\boxed{\frac{3}{5}}$

Answer: (C) $\frac{3}{5}$

5. If $\cot\theta + \csc\theta = 5$, then $\cos\theta$ =?

Use identity:

$(\csc\theta + \cot\theta)(\csc\theta - \cot\theta) = 1$

Given:

$\csc\theta + \cot\theta = 5$

So,

$\csc\theta - \cot\theta = \frac{1}{5}$

Adding:

$2\csc\theta = \frac{26}{5}$

$\csc\theta = \frac{13}{5}$

Subtracting:

$2\cot\theta = \frac{24}{5}$

$\cot\theta = \frac{12}{5}$

$\cos\theta = \frac{\cot\theta}{\csc\theta} = \frac{12/5}{13/5} = \frac{12}{13}$

$\boxed{\frac{12}{13}}$

Answer: (D) $\frac{12}{13}$

6. Evaluate $2(\cos^2 45° + \tan^2 60°) - 6(\sin^2 45° - \tan^2 30°)$

Use standard values:

$\cos^2 45° = \frac{1}{2}, \tan^2 60° = 3$

$\sin^2 45° = \frac{1}{2}, \tan^2 30° = \frac{1}{3}$

Substitute:

$2\left(\frac{1}{2}+3\right) - 6\left(\frac{1}{2}-\frac{1}{3}\right)$

$= 2\left(\frac{7}{2}\right) - 6\left(\frac{1}{6}\right)$

$= 7 - 1$

$= 6$

$\boxed{6}$

Answer: (C) 6

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. If $\cos\theta = \frac{5}{13}$, then select the correct statements.

Using:

$\sin^2\theta + \cos^2\theta = 1$

$\sin\theta = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \sqrt{\frac{144}{169}} = \frac{12}{13}$

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{12/13}{5/13} = \frac{12}{5}$

$\sec\theta = \frac{1}{\cos\theta} = \frac{13}{5}$

$\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{13}{12}$

Hence all are correct.

Correct answers: A, B, C and D

8. Select the correct values for 60°.

Standard values:

$\sin 60° = \frac{\sqrt{3}}{2}$

$\cos 60° = \frac{1}{2}$

$\tan 60° = \sqrt{3}$

$\sec 60° = \frac{1}{\cos 60°} = 2$

So option D is incorrect.

Correct answers: A, B and C

9. If $\tan\theta = \frac{3}{4}$, then which are correct?

$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{4}$

Take sides 3,4,5.

$\sin\theta = \frac{3}{5}$

$\cos\theta = \frac{4}{5}$

$\operatorname{cosec}\theta = \frac{5}{3}$

$\sec\theta = \frac{5}{4}$

Hence all are correct.

Correct answers: A, B, C and D

COMPREHENSION TYPE

PASSAGE

10. If $\sin\theta = \frac{3}{5}$, then find $\cos\theta$.

Using:

$\sin^2\theta + \cos^2\theta = 1$

$\cos\theta = \sqrt{1 - \left(\frac{3}{5}\right)^2}$

$$= \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

$$\boxed{\frac{4}{5}}$$

Answer: (A) $\frac{4}{5}$

11. If $\tan\theta = \frac{5}{12}$, then find $\sec\theta$.

Take opposite = 5, adjacent = 12.

Hypotenuse:

$a^2 + b^2 = c^2$

$= \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = 13$

$\cos\theta = \frac{12}{13}$

$\sec\theta = \frac{13}{12}$

$$\boxed{\frac{13}{12}}$$

Answer: (A) $\frac{13}{12}$

12. If $\cos\theta = \frac{12}{13}$, then find $\sin\theta$.

Using:

$\sin^2\theta + \cos^2\theta = 1$

$$\sin\theta = \sqrt{1 - \left(\frac{12}{13}\right)^2}$$

$$= \sqrt{1 - \frac{144}{169}} = \sqrt{\frac{25}{169}} = \frac{5}{13}$$

$$\boxed{\frac{5}{13}}$$

Answer: (A) $\frac{5}{13}$

MATRIX MATCHING TYPE

13.

1. First Expression

$$\frac{1 - \cos\theta}{\sin\theta}$$

Multiply the numerator and denominator by$(1+\cos\theta)$

$$\frac{(1 - \cos\theta)(1 + \cos\theta)}{\sin\theta(1 + \cos\theta)}$$

Using the identity$1 - \cos^2\theta = \sin^2\theta$:

$$\frac{\sin^2\theta}{\sin\theta(1 + \cos\theta)}$$

$$= \frac{\sin\theta}{1 + \cos\theta}$$

Matches with: C

2. Second Expression

$$\frac{\sec^2\theta - 1}{\tan\theta}$$

Using the identity $\sec^2\theta - 1 = \tan^2\theta$

$$\frac{\tan^2\theta}{\tan\theta}$$

$= \tan\theta$

Matches with: D

3. Third Expression

$$\frac{1}{1-\sin^2\theta}$$

Using the identity $1 - \sin^2\theta = \cos^2\theta$

$$\frac{1}{\cos^2\theta}$$

$= \sec^2\theta$

Matches with: B

4. Fourth Expression

$$\frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta}$$

Using the identity $\sin^2\theta + \cos^2\theta = 1$

$$\frac{1}{\sin^2\theta}$$

$= \csc^2\theta$

(Note: $\csc^2\theta$ is also written as $\text{cosec}^2\theta$)

Matches with: A

INTEGER TYPE

14. The value of $(\sin A - \cos A)^2 + (\sin A + \cos A)^2 =$ _____

Expand both the squares:

$(\sin^2 A - 2\sin A\cos A + \cos^2 A) + (\sin^2 A + 2\sin A\cos A + \cos^2 A)$

The middle terms cancel out:

$= 2\sin^2 A + 2\cos^2 A$

Using $\sin^2 A + \cos^2 A = 1$

$= 2(1) = 2$

Answer: 2

15. The value of $\sin^6\theta + \cos^6\theta + 3\sin^2\theta\cos^2\theta =$ _____

Use the identity:

$a^3 + b^3 = (a+b)^3 - 3ab(a+b)$

Let

$a = \sin^2\theta, b = \cos^2\theta$

Then,

$\sin^6\theta + \cos^6\theta = (\sin^2\theta + \cos^2\theta)^3 - 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta)$

Using $\sin^2\theta + \cos^2\theta = 1$

$= 1^3 - 3\sin^2\theta\cos^2\theta(1)$

$= 1 - 3\sin^2\theta\cos^2\theta$

Now add $3\sin^2\theta\cos^2\theta$:

$= 1$

Answer:1

3. Polynomials and Pair of Linear Equations

LEVEL-I

1. **Correct Answer: C)** $\frac{x^2+x^3}{x}$

Explanation:

$\frac{x^2+x^3}{x} = x + x^2$

All powers of $x$are non-negative integers, so it is a polynomial.

2. **Correct Answer: A)** $\frac{1}{x^{-2}} + \frac{1}{x^{-1}}$

Explanation:

$\frac{1}{x^{-2}} + \frac{1}{x^{-1}} = x^2 + x$

Since all powers of $x$are non-negative integers, $x^2 + x$is a polynomial.

3. **Correct Answer: C) 5**

Explanation:

Polynomial:

$x - x^5 + 1$

Highest power of $x$is 5. Hence degree = 5.

4. **Correct Answer**: **D) Not defined**

Explanation:

Given expression:

$\frac{\sqrt{2}x^5 - x^{10} + 1}{x}$

Dividing each term by x:

$\sqrt{2}x^4 - x^9 + \frac{1}{x}$

Since $\frac{1}{x} = x^{-1}$has a negative exponent, the expression is **not a polynomial**.

Therefore, its degree is **not defined**.

5. **Correct Answer: B) 3**

Explanation:

Polynomial:

$4x^2 + 5x + 3$

It has three terms: $4x^2$, $5x$, 3.

6. **Correct Answer: C)** $\boldsymbol{x^2 + 2x + 10}$

Explanation:

A trinomial has exactly three terms.

$x^2 + 2x + 10$

contains 3 terms.

7. **Correct Answer: B)** $\boldsymbol{x^4 - 30x^3 + 3x^2}$

Explanation:

A bi-quadratic polynomial has degree 4.

Highest power here is 4.

8. **Correct Answer: C) 6**

Explanation:

Given polynomial: $4 + x^3 - x + 2x^2$

At $x = -2$: $4 + (-2)^3 - (-2) + 2(-2)^2$
$= 4 - 8 + 2 + 8$
$= 6$

9. Correct Answer: B) $\frac{1}{\sqrt{3}}$

Explanation:
Given polynomial: $3x^2 - 1$
For zeroes: $3x^2 - 1 = 0$
$3x^2 = 1$
$x^2 = \frac{1}{3}$
$x = \pm\frac{1}{\sqrt{3}}$
From the options, the correct answer is:
$\frac{1}{\sqrt{3}}$

10. Correct Answer: C) $5a$

Explanation:
By Remainder Theorem, remainder when divided by $x - a$is $f(a)$.
Given:
$f(x) = x^3 - ax^2 + 6x - a$
Substitute $x = a$:
$f(a) = a^3 - a(a^2) + 6a - a$
$= a^3 - a^3 + 5a$
$= 5a$
So the correct answer is:

11. Correct Answer: A) 0

Explanation:
If $7 + 3x$divides $3x^2 + 7x$, then remainder is always 0.

12. Correct Answer: D) 2020

Explanation:
For division by $x - 1$, remainder is $f(1)$. $1 + 1 + 1 + \cdots + 1$
Number of terms from $x^0$to x^{2019}: $2019 + 1 = 2020$
Hence remainder = 2020.

13. Correct Answer: C) -x

Explanation:
Given: $x^{19} + x^{17} + x^{13} + x^{11} + x^7 + x^5 + x^3$
Dividing by $x^2 + 1$, use:
$x^2 = -1$
Reducing powers: $x^{19} \to -x, x^{17} \to x, x^{13} \to x, x^{11} \to -x,$
$x^7 \to -x, x^5 \to x, x^3 \to -x$
Adding: $-x + x + x - x + x - x = -x$
Hence remainder is:
$-x$

14. Correct Answer: C) $x^3 - 6x^2 + 11x - 6$

is divided by
$x + 3$,
then the quotient is:
Using synthetic division with $k = -3$:
Coefficients: $1, -3, -7, 27, -18$

Drop the 1.
$1 \times (-3) = -3 \rightarrow (-3) + (-3) = -6$
$(-6) \times (-3) = 18 \rightarrow (-7) + 18 = 11$
$11 \times (-3) = -33 \rightarrow 27 + (-33) = -6$
$(-6) \times (-3) = 18 \rightarrow -18 + 18 = 0$(Remainder)
The resulting coefficients $1, -6, 11, -6$correspond to:
$x^3 - 6x^2 + 11x - 6$

15. Correct Answer: B) $9x + 10$
Dividend:$3x^3 + x^2 + 2x + 5$
Divisor:$x^2 + 2x + 1$
Since divisor is quadratic, remainder is of form $ax + b$.
After polynomial division:
$3x^3 + x^2 + 2x + 5 = (x^2 + 2x + 1)(3x - 5) + (9x + 10)$
So remainder is:$9x + 10$
Hence, the correct option is **B**.

16. Correct Answer: C) 3
Using Remainder Theorem for divisor $x + 2$:$f(x) = 2x^2 + 3x + 1$
Remainder:$f(-2) = 2(-2)^2 + 3(-2) + 1$
$= 8 - 6 + 1 = 3$
So remainder is 3.
Hence, the correct option is **C**.

17. Correct Answer: A) $-x^2 + 2$; $-5x + 10$
Divide:$x^4 - 5x + 6$
by$2 - x^2$
Quotient obtained:$-x^2 - 2$
Remainder obtained:$-5x + 10$
Hence, the correct option is **A**.

18. Correct Answer: A) 0
Using Remainder Theorem for divisor:$x + \frac{2}{3}$
So substitute $x = -\frac{2}{3}$.$f(x) = 3x^4 + 2x^3 - \frac{x^2}{3} - \frac{x}{9} + \frac{2}{27}$
Substituting $x = -\frac{2}{3}$gives:$f\left(-\frac{2}{3}\right) = 0$
Therefore remainder is 0.
Hence, the correct option is **A**.

19. Correct Answer: B)- 1
For same remainder on division by $x - 3$:$p(3)^3 + 4(3)^2 + 3(3) - 4$
$= 27p + 36 + 9 - 4$
$= 27p + 41$
Second polynomial:$p + 3^3 - 4(3) = p + 27 - 12 = p + 15$
Equating remainders:$27p + 41 = p + 15$
$26p = -26$
$p = -1$

20. Correct Answer: C) 1
For same remainder when divided by $x - 2$, substitute $x = 2$.
First polynomial: $3(2)^3 + a(2)^2 + 3(2) + 5$
$= 24 + 4a + 6 + 5$
$= 35 + 4a$
Second polynomial: $a - 2(2) + (2)^2 + 4(2)^3$

$= a - 4 + 4 + 32$
$= a + 32$
Equating:$35 + 4a = a + 32$
$3a = -3$
$a = -1$
Now,$2 + a = 2 + (-1) = 1$
Hence, the correct option is C.
Top of Form

LEVEL - II

SINGLE OPTION CORRECT TYPEBottom of Form

1. Correct Answer: B) 98

Explanation:
$\frac{x^{100} - 3x^2 + x^6}{x^2} = x^{98} - 3 + x^4$
The highest power of $x$is 98. Hence, degree = 98.

2. Correct Answer: B) $\mathbf{x^{10} - \frac{1}{3}x^8}$

Explanation:
A binomial has exactly two terms.
A has 3 terms
B has 2 terms
C has 3 terms
D has 1 term
But among the given options, B contains fractional coefficient and higher power expression; the intended simple binomial form is not proper

3. Correct Answer: C) $x^3 - 3x$

Explanation:
A cubic polynomial has degree 3.
$x^3 - 3x$
has highest power 3, so it is cubic.

4. Correct Answer: D) Not defined

Explanation:
The zero polynomial has no defined degree because all coefficients are zero.

5. Correct Answer: C) $x^2 - x + 1$

Explanation:
Using:
Dividend = Divisor × Quotient + Remainder
$x^3 - 3x^2 + x + 2 = g(x)(x - 2) + (-2x + 4)$
$x^3 - 3x^2 + 3x - 2 = g(x)(x - 2)$
Factorizing: $x^3 - 3x^2 + 3x - 2 = (x - 2)(x^2 - x + 1)$
Hence, $g(x) = x^2 - x + 1$

6. Correct Answer: B) 9

Explanation:
For divisor $4x + 2$, remainder theorem gives:
$x = -\frac{1}{2}$
Substitute in:
$8x^4 + 12x^3 + 16x^2 + 8x + 10$

$= 8\left(\frac{1}{16}\right) + 12\left(-\frac{1}{8}\right) + 16\left(\frac{1}{4}\right) + 8\left(-\frac{1}{2}\right) + 10$

$= \frac{1}{2} - \frac{3}{2} + 4 - 4 + 10 = 9$

Hence, remainder = 9.

7. Correct Answer: A) -1

Given polynomial: $3x^4 + 6x^3 - 2x^2 - 10x - 5$

Two zeroes are: $\sqrt{\frac{5}{3}}, -\sqrt{\frac{5}{3}}$

So, $x^2 - \frac{5}{3}$

is a factor. $x^2 - \frac{5}{3} = \frac{1}{3}(3x^2 - 5)$

Divide the polynomial by $3x^2 - 5$: $3x^4 + 6x^3 - 2x^2 - 10x - 5 = (3x^2 - 5)(x^2 + 2x + 1)$

Now, $x^2 + 2x + 1 = (x + 1)^2$

Hence the other zero is: -1

8. Correct Answer: B) 3,1

Given: $x + 2$

is a factor of $x^2 + ax + 2b$

By Factor Theorem: $(-2)^2 + a(-2) + 2b = 0$

$4 - 2a + 2b = 0$

$2 - a + b = 0$

$b = a - 2$

Also given: $a + b = 4$

Substitute: $a + (a - 2) = 4$

$2a = 6$

$a = 3$

$b = 1$

9. Correct Answer: A

To check whether $x + 2$is a factor, substitute $x = -2$.

Option A $(-2)^5 - 4(-2)^4 - 3(-2)^3 + 8(-2)^2 - 14(-2) + 12$

$-32 - 64 + 24 + 32 + 28 + 12 = 0$

So $x + 2$is a factor.

Option B $-32 + 64 + 24 + 32 + 28 + 12 = 128 \neq 0$

Not a factor.

Option C $-32 - 64 - 24 + 32 + 28 + 12 = -48 \neq 0$

Not a factor.

Option D $-32 - 64 + 24 + 32 - 28 + 12 = -56 \neq 0$

Not a factor.

10. Correct Answer: B) x−2

Given polynomials: $3x^2 - x - 10$

and $2x^2 - x - 6$

Factorize: $3x^2 - x - 10 = (x - 2)(3x + 5)$

$2x^2 - x - 6 = (x - 2)(2x + 3)$

Common factor: $x - 2$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B) $-\frac{3}{2}$ A) $-\frac{9}{4}$

Real roots of $2x^{99} + 3x^{98} + 2x^{97} + 3x^{96} + \cdots + 2x + 3 = 0$

Group the terms: $x^{98}(2x + 3) + x^{96}(2x + 3) + \cdots + 1(2x + 3) = 0$

$(2x+3)(x^{98}+x^{96}+\cdots+1)=0$

1. $2x+3=0 \Rightarrow x=-\frac{3}{2}$
2. The second bracket $(x^{98}+\cdots+1)$ is always $\geq 1$for real x.

12. Correct Answer: B and C

Polynomial:
$11x^2-88x^3+14x^4$
Highest degree is 4→ bi-quadratic polynomial.
It has 3 terms → trinomial.
Correct Options: B) Bi-quadratic and C) Trinomial

13. Correct Answer: B) 12

Given:
$2x^{3m+2n}+4x^{5m+2n}+2x^{m+n}$
For $m=2, n=1$:
Powers become:
$3(2)+2(1)=8$
$5(2)+2(1)=12$
$2+1=3$
Highest power = 12

14. Correct Answer: A

x^n-y^n
always has factor: $x-y$
for every natural number n.

COMPREHENSION TYPE

Passage – I

Given polynomials ax^3+3x^2-3
and $2x^3-5x+a$
When divided by $x-4$, remainders are $R_1$and R_2.
By Remainder Theorem:$R_1=p(4)=a(4)^3+3(4)^2-3$
$R_1=64a+48-3$
$R_1=64a+45$
Also,$R_2=2(4)^3-5(4)+a$
$R_2=128-20+a$
$R_2=108+a$

15. If $\boldsymbol{R_1=R_2}$, then $\boldsymbol{a}$is
$64a+45=108+a$
$63a=63$
$a=1$
Answer: B) 1

16. If $\boldsymbol{R_1+R_2=0}$, then $\boldsymbol{a}$is
$(64a+45)+(108+a)=0$
$65a+153=0$
$65a=-153$
$a=-\frac{153}{65}$
Answer: A) $\boldsymbol{-\frac{153}{65}}$

Matrix Matching Type

a) If 2 is the root of$x^2 + px + 26 = 0$
Substitute $x = 2$:$2^2 + 2p + 26 = 0$
$4 + 2p + 26 = 0$
$2p = -30$
$p = -15$
a → S

b) If 5 is root of$x^2 - (k-1)x + 10 = 0$
Substitute $x = 5$: $25 - 5(k-1) + 10 = 0$
$25 - 5k + 5 + 10 = 0$
$40 - 5k = 0$
$k = 8$
b → P

c) Zeroes of polynomial $4x^2 - 9$
Factorize:$4x^2 - 9 = (2x-3)(2x+3)$
Zeroes are:$x = \frac{3}{2}, x = -\frac{3}{2}$
c → Q

d) Degree of $(x-a)(x-b)(x-c)(x-d)$
Each factor has degree 1.
Total degree: $1 + 1 + 1 + 1 = 4$
d → R
Final Matchinga → S, b → P, c → Q, d → R

INTEGER TYPE

18. Given:$1 + x + x^2 = 0$
Expression:$(ax^2 + bx + c) + (a + bx^2 + cx) + (ax + b + cx^2)$
Combine like terms:$(a + b + c)x^2 + (a + c)x + (a + b + c)$
Using$1 + x + x^2 = 0$
we get: $(a + b + c)(x^2 + x + 1) = 0$
Hence value is:0
Answer: 0

19. Find remainder when $x^5 - x^4 + x^3 - x^2 + x - 1$
is divided by $x - 1$.
By Remainder Theorem: $f(1) = 1 - 1 + 1 - 1 + 1 - 1$
$= 0$
Answer: 0

20. Given $(x-2)$is a common factor of $x^2 + ax + b$
and $x^2 + cx + d$
Using Factor Theorem: For first polynomial:$2^2 + 2a + b = 0$
$4 + 2a + b = 0$
$b = -4 - 2a$
For second polynomial:$4 + 2c + d = 0$
$d = -4 - 2c$
Now, $b - d = (-4 - 2a) - (-4 - 2c)$
$= 2c - 2a$
$= 2(c - a)$ Therefore, $\frac{b-d}{c-a} = 2$
Answer: 2

Student Practice corner

Single option correct type

1. Correct Answer: D) $(4x-5)(x-2)$

Factorize:

$4x^2-13x+10$

$=4x^2-8x-5x+10$

$=4x(x-2)-5(x-2)$

$=(4x-5)(x-2)$

Correct Option: D

2. Correct Answer: B) $-9\sqrt{3}, 5$

Given equation:

$\sqrt{3}x^2+27x+5\sqrt{3}=0$

For $ax^2+bx+c=0$: Sum of zeroes $=-\frac{b}{a}$

$=-\frac{27}{\sqrt{3}}=-9\sqrt{3}$

Product of zeroes $=\frac{c}{a}$

$=\frac{5\sqrt{3}}{\sqrt{3}}=5$

3. Correct Answer: D) 6

Given $x-2$is a factor of $x^3-kx^2+3x+10$

By Factor Theorem:

$2^3-k(2^2)+3(2)+10=0$

$8-4k+6+10=0$

$24-4k=0$

$k=6$

4. Correct Answer: B) x−2

Factorize: $3x^2-x-10=(x-2)(3x+5)$

$2x^2-x-6=(x-2)(2x+3)$

Common factor: $x-2$

Correct Option: B

5. Value of $\frac{x^2+11x+28}{x^2+13x+40}\div\frac{x^2+6x+8}{x^2+11x+24}$

First, factor every expression: $\frac{(x+7)(x+4)}{(x+8)(x+5)}\div\frac{(x+4)(x+2)}{(x+8)(x+3)}$

Invert the second fraction to multiply: $\frac{(x+7)(x+4)}{(x+8)(x+5)}\times\frac{(x+8)(x+3)}{(x+4)(x+2)}$

Cancel $(x+4)$and $(x+8)$: $\frac{(x+7)(x+3)}{(x+5)(x+2)}$

Correct Option: D

6. Correct Answer: B) $(9+4x)(9-4x)$

Use identity:

$a^2-b^2=(a+b)(a-b)$

$81-16x^2=9^2-(4x)^2$

$=(9+4x)(9-4x)$

7. Correct Answer: A) $x+3\sqrt{2}$

$x^2+4\sqrt{2}x+6$

Find factors: $=(x+\sqrt{2})(x+3\sqrt{2})$

One factor is: $x+3\sqrt{2}$

8. **Correct Answer: B) $36a^2 + 30a + 25$**

$216a^3 - 125 = (6a)^3 - 5^3$

Using identity:

$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$= (6a - 5)(36a^2 + 30a + 25)$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. If $x - 2$and $x + 3$are both factors of $x^3 + ax^2 + bx - 12$

then $a, b$are

Since $x - 2$is a factor: $2^3 + a(2^2) + 2b - 12 = 0$

$8 + 4a + 2b - 12 = 0$

$2a + b = 2 \ldots (1)$

Since $x + 3$is a factor: $(-3)^3 + a(-3)^2 + b(-3) - 12 = 0$

$-27 + 9a - 3b - 12 = 0$

$3a - b = 13 \ldots (2)$

Add (1) and (2): $5a = 15$

$a = 3$

Substitute in (1): $2(3) + b = 2$

$b = -4$

Correct Options: A) 3 and B) -4

10. The factor of $x^3 + 6x^2 + 11x + 6$ is

Factorize: $x^3 + 6x^2 + 11x + 6 = (x + 1)(x + 2)(x + 3)$

Hence factors are: $x + 1, x + 2, x + 3$

Correct Options: A, B, C

COMPREHENSION TYPE

Passage

11. The factors of $4a^2 - 4\sqrt{3}a + 3$

Is $4a^2 - 4\sqrt{3}a + 3 = (2a - \sqrt{3})^2$

$= (2a - \sqrt{3})(2a - \sqrt{3})$

Hence factor is: $2a - \sqrt{3}$

Correct Option: B) $2a - \sqrt{3}$

12. The factors of

$x^2 - 2x - 15$

is

Find two numbers whose product is $-15$and sum is -2: $-5, 3$

So, $x^2 - 2x - 15 = (x - 5)(x + 3)$

One factor is: $x + 5$

Correct Option: C) $5 + x$

INTEGER TYPE

14. Given: $x^2 - 25x + 126 = 0$

Factorize:$x^2 - 25x + 126 = (x - 7)(x - 18)$

One factor is $x - a$.

So,$a = 7$ or 18

Answer: 7 or 18

15. Given equation:$2x^2 - 179x + 325 = 0$

Sum of roots of quadratic equation:$-\frac{b}{a}$

$= \frac{179}{2}$

Given: $\frac{179}{b} = \frac{179}{2}$

Hence, $b = 2$

Answer: 2

Linear Equations in Two Variables

LEVEL-I

1. Correct Answer: B) (3, 2)

Explanation:

Substitute $x = y + 1$from $x - y = 1$into $2x + 3y = 12$:

$2(y + 1) + 3y = 12 \Rightarrow 5y = 10 \Rightarrow y = 2,\ x = 3.$

2. Correct Answer: C) $3x + 2y = 7$

Explanation:

A linear equation in two variables has variables only of degree 1.

3. Correct Answer: C) 7

Explanation:

Add the equations:

$(x + y) + (x - y) = 10 + 4$

$2x = 14 \Rightarrow x = 7.$

(None of the options match; correct value is 7.)

4. Correct Answer: A) $(1, 4)$.

Explanation:

From $x + y = 5,\ y = 5 - x.$

Substitute into $3x + 2y = 11$:

$3x + 2(5 - x) = 11 \Rightarrow x = 1,\ y = 4.$

5. Correct Answer: B) (2,0).

Explanation:

Substitute $(3, 1)$:

$4(3) - 2(1) = 12 - 2 = 10 \neq 8.$

Checking $(2, 0)$:

$4(2) - 2(0) = 8.$

6. Correct Answer: C) Infinite solutions

Explanation:

$4x + 6y = 14$is exactly $2(2x + 3y = 7)$, so both equations represent the same line.

7. Correct Answer: C) No solution

Explanation:

$x + y = 5 \Rightarrow 2x + 2y = 10$, but given equation is $2x + 2y = 12.$

Same left side but different constants ⇒ parallel lines ⇒ no solution.

8. If $5x + y = 16$and $x - y = 2$, then y = ?

Adding the equations:

$5x + y + x - y = 16 + 2$

$6x = 18 \Rightarrow x = 3$

Substitute in $x - y = 2$:

$3 - y = 2 \Rightarrow y = 1$

Correct Answer: A) 1

9. The method in which one variable is expressed in terms of another is called
Correct Answer: C) Substitution

10. If $3x + y = 11$and $x + y = 7$, then $x =$
Subtract the second equation from the first:
$(3x + y) - (x + y) = 11 - 7$
$2x = 4 \Rightarrow x = 2$
Correct Answer: C) 2

11. A pair of equations having one common solution is called
Correct Answer: C) Consistent
Top of Form
Bottom of Form

12. If $x = 2$and $y = 3$, then $2x + 5y =$
$2(2) + 5(3) = 4 + 15 = 19$
Correct Answer: B) 19

13. The equations $x + 2y = 5$and $2x + 4y = 10$represent
Since the second equation is exactly twice the first, both represent the same line.
Correct Answer: C) Coincident lines

14. If $2x - y = 3$and $x + y = 6$, then $x + y =$
Given directly:
$x + y = 6$
Correct Answer: C) 6

15. The sum of two numbers is 20 and their difference is 4. The larger number is
Let numbers be $x$and y.
$x + y = 20, x - y = 4$
Adding:
$2x = 24 \Rightarrow x = 12$
Correct Answer: C) 12

16. The equations $3x + 2y = 5$and $6x + 4y = 9$have
$\frac{a_1}{a_2} = \frac{3}{6} = \frac{1}{2}, \frac{b_1}{b_2} = \frac{2}{4} = \frac{1}{2}$
But
$\frac{c_1}{c_2} = \frac{5}{9} \neq \frac{1}{2}$
Hence, the lines are parallel.
Correct Answer: C) No solution

17. If $x + y = 8$and $x - y = 2$, then $y =$
Adding equations:
$2x = 10 \Rightarrow x = 5$
Substitute:
$5 + y = 8 \Rightarrow y = 3$
Correct Answer: B) 3

18. Let the two-digit number be:
$10x + y$
where $x$is the tens digit and $y$is the units digit.
Given:
$x + y = 10$
After interchanging digits, number becomes:
$10y + x$
According to question:

$10y + x = (10x + y) + 36$
$10y + x = 10x + y + 36$
$9y - 9x = 36$
$y - x = 4$
Now solve:
$x + y = 10$
$y - x = 4$
Add equations:
$2y = 14$
$y = 7$
$x = 3$
Number:
$10x + y = 37$
Correct Answer: B) 37

19. The solution of $x + y = 9$and $x - y = 1$is
Adding:
$2x = 10 \Rightarrow x = 5$
Then:
$5 + y = 9 \Rightarrow y = 4$
Correct Answer: A) (5, 4)

20. If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the pair of equations has
When
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
the lines intersect at one point.
Correct Answer: C) Unique solution

Level – II

1. For infinite many solutions:
$3x + 2y = 11, 6x + 4y = k$
Second equation must be twice the first:
$2(3x + 2y = 11) \Rightarrow 6x + 4y = 22$
Correct Answer: C) 22

2. Equations:
$2x + 3y = 7, 4x + 6y = 15$
Doubling first equation gives:
$4x + 6y = 14$
But second equation is:
$4x + 6y = 15$
Same coefficients but different constants ⇒ parallel lines.
Correct Answer: C) No solution

3. Given:
$x + y = 7, 2x - 3y = -1$
From $x + y = 7$:
$x = 7 - y$
Substitute into second equation:
$2(7 - y) - 3y = -1$
$14 - 2y - 3y = -1$
$14 - 5y = -1$

$-5y = -15$
$y = 3$
Then:
$x = 7 - 3 = 4$
So, $(a, b) = (4,3)$
Now,
$a^2 + b^2 = 4^2 + 3^2 = 16 + 9 = 25$
Correct Answer: B) 25

4. Given:
$5x - 3y = 7, x = 2$
Substitute:
$10 - 3y = 7$
$-3y = -3$
$y = 1$
Correct Answer: A) 1

5. Equations:
$(k - 1)x + 3y = 2,4x + 6y = 5$
For unique solution:
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
$\frac{k-1}{4} \neq \frac{3}{6} = \frac{1}{2}$
$2(k - 1) \neq 4$
$2k - 2 \neq 4$
$k \neq 3$
Correct Answer: D) $k \neq 3$

6. Equations:
$x + y = 4, x - y = 0$
Add:
$2x = 4 \Rightarrow x = 2$
Then:
$2 - y = 0 \Rightarrow y = 2$
Correct Answer: B) $x = 2,\ y = 2$

7. Equations:
$x + y = 5, x - y = 1$
Add:
$2x = 6 \Rightarrow x = 3$
Then:
$3 + y = 5 \Rightarrow y = 2$
Correct Answer: B) (3, 2)

8. Equations:
$ax + by = c, bx + ay = c$
Subtract:
$(a - b)x + (b - a)y = 0$
$(a - b)(x - y) = 0$
Since $a \neq b$:
$x - y = 0$
$x = y$
Correct Answer: A) $x = y$

9. If $x = 2$, $y = 3$:
$2x + y = 2(2) + 3 = 7$
Correct Answer: 7

10. Which of the following equations is linear?
A linear equation has variables only to power 1 and no product of variables.
A) $x^2 + y = 6 \rightarrow$ not linear
B) $xy = 8 \rightarrow$ not linear
C) $3x - 2y = 5 \rightarrow$ linear
D) $y^2 = 9 \rightarrow$ not linear
Correct Answer: C) $3x - 2y = 5$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. If $\boldsymbol{x + y = 12}$, $\boldsymbol{xy = 35}$, then the possible values of $\boldsymbol{x}$are
Using:
$t^2 - 12t + 35 = 0$
$(t - 5)(t - 7) = 0$
$t = 5$ or 7
Correct Answers: A) 5 , B) 7

12. For the equations
$2x + 3y = 7, 4x + 6y = 14$
Second equation is twice the first, so both represent the same line.
Hence:
Lines are coincident
Infinitely many solutions
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
Correct Answers: A, B, D

13. If $x - y = 4, xy = 45$
Check options:
A) $(9, 5)$
$9 - 5 = 4, 9 \times 5 = 45$
C) $(-5, -9)$
$-5 - (-9) = 4, (-5)(-9) = 45$
Correct Answers: A, C

Comprehension type

Passage – I

Given equations:
$x + y = 16$
$x - y = 10$
Add both equations:
$2x = 26$
$x = 13$
Substitute in $x + y = 16$:
$13 + y = 16$
$y = 3$
So,
Speed of boat in still water = x = 13km/h
Speed of current = y = 3km/h

14. The speed of the boat in still water is
Correct Answer: C) 13 km/h

15. The speed of the current is
Correct Answer: B) 3 km/h

MATRIX MATCHING TYPE

16. Match the Columns

A) $\boldsymbol{2x + 3y = 12}$, $\boldsymbol{x - y = 1}$
From $x - y = 1$:
$x = y + 1$
Substitute:
$2(y + 1) + 3y = 12$
$5y = 10 \Rightarrow y = 2,\ x = 3$
Matches with P) (3,2)

B) $\boldsymbol{x + y = 9}$, $\boldsymbol{x - y = 3}$
Add equations:
$2x = 12 \Rightarrow x = 6$
$y = 3$
Matches with S) (6,3)

C) $\boldsymbol{3x + 6y = 12}$, $\boldsymbol{x + 2y = 4}$
First equation simplifies to:
$x + 2y = 4$
Same equations ⇒ infinitely many solutions.
Matches with Q) Infinite solutions

D) $\boldsymbol{2x + 4y = 8}$, $\boldsymbol{4x + 8y = 20}$
Doubling first equation gives:
$4x + 8y = 16$
But second equation is:
$4x + 8y = 20$
Parallel lines ⇒ no solution.
Matches with R) No solution
Final Matching: A → P, B → S, C → Q, D → R

INTEGER TYPE

17. Given: $2x + y = 11, x - y = 1$
Add equations: $3x = 12$
$x = 4$
Answer: 4

18. $x + y = 9, x - y = 3$
Add: $2x = 12$
$x = 6$
Answer: 6

19. $x + y = 8, x - y = 2$
Subtract second from first: $2y = 6$
$y = 3$
Answer: 3

20. $3x - y = 7, x = 3$
Substitute:
$9 - y = 7$
$y = 2$
Answer: 2

Student practice corner

SINGLE OPTION CORRECT TYPE

1. Which ordered pair satisfies both equations $x + y = 9$and $x - y = 1$?

Add equations:

$2x = 10 \Rightarrow x = 5$

Then: $5 + y = 9 \Rightarrow y = 4$

Correct Answer: B) (5, 4)

2. Which of the following is a linear equation in two variables?

A linear equation has variables only to power 1.

$2x - 3y = 5$

Correct Answer: C) $2x - 3y = 5$

3. If $x = 2$and $y = 5$, then the value of $3x + y$is

$3(2) + 5 = 6 + 5 = 11$

Correct Answer: B) 11

4. The solution of $x + y = 8$and $x - y = 2$is

Add equations:

$2x = 10 \Rightarrow x = 5$

Then: $5 + y = 8 \Rightarrow y = 3$

Correct Answer: A) (5, 3)

5. Solve for $x$and y:

$2x + 3y = 13, x - y = 1$

From $x - y = 1$:

$x = y + 1$

Substitute:

$2(y + 1) + 3y = 13$

$5y + 2 = 13$

$5y = 11$

$y = \frac{11}{5}$

$x = \frac{16}{5}$

Correct Answer: A)

6. The equations $x + y = 5$and $x + y = 9$represent

Same coefficients but different constants ⇒ parallel lines.

Correct Answer: C) Parallel lines

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. A pair of linear equations has no solution when

For no solution:

$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

This represents parallel lines and inconsistent equations.

Correct Answers: A, B, D

10. If $2x + 3y = 18, x - y = 2$

From:

$x - y = 2$

$x = y + 2$

Substitute into first equation:

$2(y + 2) + 3y = 18$

$2y + 4 + 3y = 18$

$5y = 14$

$y = 2.8$

Then:

$x = 2.8 + 2 = 4.8$

Check options:

A) $x = 4.8$

B) $y = 2.8$

Correct Answers: A, B

COMPREHENSION TYPE

Given:

$3x + 2y = 94$

$5x + 4y = 166$

Multiply first equation by 2:

$6x + 4y = 188$

Subtract second equation:

$x = 22$

Substitute into first equation:

$3(22) + 2y = 94$

$66 + 2y = 94$

$2y = 28$

$y = 14$

11. The value of ***x*** is

Correct Answer: D) 22

12. The value of ***y*** is

Correct Answer: B) 14

13. Match the Columns

Column I → Column II

A) Pair of equations having one solution
→ Intersecting lines and Unique solution → Q

B) Pair of equations having no solution
→ Parallel lines → R

C) Pair of equations having infinitely many solutions
→ Coincident lines → P

D) $2x + 6 = 10$ then $x = 2$
→ S

14. If

$3x + 5 = 20$

$3x = 15$

$x = 5$

Answer: 5

15. The pair of equations

$x + y = 6, x - y = 0$

Add equations:

$2x = 6$

$x = 3$

Answer: 3

4. Quadratic Equations

LEVEL – I

1. For $6x^2 - 5x + 1 = 0$

Using factorization:

$6x^2 - 5x + 1 = (3x - 1)(2x - 1)$

So,

$x = \frac{1}{3}, \frac{1}{2}$

Correct Answer: C) $\frac{1}{2}$or $\frac{1}{3}$

2. For $x^2 + 6x + 36 = 0$

Discriminant:

$D = b^2 - 4ac$

$= 6^2 - 4(1)(36)$

$= 36 - 144 = -108$

Correct Answer: B) -108

3. Let the consecutive integers be x and $x + 1$.

$x^2 + (x + 1)^2 = 613$

$2x^2 + 2x - 612 = 0$

$x^2 + x - 306 = 0$

$(x - 17)(x + 18) = 0$

So,$x = 17$

Numbers are 17,18.

Correct Answer: D) 17, 18

4. For $12x^2 + mx + 5 = 0$

Roots are in ratio 3: 2.

Let roots be $3k$and $2k$.

Product of roots:

$3k \cdot 2k = \frac{5}{12}$

$6k^2 = \frac{5}{12}$

$k^2 = \frac{5}{72}$

Sum of roots:

$5k = -\frac{m}{12}$

$m = -60k$

$m^2 = 3600 \cdot \frac{5}{72} = 250$

$m = \pm 5\sqrt{10}$

Correct Answer: C) $\pm 5\sqrt{10}$

5. Since $\alpha, \beta$are roots of

$ax^2 + bx + c = 0$

Then,

$a\alpha + b = -\frac{c}{\alpha}, a\beta + b = -\frac{c}{\beta}$

So,

$(a\alpha + b)^{-3} + (a\beta + b)^{-3}$

$= \left(\frac{\alpha}{c}\right)^3 + \left(\frac{\beta}{c}\right)^3$

$= -\frac{\alpha^3 + \beta^3}{c^3}$

Using identities:

$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= \left(\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(\frac{b}{a}\right)$

$= \frac{-b^3 + 3abc}{a^3}$

Hence,

$\left(a\alpha + b)^{-3} + (a\beta + b\right)^{-3} = \frac{b^3 - 3abc}{a^3c^3}$

Correct Answer: D) $\frac{b^3-3abc}{a^3c^3}$

6. For equal roots of

$(3m + 1)x^2 + 2(m + 1)x + m = 0$

Discriminant $D = 0$

$[2(m + 1)]^2 - 4(3m + 1)(m) = 0$

$4(m + 1)^2 - 4m(3m + 1) = 0$

$(m + 1)^2 - m(3m + 1) = 0$

$m^2 + 2m + 1 - 3m^2 - m = 0$

$2m^2 - m - 1 = 0$

$(2m + 1)(m - 1) = 0$

$m = -\frac{1}{2}, 1$

Correct Answer: B) $-\frac{1}{2}, 1$

7. For

$(q - r)x^2 + (r - p)x + (p - q) = 0$

Put $x = 1$:

$(q - r) + (r - p) + (p - q) = 0$

Hence $x = 1$is a root.

Correct Answer: B) 1

8. Given:

$x^2 + 2x + 3 = 0$

$\alpha + \beta = -2, \alpha\beta = 3$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (-2)^2 - 2(3)$

$= 4 - 6 = -2$

Correct Answer: A) -2

9. Since $\alpha, \beta$are roots of

$ax^2 + bx + c = 0$

$a\alpha + b = -\frac{c}{\alpha}, a\beta + b = -\frac{c}{\beta}$

So,

$(a\alpha + b)^{-2} + (a\beta + b)^{-2}$

$= \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2}$

$= \frac{\alpha^2 + \beta^2}{c^2}$

Now,

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(\frac{b}{a}\right)^2 - 2\frac{c}{a}$

$= \frac{b^2 - 2ac}{a^2}$

Hence,

$\frac{\alpha^2 + \beta^2}{c^2} = \frac{b^2 - 2ac}{a^2c^2}$

Correct Answer: C) $\frac{b^2-2ac}{a^2c^2}$

10. Given:

$8x^2 - 3x + 27 = 0$

$\alpha\beta = \frac{27}{8}$

Expression:

$\left(\frac{\alpha^2}{\beta}\right)^{1/3} + \left(\frac{\beta^2}{\alpha}\right)^{1/3}$

$= \alpha^{2/3}\beta^{-1/3} + \beta^{2/3}\alpha^{-1/3}$

Taking common factor:

$= (\alpha\beta)^{-1/3}(\alpha + \beta)$

Now,$\alpha + \beta = \frac{3}{8}$

and$\left(\alpha\beta)^{-1/3} = \left(\frac{27}{8}\right)^{-1/3} = \frac{2}{3}\right.$

Therefore,

$\frac{2}{3} \times \frac{3}{8} = \frac{1}{4}$

Correct Answer: B) $\frac{1}{4}$

11. Given:

α, β are roots of $ax^2 + bx + c = 0$

Expression:

$\alpha^5\beta^8 + \alpha^8\beta^5$

$= \alpha^5\beta^5(\alpha^3 + \beta^3)$

Now,

$\alpha\beta = \frac{c}{a}, \alpha + \beta = -\frac{b}{a}$

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= \left(\frac{b}{a}\right)^3 - 3\left(\frac{c}{a}\right)\left(\frac{b}{a}\right)$

$= \frac{3abc - b^3}{a^3}$

Also, $\alpha^5\beta^5 = \left(\frac{c}{a}\right)^5$

Therefore,

$\alpha^5\beta^8 + \alpha^8\beta^5 = \left(\frac{c}{a}\right)^5 \cdot \frac{3abc - b^3}{a^3}$

$= \frac{c^5(3abc - b^3)}{a^8}$

Correct Answer: A) $\frac{c^5(3abc-b^3)}{a^8}$

12. Given:

$x^2 + kx + 2 = 0$

$\alpha + \beta = -k, \alpha\beta = 2$

Also,

$\alpha - \beta = 1$

Using:

$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$1 = k^2 - 8$

$k^2 = 9$

$k = \pm 3$

Correct Answer: C) ± 3

13. Equation:

$x^2 - 3kx + 2e^{2\log k} - 1 = 0$

Since,

$e^{2\log k} = k^2$

Equation becomes:

$x^2 - 3kx + 2k^2 - 1 = 0$

Product of roots:

$2k^2 - 1 = 7$

$2k^2 = 8$

$k^2 = 4$

Given $k > 0$,

$k = 2$

Sum of roots:

$3k = 3(2) = 6$

Correct Answer: C) 6

14. Since α, β are roots of

$ax^2 + bx + c = 0$

Then, $a\alpha + b = -\frac{c}{\alpha}$

So,$a\beta + b = -\frac{c}{\beta}$

Now,

$\left(\frac{\alpha}{a\beta + b}\right)^3 - \left(\frac{\beta}{a\alpha + b}\right)^3$

$= \left(\frac{\alpha}{-c/\beta}\right)^3 - \left(\frac{\beta}{-c/\alpha}\right)^3$

$= \left(\frac{\alpha\beta}{c}\right)^3 - \left(\frac{\alpha\beta}{c}\right)^3$

$= 0$

Correct Answer: A) 0

15. Let

$x = 2 - \frac{1}{x}$

Then,

$x^2 = 2x - 1$

$x^2 - 2x + 1 = 0$

$(x-1)^2 = 0$

$x = 1$

Correct Answer: B) 1

16. For equal roots, discriminant $D = 0$.

$x^2 - 2(5+2k)x + 3(7+10k) = 0$

Here,

$a = 1, b = -2(5+2k), c = 3(7+10k)$

$b^2 - 4ac = 0$

$[-2(5+2k)]^2 - 4(1)\cdot 3(7+10k) = 0$

$4(5+2k)^2 - 12(7+10k) = 0$

$(5+2k)^2 - 3(7+10k) = 0$

$25 + 20k + 4k^2 - 21 - 30k = 0$

$4k^2 - 10k + 4 = 0$

$2k^2 - 5k + 2 = 0$

$(2k-1)(k-2) = 0$

$k = \frac{1}{2}, 2$

Correct Answer: C) $2, \frac{1}{2}$

17. $x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0$

Let

$t = x^{\frac{1}{3}}$

Then,

$t^2 + t - 2 = 0$

$(t+2)(t-1) = 0$

So,

$t = 1$ or $t = -2$

$x = 1^3 = 1, x = (-2)^3 = -8$

Correct Answer: C) $1, -8$

18. $2^x + 27(2^{-x}) = 12$

Let $2^x = t$

Then,

$2^{-x} = \frac{1}{t}$

So,

$t + \frac{27}{t} = 12$

$t^2 - 12t + 27 = 0$

$(t-3)(t-9) = 0$

$t = 3$ or 9

$2^x = 3 \Rightarrow x = \log_2 3$

$2^x = 9 \Rightarrow x = \log_2 9$

Correct Answer: A) $\log_2 3, \log_2 9$

19. Let the consecutive numbers be x and $x+1$.
Given:
$\frac{1}{x}+\frac{1}{x+1}=\frac{23}{132}$
Taking LCM:
$\frac{2x+1}{x(x+1)}=\frac{23}{132}$
Cross multiply:
$132(2x+1)=23x(x+1)$
$264x+132=23x^2+23x$
$23x^2-241x-132=0$
Solving:
$23x^2-253x+12x-132=0$
$23x(x-11)+12(x-11)=0$
$(x-11)(23x+12)=0$
$x=11$
So the numbers are 11 and 12.
Correct Answer: D) 11

20. Sum of roots:
$\alpha+\beta=12$
Sum of cubes:
$\alpha^3+\beta^3=468$
Using,
$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$
$468=12^3-3(\alpha\beta)(12)$
$468=1728-36\alpha\beta$
$36\alpha\beta=1260$
$\alpha\beta=35$
Quadratic equation:
$x^2-(\alpha+\beta)x+\alpha\beta=0$
$x^2-12x+35=0$
Correct Answer: C) $x^2-12x+35=0$

LEVEL-II

Single Answer Questions

1. Which of the following is a quadratic equation?
A quadratic equation has degree 2.
A) $x^2+\frac{1}{x}=0\rightarrow$ not polynomial
B) $x^2+3=0\rightarrow$ degree 2 ✓
C) $x^2-x^{2/3}+1=0\rightarrow$ fractional power
D) $x+\frac{1}{x}=0\rightarrow$ not quadratic
Correct Answer: B) $x^2+3=0$

2. Roots of $3a^2x^2+8abx+4b^2=0$
Using quadratic formula:
$x=\frac{-8ab\pm\sqrt{(8ab)^2-4(3a^2)(4b^2)}}{2(3a^2)}$
$=\frac{-8ab\pm\sqrt{64a^2b^2-48a^2b^2}}{6a^2}$

$= \frac{-8ab \pm 4ab}{6a^2}$

Roots are:

$x = -\frac{2b}{3a}, x = -\frac{b}{a}$

From options:

Correct Answer: B) $-\frac{2b}{3a}$

3. Roots of $25x^2 + 30x + 7 = 0$

Using quadratic formula:

$x = \frac{-30 \pm \sqrt{30^2 - 4(25)(7)}}{50}$

$= \frac{-30 \pm \sqrt{900 - 700}}{50}$

$= \frac{-30 \pm 10\sqrt{2}}{50}$

$= \frac{-3 \pm \sqrt{2}}{5}$

From options:

Correct Answer: D) $\frac{-3-\sqrt{2}}{5}$

4. Roots of $2^{2x+3} - 57 = 65(2^x - 1)$

Let $y = 2^x$

$2^{2x+3} = 8(2^x)^2 = 8y^2$

Equation becomes:

$8y^2 - 57 = 65y - 65$

$8y^2 - 65y + 8 = 0$

Factorizing:

$(8y - 1)(y - 8) = 0$

$y = \frac{1}{8}$ or 8

Since $y = 2^x$,

$2^x = \frac{1}{8} = 2^{-3} \Rightarrow x = -3$

or

$2^x = 8 = 2^3 \Rightarrow x = 3$

Correct Answer: A) 3

5. If $\alpha + \beta = 2$, $\alpha\beta = 3$, find $\alpha^3 + \beta^3$

Formula:

$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

$= 2^3 - 3(3)(2)$

$= 8 - 18 = -10$

Correct Answer: B) -10

6. If the roots of $12x^2 + mx + 5 = 0$ are in the ratio $5:4$, find m

Let roots be $5k$ and $4k$.

Product of roots:

$5k \cdot 4k = \frac{5}{12}$

$20k^2 = \frac{5}{12}$

$k^2 = \frac{1}{48}$

$k = \frac{1}{4\sqrt{3}}$

Sum of roots:

$5k + 4k = 9k = \frac{-m}{12}$

$m = -108k$

$m = -108\left(\frac{1}{4\sqrt{3}}\right) = -\frac{27}{\sqrt{3}} = -9\sqrt{3}$

Similarly, positive value is also possible.

Correct Answer: B) $\pm 9\sqrt{3}$

7. If one root of $x^2 - 10x + 3(7 + 10k) = 0$is double the other, find k

Let roots be a and $2a$.

Sum of roots:

$a + 2a = 10$

$3a = 10$

$a = \frac{10}{3}$

Product of roots:

$a(2a) = 3(7 + 10k)$

$2\left(\frac{10}{3}\right)^2 = 3(7 + 10k)$

$\frac{200}{9} = 21 + 30k$

$200 = 189 + 270k$

$11 = 270k$

$k = \frac{11}{270}$

Correct Answer: C) $\frac{11}{270}$

8. If $\frac{3}{x+2} - \frac{2}{x-1} = 5$, find x

Taking LCM:

$\frac{3(x-1) - 2(x+2)}{(x+2)(x-1)} = 5$

$\frac{3x - 3 - 2x - 4}{(x+2)(x-1)} = 5$

$\frac{x - 7}{(x+2)(x-1)} = 5$

$x - 7 = 5(x^2 + x - 2)$

$x - 7 = 5x^2 + 5x - 10$

$5x^2 + 4x - 3 = 0$

Using quadratic formula:

$x = \frac{-4 \pm \sqrt{16 + 60}}{10}$

$= \frac{-4 \pm \sqrt{76}}{10} = \frac{-2 \pm \sqrt{19}}{5}$

Correct Answer: A) $\frac{-2}{5} \pm \frac{\sqrt{19}}{5}$

9. Given,
$3^{2x} - 3^{x+1} + 1 = 0$
Let
$3^x = t(t > 0)$
Then,
$3^{2x} = (3^x)^2 = t^2, 3^{x+1} = 3 \cdot 3^x = 3t.$

So the equation becomes
$t^2 - 3t + 1 = 0.$

Using the quadratic formula,
$t = \frac{3 \pm \sqrt{9-4}}{2} = \frac{3 \pm \sqrt{5}}{2}.$
Since $t = 3^x$,
$x = \log_3\left(\frac{3 \pm \sqrt{5}}{2}\right).$

One or More Than One Correct Answer Questions

10. If α, β are roots of $x^2 + x + 2 = 0$
Given:
$\alpha + \beta = -1, \alpha\beta = 2$

A) $\alpha^3 + \beta^3 = 5$
$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
$= (-1)^3 - 3(2)(-1) = -1 + 6 = 5$
True

B) $\alpha^3\beta^3 + \alpha^2\beta^3 + \alpha^3\beta^2 = 3$
$= (\alpha\beta)^3 + \alpha^2\beta^2(\alpha + \beta)$
$= 2^3 + (2^2)(-1) = 8 - 4 = 4$
False

C) $\alpha - \beta = \sqrt{-7}$
$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$
$= 1 - 8 = -7$
$\alpha - \beta = \pm\sqrt{-7}$
True

D) $\alpha^2 + \beta^2 = -3$
$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
$= 1 - 4 = -3$
True

Correct Answers: A, C, D

11. If $6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x - \frac{1}{x}\right) + 12 = 0$
Let
$y = x - \frac{1}{x}$
Then,
$x^2 + \frac{1}{x^2} = y^2 + 2$
Substitute:
$6(y^2 + 2) - 25y + 12 = 0$

$6y^2 - 25y + 24 = 0$

$(3y - 8)(2y - 3) = 0$

$y = \frac{8}{3}$ or $\frac{3}{2}$

Checking options:

For $x = 2$,

$2 - \frac{1}{2} = \frac{3}{2}$

For $x = -\frac{1}{2}$,

$-\frac{1}{2} - (-2) = \frac{3}{2}$

For $x = 3$,

$3 - \frac{1}{3} = \frac{8}{3}$

For $x = -\frac{1}{3}$,

$-\frac{1}{3} - (-3) = \frac{8}{3}$

Correct Answers: A, B, C, D

12. If the product of roots of

$\frac{1}{k}(5x^2 - 4x + 38) + (-4x^2 - 2x - 1) = 0$

is -5, find k

Simplify: $\left(\frac{5}{k} - 4\right)x^2 + \left(\frac{4}{k} - 2\right)x + \left(\frac{38}{k} - 1\right) = 0$

Product of roots:

$\frac{c}{a} = -5$

$\frac{\frac{38}{k} - 1}{\frac{5}{k} - 4} = -5$

Cross multiply:

$38 - k = -25 + 20k$

$63 = 21k$

$k = 3$

Correct Answer: A) 3

Comprehension Type Questions

Passage – I

13. The value of

$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}}$

Given:

$\alpha\beta = \frac{c}{a}$

Now, $\alpha^{-3} + \beta^{-3} = \frac{\alpha^3 + \beta^3}{(\alpha\beta)^3}$

Therefore,

$\frac{\alpha^3 + \beta^3}{\alpha^{-3} + \beta^{-3}} = (\alpha\beta)^3 = \left(\frac{c}{a}\right)^3 = \frac{c^3}{a^3}$

Correct Answer: C) $\frac{c^3}{a^3}$

14. The value of

$\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}$

Take LCM:

$=\frac{\alpha^3+\beta^3}{\alpha\beta}$

Now, $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$

Using

$\alpha+\beta=-\frac{b}{a},\alpha\beta=\frac{c}{a}$

$\alpha^3+\beta^3=-\frac{b^3}{a^3}+\frac{3abc}{a^3}$

$=\frac{3abc-b^3}{a^3}$

Divide by $\alpha\beta=\frac{c}{a}$:

$\frac{3abc-b^3}{a^3}\times\frac{a}{c}=\frac{3abc-b^3}{a^2c}$

Correct Answer: A) $\frac{3abc-b^3}{a^2c}$

15. The value of

$\frac{1}{\alpha^2}+\frac{1}{\beta^2}$

$=\frac{\alpha^2+\beta^2}{(\alpha\beta)^2}$

Now, $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=\left(\frac{b}{a}\right)^2-2\left(\frac{c}{a}\right)$

$=\frac{b^2-2ac}{a^2}$

Also, $\left(\alpha\beta\right)^2=\left(\frac{c}{a}\right)^2=\frac{c^2}{a^2}$

Therefore,

$\frac{\frac{b^2-2ac}{a^2}}{\frac{c^2}{a^2}}=\frac{b^2-2ac}{c^2}$

Correct Answer: B) $\frac{b^2-2ac}{c^2}$

Matching Type Questions

16. Match Column-I with Column-II

a) $12+\sqrt{12+\sqrt{12+\cdots}}$

Let $x=12+\sqrt{12+\sqrt{12+\cdots}}$

Then,

$x=12+\sqrt{x}$

$x-12=\sqrt{x}$

Squaring:

$x^2-24x+144=x$

$x^2-25x+144=0$

$(x-16)(x-9)=0$

Taking valid value:

$x = 16$

a → Q

b) $\sqrt{3-\sqrt{3-\sqrt{3-\cdots}}}$

Let $x = \sqrt{3-x}$

Squaring:

$x^2 = 3 - x$

$x^2 + x - 3 = 0$

$x = \frac{-1 \pm \sqrt{13}}{2}$

Positive value:

$x = \frac{-1 + \sqrt{13}}{2}$

b → R

c) $1 + \frac{1}{1+\frac{1}{1+\cdots}}$

Let

$x = 1 + \frac{1}{x}$

$x^2 = x + 1$

$x^2 - x - 1 = 0$

$x = \frac{1 \pm \sqrt{5}}{2}$

Positive value:

$x = \frac{\sqrt{5}+1}{2}$

c → P

d) $\sqrt{6+\sqrt{6+\sqrt{6+\cdots}}}$

Let

$x = \sqrt{6+x}$

Squaring:

$x^2 = 6 + x$

$x^2 - x - 6 = 0$

$(x-3)(x+2) = 0$

Positive value: $x = 3$

d → S

Integer Type Questions

17. If $(a^2-1)x^2 + (a-1)x + (a^2-4a+3) = 0$ is an identity in x, then all coefficients must be zero.

$a^2 - 1 = 0$

$(a-1)(a+1) = 0$

$a = 1$ or -1

Also, $a - 1 = 0 \Rightarrow a = 1$

and

$a^2 - 4a + 3 = 0$

$(a-1)(a-3) = 0$

Common value is: $a = 1$

Answer: 1

18. Product of the roots of $2x^2 - 7x + 3 = 0$

For $ax^2 + bx + c = 0$,

$\alpha\beta = \frac{c}{a}$

$= \frac{3}{2}$

Answer: $\frac{3}{2}$

19. Let the consecutive positive integers be $x$and $x + 1$.

Given:

$x^2 + (x + 1)^2 = 145$

$x^2 + x^2 + 2x + 1 = 145$

$2x^2 + 2x - 144 = 0$

$x^2 + x - 72 = 0$

$(x + 9)(x - 8) = 0$

So, $x = 8$.

The integers are 8and 9.

Their difference is:

$9 - 8 = 1$

Answer: 1

20. Number of real roots of

$2^{2x^2-7x+5} = 1$

Since $2^y = 1$only when $y = 0$,

$2x^2 - 7x + 5 = 0$

Factorizing:

$(2x - 5)(x - 1) = 0$

$x = \frac{5}{2}, 1$

Hence there are 2real roots.

Answer: 2

STUDENT PRACTICE QUESTIONS

Single Answer Questions

1. The roots of

$\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$

Using quadratic formula:

$x = \frac{-10 \pm \sqrt{100 - 4(\sqrt{3})(-8\sqrt{3})}}{2\sqrt{3}}$

$= \frac{-10 \pm \sqrt{100 + 96}}{2\sqrt{3}}$

$= \frac{-10 \pm 14}{2\sqrt{3}}$

Roots:

$x = \frac{4}{2\sqrt{3}} = \frac{2}{\sqrt{3}}$

$x = \frac{-24}{2\sqrt{3}} = -4\sqrt{3}$

Correct Answer: C) $\frac{2}{\sqrt{3}}, -4\sqrt{3}$

2. If one root of $x^2 - 5x + k = 0$ is 2
Sum of roots: $\alpha + \beta = 5$
If one root is 2,
$2 + \beta = 5$
$\beta = 3$
Correct Answer: B) 3

3. Quadratic equation whose roots are 3,5
Sum of roots: $3 + 5 = 8$
Product: $3 \times 5 = 15$
Equation: $x^2 - (\text{sum})x + (\text{product}) = 0$
$x^2 - 8x + 15 = 0$
Correct Answer: A) $x^2 - 8x + 15 = 0$

4. If roots of $x^2 + px + 4 = 0$ are equal
For equal roots: $b^2 - 4ac = 0$
$p^2 - 4(1)(4) = 0$
$p^2 = 16$
$p = \pm 4$
From options:
Correct Answer: D) 4

5. The equation whose roots are $2\sqrt{3} - 5,\ -2\sqrt{3} - 5$
Sum of roots: $(2\sqrt{3} - 5) + (-2\sqrt{3} - 5) = -10$
Product: $(2\sqrt{3} - 5)(-2\sqrt{3} - 5) = 25 - 12 = 13$
Equation:
$x^2 - (\text{sum})x + (\text{product}) = 0$
$x^2 + 10x + 13 = 0$
Correct Answer: C) $x^2 + 10x + 13 = 0$

6. The equation $9x^2 + kx + 1 = 0$has equal roots
For equal roots: $b^2 - 4ac = 0$
$k^2 - 4(9)(1) = 0$
$k^2 = 36$
$k = \pm 6$
Correct Answer: C) ± 6

7. The solution set of
$7^{1+x} + 7^{1-x} = 50$
$7 \cdot 7^x + 7 \cdot 7^{-x} = 50$
$7\left(7^x + \frac{1}{7^x}\right) = 50$
$7^x + \frac{1}{7^x} = \frac{50}{7}$
Let $y = 7^x$
$y + \frac{1}{y} = \frac{50}{7}$
$7y^2 - 50y + 7 = 0$
$(7y - 1)(y - 7) = 0$
$y = \frac{1}{7}$ or 7
So, $7^x = 7 \Rightarrow x = 1$
$7^x = \frac{1}{7} \Rightarrow x = -1$

Correct Answer: B) $\mathbf{1, -1}$

One or More Than One Correct Answer Questions

8. Which of the following equations have equal roots?

For equal roots:

$b^2 - 4ac = 0$

A) $x^2 - 6x + 9 = 0$

$(-6)^2 - 4(1)(9) = 36 - 36 = 0$

Equal roots

B) $x^2 + 4x + 4 = 0$

$4^2 - 4(1)(4) = 16 - 16 = 0$

Equal roots

C) $x^2 - 5x + 6 = 0$

$25 - 24 = 1 \neq 0$

Not equal roots

D) $x^2 + 1 = 0$

$0^2 - 4(1)(1) = -4 \neq 0$

Not equal roots

Correct Answers: A, B

9. If α, β are roots of $x^2 + x + 1 = 0$

Given:

$\alpha + \beta = -1, \alpha\beta = 1$

A) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = 1$

$= \frac{\alpha^2 + \beta^2}{\alpha\beta}$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= 1 - 2 = -1$

$\frac{-1}{1} = -1$

False

B) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 1$

$= \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2}$

$= \frac{-1}{1} = -1$

False

C) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = -1$

From above value:

$= -1$

True

D) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = -1$

From above:

$= -1$

True

Correct Answers: C, D

Comprehension Type Questions

Passage – I

10. The roots of the equation $x^2 - 5x + 6 = 0$are

Factorizing:
$x^2 - 5x + 6 = (x - 2)(x - 3)$
$x = 2, 3$
Correct Answer: A) 2, 3

11. Find the value of k if one root of $x^2 - 7x + k = 0$ is 3
Substitute $x = 3$: $3^2 - 7(3) + k = 0$
$9 - 21 + k = 0$
$k = 12$
Correct Answer: B) 12

12. The discriminant of $2x^2 - 4x + 1 = 0$ is
$D = b^2 - 4ac$
Here, $a = 2$, $b = -4$, $c = 1$
$D = (-4)^2 - 4(2)(1)$
$= 16 - 8$
$= 8$
Correct Answer: B) 8

Matching Type Questions

13.

1. $x^2 - 5x + 6 = 0$
Factorizing: $(x - 2)(x - 3) = 0$
Roots are $(2, 3)\rightarrow$ C
2. $x^2 + 4x + 4 = 0$
Factorizing: $(x + 2)^2 = 0$
Roots are equal → A
3. $x^2 + 1 = 0$
$x^2 = -1$gives imaginary roots → B
4. $x^2 - 9 = 0$
$(x - 3)(x + 3) = 0$
Roots are $(-3,3)\rightarrow$ D

Integer Type Questions

14. Let the numbers be $x$and y.
Given: $x + y = 27$ and $xy = 182$
$(x - y)^2 = (x + y)^2 - 4xy$
$(x - y)^2 = (x + y)^2 - 4xy$
$= (27)^2 - 4(182) = 729 - 728 = 1$
$x - y = 1$
Answer: 1

15. For $x^2 - 10x + 21 = 0$,
Discriminant:
$D = b^2 - 4ac$
Here, $a = 1$, $b = -10$, $c = 21$
$D = (-10)^2 - 4(1)(21)$
$= 100 - 84 = 16$
Answer: 16

5. Plane Geometry

LEVEL – I

1. In a quadrilateral, sum of angles = 360°
$55° + 65° + 105° = 225°$
Fourth angle
$360° - 225° = 135°$
Answer: B) 135°

2. Let each of the remaining angles be x.
$80° + 120° + x + x = 360°$
$200° + 2x = 360°$
$2x = 160°$
$x = 80°$
Answer: C) 80°

3. Angles are:
$x, x - 10°, x + 30°, 2x$
Sum of angles of quadrilateral:
$x + (x - 10) + (x + 30) + 2x = 360$
$5x + 20 = 360$
$5x = 340$
$x = 68°$
Angles:
68°, 58°, 98°, 136°
Smallest angle = 58°
Answer: C) 58°

4. Ratio $1:2:3:6$
Total parts:
$1 + 2 + 3 + 6 = 12$
Each part:
$\frac{360°}{12} = 30°$
Largest angle: $6 \times 30° = 180°$
Answer: A) 180°

5. In trapezium $ABCD$, $AB \parallel CD$
Consecutive interior angles are supplementary.
$\angle A + \angle D = 180°$
$\angle A = 180° - 50° = 130°$
Answer: D) 130°

6. In an isosceles trapezium, diagonals are equal.
$AC = BD$
Given $AC = 10$ cm
$BD = 10$ cm
Answer: A) 10 cm

7. In a parallelogram, diagonals bisect each other.
So, $AO = \frac{AC}{2}$
Given $AO = 5$ cm
$AC = 2 \times 5 = 10$ cm
Answer: B) 10 cm

8. In a rhombus, diagonals intersect at right angles.
$\angle AOB = 90°$
Answer: D) 90°

9. In a rectangle, diagonals are equal.
$AC = BD$
Given $AC = 10$ cm
$BD = 10$ cm
Answer: B) 10 cm

10. In a square, diagonal bisects the 90°angle.
$\angle BAC = \frac{90°}{2} = 45°$
Answer: D) 45°

11. In a parallelogram, adjacent angles are supplementary.
$\angle A + \angle B = 180°$
Given $\angle A = 50°$
$\angle B = 180° - 50° = 130°$
Answer: B) 130°

12. In a parallelogram, diagonals bisect each other.
So,
$OA = OC$
Given:
$OA = 2x + 15, OC = 10x - 9$
Equating,
$2x + 15 = 10x - 9$
$24 = 8x$
$x = 3$
Then, $OA = 2(3) + 15 = 21$
Since $OA = \frac{AC}{2}$,
$AC = 2 \times 21 = 42$
Answer: D) 42

13. In rectangle $ABCD$,
$AB = 4$ cm, $BC = 3$ cm
Using Pythagoras theorem for diagonal AC:
$a^2 + b^2 = c^2$
$AC^2 = 4^2 + 3^2$
$AC^2 = 16 + 9 = 25$
$AC = 5$ cm
Answer: B) 5 cm

14. Let
$PQ = x$ cm
Then
$QR = x + 3$
Perimeter of parallelogram:
$2(PQ + QR) = 70$
$2(x + x + 3) = 70$
$2(2x + 3) = 70$
$4x + 6 = 70$
$4x = 64$

$x = 16$
Therefore,
$QR = x + 3 = 19$ cm
Answer: D) 19 cm

15. Angles are in the ratio $2:3:7:6$.
Total parts:
$2 + 3 + 7 + 6 = 18$
Sum of angles of a quadrilateral:
$360°$
One part:
$\frac{360°}{18} = 20°$
Largest angle:
$7 \times 20° = 140°$
Answer: C) $140°$

16. Let adjacent sides be $x$cm and $x + 25$cm.
Perimeter of parallelogram:
$2(x + x + 25) = 150$
$2(2x + 25) = 150$
$4x + 50 = 150$
$4x = 100$
$x = 25$
Other side:
$25 + 25 = 50$
Thus, adjacent sides are 25 cmand 50 cm.
Answer: A) 25 cm, 50 cm

17. In a parallelogram, opposite angles are equal.
$\angle A = \angle C$
Given:
$2x + 30 = x + 120$
$x = 90$
Answer: B) 90

18. In a parallelogram, adjacent angles are supplementary.
$\angle A + \angle B = 180°$
Given:
$(3x - 10) + (5x + 30) = 180$
$8x + 20 = 180$
$8x = 160$
$x = 20$
Then,
$\angle B = 5(20) + 30 = 130°$
Opposite angles are equal, so
$\angle D = 130°$
Answer: B) $130°$

19. In rectangle $ABCD$,
$AB = 5$ cm, $BC = 12$ cm
Using Pythagoras theorem:
$a^2 + b^2 = c^2$
$AC^2 = 5^2 + 12^2$

$AC^2 = 25 + 144 = 169$
$AC = 13$ cm
Answer: D) 13 cm

20. Let
$PQ = x$
Then
$QR = x + 3$
Perimeter of parallelogram:
$2(PQ + QR) = 78$
$2(x + x + 3) = 78$
$2(2x + 3) = 78$
$4x + 6 = 78$
$4x = 72$
$x = 18$
Therefore,
$QR = x + 3 = 21$ cm
Answer: A) 21 cm

LEVEL – II

Single option type

1. Sum of interior angles of a quadrilateral:
360°
Answer: B) 360°

2. In a parallelogram:
Opposite angles are equal
Adjacent angles are supplementary
Given:
$\angle A = 115°$
Then,
$\angle C = 115°$
$\angle B = \angle D = 180° - 115° = 65°$
Answer: A) $\angle B = 65°, \angle C = 115°, \angle D = 65°$

3. In parallelogram $PQRS$,
$\angle SPQ = 65°$
Adjacent angles are supplementary:
$\angle SQR = 50°$
At vertex Q,
$\angle PQR = 180° - 65° = 115°$
Diagonal QSdivides angle PQR:
$\angle PQS + \angle SQR = 115°$
$\angle PQS = 115° - 50° = 65°$
Since opposite sides are parallel,
$\angle RSQ = \angle PQS = 65°$
Answer: A) 65°

4. Two angles are 72°and 56°.
Remaining two angles are in ratio 3: 5.
Sum of known angles:
$72° + 56° = 128°$
Remaining sum:

$360° - 128° = 232°$
Let remaining angles be $3x$and $5x$.
$3x + 5x = 232°$
$8x = 232°$
$x = 29°$
Angles:
$3x = 87°, 5x = 145°$
Largest angle:
$145°$
Answer: D) $145°$

5. Given $AB \parallel DC$
Consecutive interior angles are supplementary.
$\angle A : \angle D = 2:3$
Let
$\angle A = 2x, \angle D = 3x$
$2x + 3x = 180°$
$5x = 180°$
$x = 36°$
$\angle D = 3x = 108°$
Answer: B) $108°$

6. In a rhombus, diagonals bisect each other at right angles.
Half diagonals:
$\frac{6}{2} = 3, \frac{8}{2} = 4$
Side of rhombus is hypotenuse:
$a^2 + b^2 = c^2$
$\text{side}^2 = 3^2 + 4^2$
$= 9 + 16 = 25$
side = 5 cm
Answer: B) 5 cm

7. In a parallelogram, adjacent angles are supplementary.
Given: $\angle A : \angle B = 2:7$
Let $\angle A = 2x, \angle B = 7x$
$2x + 7x = 180°$
$9x = 180°$
$x = 20°$
$\angle A = 40°$
Opposite angles are equal, so
$\angle C = \angle A = 40°$
Answer: A) $40°$

8. In a rhombus, diagonals bisect each other at right angles.
Half diagonals:
$\frac{24}{2} = 12, \frac{10}{2} = 5$
Side of rhombus:
$a^2 + b^2 = c^2$
$\text{side}^2 = 12^2 + 5^2$
$= 144 + 25 = 169$
side = 13 cm

Perimeter:
$4 \times 13 = 52$ cm
Answer: C) 52 cm

9. Median of a trapezium:
$$\text{Median} = \frac{a+b}{2}$$
$$= \frac{18+32}{2}$$
$$= \frac{50}{2} = 25 \text{ cm}$$
Answer: C) 25 cm

10. In rectangle $ABCD$,
$AC = 26$ cm, $AB = 10$ cm
Using Pythagoras theorem:
$a^2 + b^2 = c^2$
$AB^2 + BC^2 = AC^2$
$10^2 + BC^2 = 26^2$
$100 + BC^2 = 676$
$BC^2 = 576$
$BC = 24$ cm
Answer: B) 24 cm

One or more than one type

11. For a parallelogram:
Opposite sides are equal
Opposite angles are equal
Diagonals bisect each other
Correct Answers: A, C, D

12. For a rectangle:
Diagonals are equal
Opposite sides are parallel
Diagonals bisect each other
Correct Answers: B, C, D

13. For a rhombus:
All sides are equal
Diagonals are perpendicular
Opposite angles are equal
Correct Answers: A, B, D

14. For a square:
Diagonals are equal
Diagonals bisect each other at right angles
All angles are 90°
Correct Answers: A, B, C

COMPREHENSION TYPE QUESTIONS

Passage–1

Given angles of quadrilateral:
$A = 5x^\circ, B = 5(x+2)^\circ, C = (6x-20)^\circ, D = 6(x+3)^\circ$
Sum of angles of quadrilateral:
$5x + 5(x+2) + (6x-20) + 6(x+3) = 360$

$5x + 5x + 10 + 6x - 20 + 6x + 18 = 360$

$22x + 8 = 360$

$22x = 352$

$x = 16$

Now,

15. $\angle B = 5(x + 2)$

$= 5(16 + 2) = 5 \times 18 = 90°$

Answer: B) $90°$

16. $\angle C = 6x - 20$

$= 6(16) - 20$

$= 96 - 20 = 76°$

Answer: C) $76°$

17. Matrix Matching

a) In a quadrilateral $ABCD$, if

$\angle A + \angle C = 180°, \angle B + \angle D = 180°$

then it represents a cyclic quadrilateral property.

Match → **Q)** $180°$

b) Sum of interior angles in a quadrilateral

$= 360°$

Match → **R)** $360°$

c) If every interior angle is less than 180°, then the quadrilateral is convex.

Match → **S) Convex quadrilateral**

d) If an interior angle is greater than 180°, then the quadrilateral is concave.

Match → **P) Concave**

Final Matching $\boxed{a \to Q, b \to R, c \to S, d \to P}$

INTEGER ANSWER TYPE

18. Median of a trapezium:

$$\text{Median} = \frac{a + b}{2}$$

Given: $EF = 4, CD = 6$

$$4 = \frac{AB + 6}{2}$$

$8 = AB + 6$

$AB = 2$

Answer: 2

19. Sum of angles of a quadrilateral:

$A + B + C + D = 360°$

Given:

$A = 60°, B = 120°$

$C + D = 360° - (60° + 120°)$

$C + D = 180°$

$$\frac{C + D}{90°} = \frac{180°}{90°} = 2$$

Answer: 2

20. Sum of angles of a quadrilateral:

$360°$

Given angles:

$85°, 95°, 110°$

Sum:

$85 + 95 + 110 = 290°$
Fourth angle:
$360° - 290° = 70°$
Answer: 70°

Student Practice corner

Single option type

1. Angles are in the ratio $3:4:5:6$.
Total parts:
$3 + 4 + 5 + 6 = 18$
One part:
$\frac{360°}{18} = 20°$
Largest angle:
$6 \times 20° = 120°$
Answer: B) $120°$

2. In a parallelogram, adjacent angles are supplementary.
Let smaller angle $= x$
Other angle $= 2x$
$x + 2x = 180°$
$3x = 180°$
$x = 60°$
Answer: B) $60°$

3. Area of a kite:
$\text{Area} = \frac{1}{2} d_1 d_2$
Given diagonals:
$d_1 = 16, d_2 = 30$
$\text{Area} = \frac{1}{2} \times 16 \times 30$
$= 8 \times 30 = 240 \text{ cm}^2$
Answer: B) 240 cm^2

4. In a square,
Diagonal:
$d = s\sqrt{2}$
Given:
$d = 8\sqrt{2}$
So,
$s = 8$
Area:
$s^2 = 8^2 = 64 \text{ cm}^2$
Answer: B) 64 cm^2

5. In a parallelogram, diagonals bisect each other.
$AO = \frac{AC}{2} = \frac{28}{2} = 14$
$OD = \frac{BD}{2} = \frac{36}{2} = 18$
$AO + OD = 14 + 18 = 32 \text{ cm}$
Answer: C) 32 cm

6. Perimeter of rhombus:
$4 \times 13 = 52$ cm
Square has same perimeter.
Let side of square $= s$
$4s = 52$
$s = 13$ cm
Answer: C) 13 cm

7. In a rhombus, diagonals bisect each other at right angles.
Given:
One diagonal = 16 cm
Side = 10 cm
Half of given diagonal:
$\frac{16}{2} = 8$ cm
Let half of other diagonal $= x$.
Using Pythagoras theorem:
$a^2 + b^2 = c^2$
$8^2 + x^2 = 10^2$
$64 + x^2 = 100$
$x^2 = 36$
$x = 6$
Other diagonal:
$2x = 12$ cm
Answer: B) 12 cm

8. In a rectangle, diagonal forms a right triangle.
Given: Diagonal = 34 cm
One side = 16 cm
Let other side $= x$.
Using Pythagoras theorem: $a^2 + b^2 = c^2$
$16^2 + x^2 = 34^2$
$256 + x^2 = 1156$
$x^2 = 900$
$x = 30$
Perimeter: $2(16 + 30) = 2(46) = 92$ cm
Since 92 cmis not in the options, the intended option appears incorrect.
Correct perimeter = 92 cm

One or more than one type

9. In a parallelogram:
Opposite angles are equal
Adjacent angles are supplementary
Given:
$\angle A = 70°$
Then,
$\angle C = 70°$
$\angle B = \angle D = 180° - 70° = 110°$
So:
A) $\angle C = 70°$
B) $\angle B = 110°$
C) $\angle D = 110°$

D) Adjacent angles are supplementary

Correct Answers: A, B, C, D

10. A quadrilateral is guaranteed to be a parallelogram if:

Opposite sides are equal

Opposite sides are parallel

Diagonals bisect each other

Correct Answers: A, B, C

COMPREHENSION TYPE QUESTIONS

Passage–1

In a parallelogram, adjacent angles are supplementary.

Given ratio: $A:B = 4:5$

Let

$A = 4x, B = 5x$

$4x + 5x = 180°$

$9x = 180°$

$x = 20°$

Therefore, $A = 4x = 80°$

$B = 5x = 100°$

11. $\angle A =$

Answer: B) $80°$

12. $\angle B =$

Answer: A) $100°$

MATRIX MATCHING TYPE

13. Matrix Matching

a) A closed figure bounded by four line segments is called→ **R) Quadrilateral**

b) A rhombus in which one angle is right angled is called→ **Q) Square**

c) A parallelogram in which two adjacent sides are equal is called→ **P) Rhombus**

d) The diagonals of a rectangle are→ **S) Equal**

Final Matching $\boxed{a \to R, b \to Q, c \to P, d \to S}$

Integer Type

14. In a rhombus, diagonals bisect each other at right angles.

Half diagonals:

$\frac{12}{2} = 6, \frac{16}{2} = 8$

Side of rhombus is the hypotenuse.

Using Pythagoras theorem: $a^2 + b^2 = c^2$

$\text{side}^2 = 6^2 + 8^2$

$= 36 + 64 = 100$

side = 10 cm

Answer: 10

15. Median of a trapezium:

$\text{Median} = \frac{a+b}{2}$

$= \frac{14+22}{2}$

$= \frac{36}{2} = 18$ cm

Answer: 18

Circles

LEVEL - I

1. A wire of length 10cm bent into a circle forms the circumference.

$C = 10$ cm

Answer: C) 10 cm

2. Circumference of circle: $C = 2\pi r$

Given:

$44 = 2 \times \frac{22}{7} \times r$

$44 = \frac{44r}{7}$

$r = 7$ cm

Answer: C) 7 cm

3. Circumference of a circle:

$C = \pi d$

Given: $d = 4.9$ cm

$C = \frac{22}{7} \times 4.9$

$C = 15.4$ cm

Answer: A) 15.4 cm

4. Given circumference = 44cm.

First find radius: $44 = 2 \times \frac{22}{7} \times r$

$r = 7$ cm

Area of circle: $A = \pi r^2$

r

$A = \pi r^2 \approx 28.27$

$C = 2\pi r \approx 18.85$

r = 3.00

$A = \frac{22}{7} \times 7 \times 7$

$A = 154$ sq.cm

Answer: B) 154 sq.cm

5. Diameter = 10.5cm

Circumference:

$C = \pi d$

$C = \frac{22}{7} \times 10.5$

$C = 33$ cm

Answer: B) 33 cm

6. Radius = 15.4cm

Circumference:

$C = 2\pi r$

$C = 2 \times \frac{22}{7} \times 15.4$

$C = 96.8$ cm

Answer: B) 96.8 cm

7. Circumference of circle:

$C = 2\pi r$

Given:

$176 = 2 \times \frac{22}{7} \times r$

$176 = \frac{44r}{7}$

$r = 28$ m

Area of circle:

$A = \pi r^2$

r

$A = \pi r^2 \approx 28.27$

$C = 2\pi r \approx 18.85$

r = 3.00

$A = \frac{22}{7} \times 28 \times 28$

$A = 2464$ sq.m

Answer: B) 2464 sq.m

8. Perimeter of a semicircle:

$\pi r + 2r$

Given:

$r = 2.1$ m

$= \frac{22}{7} \times 2.1 + 2(2.1)$

$= 6.6 + 4.2$

$= 10.8$ m

Answer: C) 10.8 m

9. Area of semicircle:

$\frac{1}{2}\pi r^2$

Given:

$r = 7$ cm

$= \frac{1}{2} \times \frac{22}{7} \times 7 \times 7$

$= 77$ sq.cm

Answer: A) 77 sq.cm

10. Circumference (perimeter) of a semicircle:

$\pi r + 2r$

Diameter:

56 cm $\Rightarrow r = 28$ cm

$= \frac{22}{7} \times 28 + 56$

$= 88 + 56$

$= 144$ cm

Answer: B) 144 cm

11. Let radius of circle = r, side of square = a.

Given:$2\pi r = 4a$

$a = \frac{\pi r}{2}$

Area of circle: πr^2

Area of square: $a^2 = \left(\frac{\pi r}{2}\right)^2 = \frac{\pi^2 r^2}{4}$

Ratio: $\pi r^2 : \frac{\pi^2 r^2}{4}$

$= 4:\pi$

Answer: C) $4:\pi$

12. Circumference of circle:

$C = \pi d$

Given:

$154 = \frac{22}{7} \times d$

$d = \frac{154 \times 7}{22}$

$d = 49$ cm

Answer: D) 49 cm

13. Area of sector:

Area $= \frac{\theta}{360°} \pi r^2$

Given:

$r = 7$ cm, $\theta = 240°$

$= \frac{240}{360} \times \frac{22}{7} \times 7 \times 7$

$= \frac{2}{3} \times 154$

$= 102.67$ sq.cm

Answer: D) 102.67 sq.cm

14. Length of arc:

Arc length $= \frac{\theta}{360°} \times 2\pi r$

Given:

$\theta = 72°, r = 14$ cm

$= \frac{72}{360} \times 2 \times \frac{22}{7} \times 14$

$= \frac{1}{5} \times 88$

$= 17.6$ cm

Answer: A) 17.6 cm

15. Arc length of sector:

Arc length $= \frac{\theta}{360°} \times 2\pi r$

Given: $\theta = 75°, r = 42$ cm

$= \frac{75}{360} \times 2 \times \frac{22}{7} \times 42$

$= \frac{5}{24} \times 264$

$= 55$ cm

Answer: D) 55 cm

16. Let diameter of circle $= d$.

Side of square $= d$.

Radius of circle:

$r = \frac{d}{2}$

Area of circle:

$\pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$

Area of square:
d^2
Ratio:
$\frac{\pi d^2}{4} : d^2$
$\pi : 4$
Answer: B) $\pi : 4$

17. Distance covered in one rotation:
$\frac{2200}{200} = 11$ m
Circumference:
$2\pi r = 11$
$2 \times \frac{22}{7} \times r = 11$
$r = \frac{11 \times 7}{44}$
$r = 1.75$ m
Answer: A) 1.75 m

18. Area of sector: 70 sq.cm
Radius:
7 cm
Using
Area $= \frac{1}{2} r \times l$
where $l$is arc length,
$70 = \frac{1}{2} \times 7 \times l$
$70 = 3.5l$
$l = 20$ cm
Perimeter of sector: $l + 2r = 20 + 14 = 34$ cm
Answer: C) 34 cm

19. Area of semicircle:
$\frac{1}{2}\pi r^2 = 1925$
$\frac{1}{2} \times \frac{22}{7} \times r^2 = 1925$
$\frac{11}{7} r^2 = 1925$
$r^2 = \frac{1925 \times 7}{11} = 1225$
$r = 35$ cm
Perimeter of semicircle:
$\pi r + 2r$
$= \frac{22}{7} \times 35 + 70$
$= 110 + 70$
$= 180$ cm
Answer: D) 180 cm

20. Area of ring:
$\pi(R^2 - r^2)$
Given:
$R = 25$ m, $r = 18$ m
$= \frac{22}{7}(25^2 - 18^2)$
$= \frac{22}{7}(625 - 324)$
$= \frac{22}{7} \times 301$
≈ 946 sq.m
Answer: C) 946 sq.m

LEVEL - II

SINGLE OPTION CORRECT TYPE

1. Area of sector:
Area $= \frac{1}{2}rl$
where $l$is arc length.
Given:
$88 = \frac{1}{2} \times 8 \times l$
$88 = 4l$
$l = 22$ cm
Answer: A) 22 cm

2. Perimeter of semicircle:
$\pi r + 2r = 144$
$\frac{22}{7}r + 2r = 144$
$\frac{36}{7}r = 144$
$r = 28$ cm
Area of semicircle:
$\frac{1}{2}\pi r^2$
$= \frac{1}{2} \times \frac{22}{7} \times 28 \times 28$
$= 1232$ sq.cm
Answer: B) 1232 sq.cm

3. Area of ring:
$\pi(R^2 - r^2)$
Given:
$R = 55, r = 41$
$= \frac{22}{7}(55^2 - 41^2)$
$= \frac{22}{7}(3025 - 1681)$
$= \frac{22}{7} \times 1344$
$= 4224$ sq.m
Answer: D) 4224 sq.m

4. Area of sector : Area of circle

$= \frac{90°}{360°}$

$= 1:4$

Answer: C) 1:4

5. Radius of field:

$r = 105$ m

Angle of sector:

$72°$

Arc length:

$\frac{72}{360} \times 2\pi r$

$= \frac{1}{5} \times 2 \times \frac{22}{7} \times 105$

$= 132$ m

Perimeter of sector:

$132 + 2(105)$

$= 132 + 210$

$= 342$ m

Answer: B) 342 m

6. In 60 minutes, the minute hand completes one full revolution.

Distance moved by tip:

$2\pi r$

Given:

$r = 14$ cm

$= 2 \times \frac{22}{7} \times 14$

$= 88$ cm

Answer: D) 88 cm

7. Side of square = 14 cm

So diameter of each semicircle = 14 cm

Radius:

$r = 7$ cm

The shaded region consists of two semicircles, which together form one full circle.

Area of shaded region:

$A = \pi r^2$

r

$A = \pi r^2 \approx 28.27$

$C = 2\pi r \approx 18.85$

r = 3.00

$= \frac{22}{7} \times 7 \times 7$

$= 154$ cm^2

Answer: D) 154 cm^2

8. Area of circular lawn:

2464 sq.m

$\pi r^2 = 2464$

$\frac{22}{7} r^2 = 2464$

$r^2 = 784$

$r = 28$ m
Path width = 7 m
Outer radius:
$R = 28 + 7 = 35$ m
Area of path:
$\pi(R^2 - r^2)$
$= \frac{22}{7}(35^2 - 28^2)$
$= \frac{22}{7}(1225 - 784)$
$= \frac{22}{7} \times 441$
$= 1386$ sq.m
Answer: B) 1386 sq.m

9. Area of circular track:
$\pi(R^2 - r^2) = 1144$
Width:
$R - r = 14$
Using $\pi = \frac{22}{7}$,
$\frac{22}{7}(R^2 - r^2) = 1144$
$R^2 - r^2 = 364$
$(R - r)(R + r) = 364$
$14(R + r) = 364$
$R + r = 26$
Solving:
$R - r = 14, R + r = 26$
$2R = 40$
$R = 20, r = 6$
Answer: C) 20 m & 6 m

10. Radii are in ratio $5:8$.
Let radii be:
$5x, 8x$
Difference between areas:
$\pi(8x)^2 - \pi(5x)^2 = 351\pi$
$64x^2 - 25x^2 = 351$
$39x^2 = 351$
$x^2 = 9$
$x = 3$
Bigger radius:
$8x = 24$
Area of bigger circle:
$\pi r^2 = \pi(24)^2 = 576\pi$
Answer: A) 576π

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Circumference of circle: $C = 2\pi r$

Given:

$176 = 2 \times \frac{22}{7} \times r$

$r = 28$ m

Area:

$A = \pi r^2$

r

$A = \pi r^2 \approx 28.27$

$C = 2\pi r \approx 18.85$

r = 3.00

$= \frac{22}{7} \times 28 \times 28$

$= 2464$ m^2

Also,

$2464 = 4 \times 616$

So:

A) 2464 m^2

C) 4×616 m^2

Correct Answers: A, C

12. Area of circle:

$A = \pi r^2$

$r = \sqrt{\frac{A}{\pi}}$

Circumference: $C = 2\pi r$

$= 2\pi \sqrt{\frac{A}{\pi}}$

$= 2\sqrt{\pi A}$

Also,

$\sqrt{4\pi A} = 2\sqrt{\pi A}$

So:

B) $2\sqrt{\pi A}$

D) $\sqrt{4\pi A}$

Correct Answers: B, D

13. Area of semicircle:

$\frac{1}{2}\pi r^2 = 1925$

$\frac{1}{2} \times \frac{22}{7} \times r^2 = 1925$

$r = 35$ cm

Perimeter of semicircle:

$\pi r + 2r$

$= \frac{22}{7} \times 35 + 70$

$= 110 + 70$

$= 180$ cm

Also,

$180 = 2 \times 90$

So:

B) 180 cm

D) 2×90 cm

Correct Answers: B, D

COMPREHENSION TYPE

14. Length of arc:

$$\text{Arc length} = \frac{\theta}{360^\circ} \times 2\pi r$$

$$= \frac{135}{360} \times 2 \times \frac{22}{7} \times 8.4$$

$$= \frac{3}{8} \times 52.8$$

$= 19.8$ cm

Answer: B) 19.8 cm

15. Area of sector:

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$= \frac{135}{360} \times \frac{22}{7} \times (8.4)^2$$

$$= \frac{3}{8} \times 221.76$$

$= 83.16$ sq.cm

Answer: C) 83.16 sq.cm

16. Matrix Matching

a) Area of sector $\frac{x^\circ}{360^\circ} \times \pi r^2$

Match → **R**

b) Length of an arc of a sector $\frac{x^\circ}{360^\circ} \times 2\pi r$

Match → **S**

c) Perimeter of a sector $l + 2r$

Match → **P**

d) If $l$is length of arc, then area of sector

$\frac{1}{2}lr$

Match → **Q**

Final Matching $\boxed{a \to R, b \to S, c \to P, d \to Q}$

INTEGER TYPE

17. Perimeter of a semicircle:

$\pi r + 2r$

Given:

$r = 28$ cm

$$= \frac{22}{7} \times 28 + 56$$

$= 88 + 56$

$= 144$ cm

Now,

$\frac{1}{72} \times 144 = 2$

Answer: 2

18. When a wire is bent into a circle, its length becomes the circumference.
Given wire length = 9units.
Circumference = 9
Answer: 9

19. Radius of wheel r = 28cm
Circumference of wheel
$2\pi r$
$= 2 \times \frac{22}{7} \times 28$
= 176 cm
Distance travelled
528 m = 52800 cm
Number of rotations
$= \frac{\text{Total distance}}{\text{Circumference}}$
$= \frac{52800}{176}$
= 300
Answer: 300

20. Area of path = $138k$ sq.m
From earlier calculation:
Area of path = 1386 sq.m
So,
$138k = 1386$
$k = 10$
Answer: 10

Student practice corner

Single option correct type

1. Area of ring:
$\pi(R^2 - r^2)$
Given:
$R = a + b, r = a - b$
$= \pi[(a + b)^2 - (a - b)^2]$
Using difference of squares: $a^2 - b^2 = (a - b)(a + b)$
a
b
aba + ba - b
$= \pi[(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2)]$
$= \pi(4ab)$
$= 4\pi ab$
Answer: D) $4\pi ab$**sq.units**

2. Area of ring:
$\pi(R^2 - r^2)$
Given:
$R = 4, r = 3$
$= \pi(16 - 9)$

$= 7\pi$

Using $\pi = \frac{22}{7}$,

$7 \times \frac{22}{7} = 22$

Answer: D) 22 sq.cm

3. Area of sector:

$A = \frac{1}{2}rl$

where $l$is arc length.

Given:

$231 = \frac{1}{2} \times 21 \times l$

$231 = 10.5l$

$l = 22$ m

Answer: A) 22 m

4. Area of sector:

$924 = \frac{1}{2} \times 42 \times l$

$924 = 21l$

$l = 44$ cm

Perimeter of sector:

$l + 2r$

$= 44 + 84$

$= 128$ cm

Answer: C) 128 cm

5. Perimeter of sector:

$l + 2r$

Given:

$l = 12$ cm, $r = 8$ cm

$= 12 + 16$

$= 28$ cm

Answer: C) 28 cm

6. Circumference of wheel:

$2\pi r$

$= 2 \times \frac{22}{7} \times 35$

$= 220$ cm

Distance travelled:

660 m = 66000 cm

Number of rotations:

$\frac{66000}{220} = 300$

The correct answer is 300

7. Circumference of circle:

$C = 2\pi r$

Given:

$264 = 2 \times \frac{22}{7} \times r$

$264 = \frac{44r}{7}$

$r = \frac{264 \times 7}{44}$

$r = 42$ cm

Answer: B) 42 cm

8. Perimeter of semicircle:

$\pi r + 2r$

Given:

$r = 28$ cm

$= \frac{22}{7} \times 28 + 56$

$= 88 + 56$

$= 144$ cm

Answer: D) 144 cm

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Length of bounding arc of sector:

Arc length $= \frac{\theta}{360°} \times 2\pi r$

Given:

$\theta = 240°, r = 7$ cm

$= \frac{240}{360} \times 2 \times \frac{22}{7} \times 7$

$= \frac{2}{3} \times 44$

$= \frac{88}{3}$ cm

$= 29\frac{1}{3}$ cm

Also,

$29\frac{1}{3} = \frac{88}{3}$

So:

A) $\frac{88}{3}$ cm

B) $29\frac{1}{3}$ cm

Correct Answers: A, B

10. Area of ring-shaped region:

$\pi(R^2 - r^2)$

Given:

$R = 20, r = 15$

$= \pi(400 - 225)$

$= 175\pi$

Using $\pi = \frac{22}{7}$,

$175 \times \frac{22}{7} = 550$

Correct Answers: C

COMPREHENSION TYPE

11. Radius of semicircular garden:

$r = 7$ m

Perimeter of semicircle:

$\pi r + 2r$

$= \frac{22}{7} \times 7 + 14$

$= 22 + 14 = 36$ m

Cost of fencing at Rs. 10 per metre:

$36 \times 10 = 360$

Answer: B) Rs. 360

12. Angle of sector:

72°

Length of arc:

$\frac{72}{360} \times$ circumference

$= \frac{1}{5} \times$ circumference

Therefore,

Circumference:Arc length = 5: 1

Answer: A) 5 : 1

MATRIX MATCHING TYPE

13. Matrix Matching

a) Circumference of a semicircle

$\pi r + 2r$

Match → **P**

b) Area of a circle

πr^2

Match → **T**

c) Diameter of a circle

$2r$

Match → **S**

d) Area of a semicircle

$\frac{\pi r^2}{2}$

Match → **Q**

Final Matching $\boxed{a \to P, b \to T, c \to S, d \to Q}$

INTEGER TYPE

14. Width of a ring:

Width = $R - r$

Given:

$R = 7$ cm, $r = 2$ cm

$7 - 2 = 5$ cm

Answer: 5

15. Radius of circle: $r = 56$ cm

Circumference: $C = 2\pi r$

$= 2 \times \frac{22}{7} \times 56$

$= 352$ cm

Now,

$\frac{1}{44} \times 352 = 8$

Answer: 8

6. Progressions

LEVEL – I

1. How many terms are there in the A.P. 20, 25, 30, …, 100?

Given:

First term $a = 20$, common difference $d = 5$, last term $l = 100$

Use:

$a_n = a + (n-1)d$

$100 = 20 + (n-1)5$

$80 = 5(n-1)$

$n - 1 = 16$

$n = 17$

Correct Answer: B) 17

2. The 11th term of an A.P. is 80 and the 16th term is 40. Find the 31st term.

Use:

$a_n = a + (n-1)d$

$a + 10d = 80$

$a + 15d = 40$

Subtract:

$5d = -40$

$d = -8$

Now,

$a + 10(-8) = 80$

$a = 160$

31st term:

$a_{31} = 160 + 30(-8)$

$= 160 - 240 = -80$

Correct Answer: A) -80

3. In a certain A.P., 5 times the 5th term equals 8 times the 8th term. Find the 13th term.

5th term:

$a + 4d$

8th term:

$a + 7d$

Given:

$5(a + 4d) = 8(a + 7d)$

$5a + 20d = 8a + 56d$

$3a + 36d = 0$

$a + 12d = 0$

But,

13th term $= a + 12d$

$= 0$

Correct Answer: D) 0

4. If the p^{th}term of an A.P. is $q$and the q^{th}term is p, then r^{th}term is

Let first term = a, common difference = d

$a + (p-1)d = q$

$a + (q-1)d = p$

Subtract:

$(p-q)d = q-p$

$d = -1$

Substitute:

$a-(p-1) = q$

$a = p+q-1$

Now r^{th} term:

$a+(r-1)d$

$= p+q-1-(r-1)$

$= p+q-r$

Correct Answer: D) $p+q-r$

5. Identify the given condition

Since

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$

are in Arithmetic Progression (A.P.), the middle term is the average of the other two.

Therefore:

$2\left(\frac{1}{b}\right) = \frac{1}{a}+\frac{1}{c}$

$\frac{2}{b} = \frac{1}{a}+\frac{1}{c}$— (Equation 1)

Rearrange the given expression

The expression is:

$E = \left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)\left(\frac{1}{b}+\frac{1}{c}-\frac{1}{a}\right)$

Rewrite it as:

$E = \left[\frac{1}{b}+\left(\frac{1}{a}-\frac{1}{c}\right)\right]\left[\frac{1}{b}-\left(\frac{1}{a}-\frac{1}{c}\right)\right]$

Using the identity $(x+y)(x-y) = x^2-y^2$

we get:

$E = \left(\frac{1}{b}\right)^2-\left(\frac{1}{a}-\frac{1}{c}\right)^2$

$E = \frac{1}{b^2}-\left(\frac{1}{a}-\frac{1}{c}\right)^2$

Express the squared term using the A.P. condition

Using the identity:

$(x-y)^2 = (x+y)^2-4xy$

$\left(\frac{1}{a}-\frac{1}{c}\right)^2 = \left(\frac{1}{a}+\frac{1}{c}\right)^2-\frac{4}{ac}$

From Equation 1,

$\frac{1}{a}+\frac{1}{c} = \frac{2}{b}$

Substitute:

$\left(\frac{1}{a}-\frac{1}{c}\right)^2 = \left(\frac{2}{b}\right)^2-\frac{4}{ac}$

$= \frac{4}{b^2}-\frac{4}{ac}$

Substitute back into the main expression

$E = \frac{1}{b^2}-\left(\frac{4}{b^2}-\frac{4}{ac}\right)$

$E = \frac{1}{b^2}-\frac{4}{b^2}+\frac{4}{ac}$

$E = \frac{4}{ac} - \frac{3}{b^2}$

Correct Answer: B) $\frac{4}{ac} - \frac{3}{b^2}$

6. If $a, b, c, d, e$are in A.P., find $a - 4b + 6c - 4d + e$

Since the terms are in A.P., let

$b = a + r, c = a + 2r, d = a + 3r, e = a + 4r$

Substitute:

$a - 4(a + r) + 6(a + 2r) - 4(a + 3r) + (a + 4r)$

$= a - 4a - 4r + 6a + 12r - 4a - 12r + a + 4r$

$= (a - 4a + 6a - 4a + a) + (-4r + 12r - 12r + 4r)$

$= 0$

Correct Answer: C) 0

7. If $S_n = 200$, $n = 10$, $a = 5$, find d

Using sum formula of A.P.:

$S_n = \frac{n}{2}[2a + (n - 1)d]$

$200 = \frac{10}{2}[2(5) + 9d]$

$200 = 5(10 + 9d)$

$40 = 10 + 9d$

$30 = 9d$

$d = \frac{10}{3}$

Correct Answer: D) $\frac{10}{3}$

8. Sum of $n$terms of the series $\sqrt{2} + \sqrt{8} + \sqrt{18} + \sqrt{32} + \cdots$

Simplify terms:

$\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}, \ldots$

So series becomes

$\sqrt{2}(1 + 2 + 3 + \cdots + n)$

Using

$1 + 2 + \cdots + n = \frac{n(n + 1)}{2}$

$S_n = \sqrt{2} \cdot \frac{n(n + 1)}{2} = \frac{n(n + 1)}{\sqrt{2}}$

Correct Answer: C) $\frac{n(n+1)}{\sqrt{2}}$

9. The sum of $n$terms of the series $1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \cdots$

General term:

$T_r = r(r + 1)(r + 2)$

Using standard result,

$\sum_{r=1}^{n} r(r + 1)(r + 2) = \frac{n(n + 1)(n + 2)(n + 3)}{4}$

Correct Answer: A) $\frac{n(n+1)(n+2)(n+3)}{4}$

10. The sum of $n$terms of the series

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \cdots$

Write:

$$\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$$

So,

$S_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$

All middle terms cancel:

$S_n = 1 - \frac{1}{n+1} = \frac{n}{n+1}$

Correct Answer: C) $\frac{n}{n+1}$

11. The number of terms in the G.P. $2, 2\sqrt{2}, 4, \ldots, 128$

First term: $a = 2$

Common ratio:

$r = \frac{2\sqrt{2}}{2} = \sqrt{2}$

Use nth term formula:

$128 = 2(\sqrt{2})^{n-1}$

$64 = (\sqrt{2})^{n-1}$

$2^6 = (2^{1/2})^{n-1}$

$2^6 = 2^{\frac{n-1}{2}}$

$6 = \frac{n-1}{2}$

$n - 1 = 12$

$n = 13$

Correct Answer: D) 13 terms

12. The first term of a G.P. is 1, the sum of the third and fifth terms is 90. Find the common ratio.

First term $a = 1$

Third term: $T_3 = r^2$

Fifth term: $T_5 = r^4$

Given:

$r^2 + r^4 = 90$

$r^2(1 + r^2) = 90$

Let $x = r^2$

$x(x + 1) = 90$

$x^2 + x - 90 = 0$

$(x - 9)(x + 10) = 0$

$x = 9$

$r^2 = 9 \Rightarrow r = \pm 3$

Correct Answer: B) ± 3

13. If 4th, 7th, and 10th terms of a G.P are p, q, r respectively, then

Let first term = a, common ratio = k

$p = ak^3, q = ak^6, r = ak^9$

Now, $q^2 = (ak^6)^2 = a^2k^{12}$

and $pr = (ak^3)(ak^9) = a^2k^{12}$

Thus, $q^2 = pr$

Correct Answer: C) $q^2 = pr$

14. In a G.P if $a_1 = 3,\ a_n = 96,\ S_n = 189$, find n

Given:

$a = 3, a_n = 96$

$3r^{n-1} = 96$

$r^{n-1} = 32$

Now use sum formula:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Try $r = 2$:

$2^{n-1} = 32 = 2^5$

$n - 1 = 5 \Rightarrow n = 6$

Check sum:

$$S_6 = \frac{3(2^6 - 1)}{2 - 1}$$

$= 3(64 - 1) = 3 \times 63 = 189$

Verified.

Correct Answer: D) 6

15. Given:

$$3 + 3x + 3x^2 + \cdots \infty = \frac{45}{8}$$

This is an infinite G.P. with

$a = 3, r = x$

Sum of infinite G.P.:

$$S = \frac{a}{1 - r}$$

$$\frac{3}{1 - x} = \frac{45}{8}$$

$24 = 45(1 - x)$

$24 = 45 - 45x$

$45x = 21$

$$x = \frac{21}{45} = \frac{7}{15}$$

Correct Answer: B) $\frac{7}{15}$

16. Given:

$x^{\frac{1}{2}} \cdot x^{\frac{1}{4}} \cdot x^{\frac{1}{8}} \cdot x^{\frac{1}{16}} \cdots \infty$

Add the exponents:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

This is an infinite G.P. with

$$a = \frac{1}{2}, r = \frac{1}{2}$$

$$S = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1$$

So, $x^1 = x$

Correct Answer: C) x

17. Geometric mean G between -9 and -16:

$G = \pm\sqrt{(-9)(-16)}$

$G = \pm\sqrt{144}$

$G = \pm 12$

For the sequence $-9, -12, -16,$

common ratio:

$\frac{-12}{-9} = \frac{4}{3}, \frac{-16}{-12} = \frac{4}{3}$

So the geometric mean is -12.

Correct Answer: B) -12

18. Insert 3 geometric means between $2\frac{1}{4}$ and $\frac{4}{9}$

Convert:

$2\frac{1}{4} = \frac{9}{4}$

So the G.P. is:

$\frac{9}{4}, G_1, G_2, G_3, \frac{4}{9}$

Let common ratio be r.

$\frac{9}{4} \cdot r^4 = \frac{4}{9}$

$r^4 = \frac{4}{9} \times \frac{4}{9} = \frac{16}{81}$

$r = \frac{2}{3}$

Now find terms:

$G_1 = \frac{9}{4} \times \frac{2}{3} = \frac{3}{2}$

$G_2 = \frac{3}{2} \times \frac{2}{3} = 1$

$G_3 = 1 \times \frac{2}{3} = \frac{2}{3}$

Thus the three means are:

$\frac{3}{2}, 1, \frac{2}{3}$

Correct Answer: A) $\frac{3}{2}, 1, \frac{2}{3}$

19. Given: $T_4 = 54, T_9 = 13122$

For G.P.:

$T_n = ar^{n-1}$

$ar^3 = 54$

$ar^8 = 13122$

Divide:

$r^5 = \frac{13122}{54} = 243 = 3^5$

$r = 3$

Now:

$a(3^3) = 54$

$27a = 54$

$a = 2$

Hence,

$T_n = 2(3^{n-1})$

Correct Answer: D) $2 \times 3^{n-1}$

20. Given: $2, a, 2^3, b, 2^5$
are in G.P.
So: $2, a, 8, b, 32$
Let common ratio be r.
$2r^2 = 8$
$r^2 = 4$
$r = 2$
Then:
$a = 2 \times 2 = 4$
$b = 8 \times 2 = 16$
$ab = 4 \times 16 = 64$
Correct Answer: B) 64

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Given A.P.: 7,11,15, … ,139
Here, $a = 7, d = 4$
Use:
$a_n = a + (n-1)d$
$139 = 7 + (n-1)4$
$132 = 4(n-1)$
$n - 1 = 33$
$n = 34$
Correct Answer: A) 34

2. Given: $2n - 5, 2n - 8, 2n - 11$
Common difference:
$d = (2n - 8) - (2n - 5) = -3$
First term:
$a = 2n - 5$
$T_{p+2} = a + (p+1)d$
$= (2n - 5) + (p+1)(-3)$
$= 2n - 5 - 3p - 3$
$= 2n - 3p - 8$
Correct Answer: A) $2n - 3p - 8$

3. First series:
3,7,11, …
$a = 3,\ d = 4$
Second series:
1,6,11, …
$a = 1,\ d = 5$
Common terms:
11,31,51, …
This is an A.P. with
$a = 11,\ d = 20$
10th common term:
$a_{10} = 11 + (10 - 1) \times 20$
$= 11 + 180$
$= 191$
Correct Answer: C) 191

4. Recall a key property of an A.P.: The sum of terms equidistant from the beginning and the end is constant and equal to the sum of the first and last terms.
$a_1 + a_{24} = a_5 + a_{20} = a_{10} + a_{15}$
Substitute this into the given equation:
$(a_1 + a_{24}) + (a_5 + a_{20}) + (a_{10} + a_{15}) = 225$
$3(a_1 + a_{24}) = 225$
$a_1 + a_{24} = 75$
We need to find the sum of the first 24terms:
$S_n = \frac{n}{2}(a_1 + a_n)$
$S_{24} = \frac{24}{2}(a_1 + a_{24})$
$S_{24} = 12 \times 75$
$S_{24} = 900$
Correct Option: C) 900

5. Since $p, q, r$are in A.P.,
$2q = p + r$
So, $q - r = -(r - q), r - p = 2(r - q)$
Also $x, y, z$are in G.P., hence
$y^2 = xz$
Expression:
$x^{q-r} \cdot y^{r-p} \cdot z^{p-q}$
Using $r - p = (r - q) + (q - p)$and A.P. relation, powers simplify to:
$\left(\frac{xz}{y^2}\right)^{q-r}$
Since $xz = y^2$,
$\frac{xz}{y^2} = 1$
Thus value:
$1^{q-r} = 1$
Correct Answer: A) 1

6. Insert four G.M.'s between 2and 486.
The G.P. is:
$2, G_1, G_2, G_3, G_4, 486$
Let common ratio be r.
$2r^5 = 486$
$r^5 = 243 = 3^5$
$r = 3$
Now find the means:
$G_1 = 2(3) = 6$
$G_2 = 6(3) = 18$
$G_3 = 18(3) = 54$
$G_4 = 54(3) = 162$
Correct Option: A) 6, 18, 54, 162

7. Given: $S_\infty = 20$
For a G.P.,
$S_\infty = \frac{a}{1 - r}$
$\frac{a}{1 - r} = 20 \cdots (1)$

Also, sum of squares:

$a^2 + a^2r^2 + a^2r^4 + \cdots = 100$

$\frac{a^2}{1-r^2} = 100 \cdots (2)$

From (1):

$a = 20(1-r)$

Substitute into (2):

$\frac{400(1-r)^2}{1-r^2} = 100$

$\frac{4(1-r)}{1+r} = 1$

$4 - 4r = 1 + r$

$5r = 3$

Thus, $5 \times r = 3$

Correct Option: B) 3

8. Find the sum of 16.2, 5.4, 1.8, ... to 5 terms.

Here,

$a = 16.2, r = \frac{5.4}{16.2} = \frac{1}{3}$

Use:

$S_n = \frac{a(1-r^n)}{1-r}$

$S_5 = \frac{16.2\left(1-\left(\frac{1}{3}\right)^5\right)}{1-\frac{1}{3}}$

$S_5 = \frac{16.2\left(1-\frac{1}{243}\right)}{\frac{2}{3}}$

$S_5 = 24.2$

Correct Option: D) 24.2

9. Evaluate:

$\sum_{n=1}^{\infty} \left(\frac{a}{a+1}\right)^n$

This is an infinite G.P. with

$a = \frac{a}{a+1}, r = \frac{a}{a+1}$

Using:

$S_\infty = \frac{a}{1-r}$

$S = \frac{\frac{a}{a+1}}{1-\frac{a}{a+1}}$

$= \frac{\frac{a}{a+1}}{\frac{1}{a+1}}$

$= a$

Correct Option: D) a

10. Evaluate:

$$\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1$$

First:$\sum_{k=1}^{j}1=j$

Then:$\sum_{j=1}^{i}j=\frac{i(i+1)}{2}$

So,

$$\sum_{i=1}^{n}\frac{i(i+1)}{2}$$

$$=\frac{1}{2}\sum_{i=1}^{n}(i^2+i)$$

Using formulas:

$$\sum i=\frac{n(n+1)}{2}$$

$$\sum i^2=\frac{n(n+1)(2n+1)}{6}$$

After simplification:

$$=\frac{n(n+1)(n+2)}{6}$$

Correct Option: A) $\frac{n(n+1)(n+2)}{6}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Let the three consecutive terms of the A.P. be:

$a-d,\ a,\ a+d$

Given:

$(a-d)+a+(a+d)=9$

$3a=9$

$a=3$

Now,

$(3-d)^2+3^2+(3+d)^2=35$

$9-6d+d^2+9+9+6d+d^2=35$

$27+2d^2=35$

$2d^2=8$

$d^2=4$

$d=\pm 2$

So the A.P. is either: 1,3,5, …

or 5,3,1, …

For 1,3,5, …

$S_n=\frac{n}{2}[2+(n-1)2]$

$=n^2$

For 5,3,1, …

$S_n=\frac{n}{2}[10+(n-1)(-2)]$

$=n(6-n)$

Correct Options: B) n^2 **and D)** $n(6-n)$

12. Let first term be a and common ratio be r.
Given:
$a + ar = \frac{5}{3}$
$a(1 + r) = \frac{5}{3} \cdots (1)$
Also,
$S_\infty = \frac{a}{1 - r} = 3$
$a = 3(1 - r) \cdots (2)$
Substitute (2) into (1):
$3(1 - r)(1 + r) = \frac{5}{3}$
$3(1 - r^2) = \frac{5}{3}$
$1 - r^2 = \frac{5}{9}$
$r^2 = \frac{4}{9}$
$r = \pm\frac{2}{3}$
Using $a = 3(1 - r)$:
If $r = \frac{2}{3}$,
$a = 1$
If $r = -\frac{2}{3}$,
$a = 5$
Correct Options: A) 1 and C) 5

13. Since $3k - 7$ is the arithmetic mean of $3k + 4$ and $k + 12$,
$2(3k - 7) = (3k + 4) + (k + 12)$
$6k - 14 = 4k + 16$
$2k = 30$
$k = 15$
Now,
$2k + 5 = 2(15) + 5$
$= 35$
Check the options:
$9 \times 5 - 10 = 45 - 10 = 35$
$7 \times 4 + 7 = 28 + 7 = 35$
$8 \times 4 + 3 = 32 + 3 = 35$
$2 \times 17 + 1 = 34 + 1 = 35$
Hence all options are correct.
Correct Options: A, B, C and D

14. Given A.P.: $54,51,48, \ldots$
Here, $a = 54, d = -3$
Sum:
$S_n = \frac{n}{2}[2a + (n - 1)d]$
$513 = \frac{n}{2}[108 - 3(n - 1)]$
$513 = \frac{n}{2}(111 - 3n)$

$1026 = n(111 - 3n)$

$3n^2 - 111n + 1026 = 0$

$n^2 - 37n + 342 = 0$

$(n - 18)(n - 19) = 0$

$n = 18$ or 19

Correct Options: B) 19 and C) 18

COMPREHENSION TYPE

Passage – I:

15. Find the value of n

Let the geometric means between 1 and 1024 be:

$1, G_1, G_2, \ldots, G_n, 1024$

This forms a G.P.

Since $1024 = 2^{10}$,

$r^{n+1} = 1024 = 2^{10}$

Product of geometric means:

$G_1 G_2 \cdots G_n = (ar)(ar^2) \cdots (ar^n)$

Here $a = 1$,

$= r^{1+2+\cdots+n} = r^{\frac{n(n+1)}{2}}$

Given:

$r^{\frac{n(n+1)}{2}} = 2^{45}$

Using $r^{n+1} = 2^{10}$,

$\left(2^{\frac{10}{n+1}}\right)^{\frac{n(n+1)}{2}} = 2^{45}$

$2^{5n} = 2^{45}$

$5n = 45 \Rightarrow n = 9$

Correct Answer: B) 9

16. Find the value of m

Arithmetic means between -2 and 1027:

$-2, A_1, A_2, \ldots, A_m, 1027$

Total terms $= m + 2$

Sum of all terms of A.P.:

$$S = \frac{m+2}{2}[1027 + (-2)]$$

$$S = \frac{m+2}{2}(1025)$$

Given sum of arithmetic means only: $A_1 + A_2 + \cdots + A_m = 1025 \times 171$

But total sum includes first and last terms:

$S = 1025 \times 171 + (-2 + 1027)$

$S = 1025 \times 171 + 1025$

$S = 1025(172)$

So,

$$\frac{m+2}{2}(1025) = 1025(172)$$

$$\frac{m+2}{2} = 172$$

$m + 2 = 344$

$m = 342$

Correct Answer: B) 342

17. Find $G_1 + G_2 + \cdots + G_n$
From Q15, $n = 9$.
Thus G.P. is:
$1, G_1, \ldots, G_9, 1024$
So total terms = 11.
$r^{10} = 1024 = 2^{10} \Rightarrow r = 2$
Hence geometric means are:
$2,4,8, \ldots,512$
Sum:
$2 + 4 + 8 + \cdots + 512$
This is a G.P. with $a = 2$, $r = 2$, $n = 9$.
$S = \frac{2(2^9 - 1)}{2 - 1}$
$= 2(511) = 1022$
Correct Answer: D) 1022

MATRIX MATCHING TYPE

18.

a) Geometric mean between 25 and 16
Geometric mean:
$\sqrt{25 \times 16} = \sqrt{400} = 20$
So,$a \rightarrow Q$
a → Q

b) If x, $2x + 2$, $3x + 3$are in G.P.
For G.P.,
$(2x + 2)^2 = x(3x + 3)$
$4x^2 + 8x + 4 = 3x^2 + 3x$
$x^2 + 5x + 4 = 0$
$(x + 1)(x + 4) = 0$
Taking $x = -4$,
Terms are:
$-4, -6, -9$
Common ratio:
$r = \frac{-6}{-4} = \frac{3}{2}$
Fourth term:
$-9 \times \frac{3}{2} = -\frac{27}{2}$
So, $b \rightarrow P$
b → P

c) Sum of series
$a - (a + d) + (a + 2d) - (a + 3d) + \cdots$
Pair terms:
$[a - (a + d)] + [(a + 2d) - (a + 3d)] + \cdots$
Each pair: $= -d$
There are $n$such pairs and one extra term:$a + 2nd$
Hence total:
$a + nd$
So, $c \rightarrow S$
c → S

d) Sum of series $2 + 5 + 8 + 11 + \cdots$

Here,

$a = 2, d = 3$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$60100 = \frac{n}{2}[4 + 3n - 3]$

$60100 = \frac{n}{2}(3n + 1)$

$120200 = n(3n + 1)$

$3n^2 + n - 120200 = 0$

Solving, $n = 200$

So, $d \rightarrow R$

d → R

INTEGER TYPE

19. Find the first term of the G.P.

Given:

$T_n = 128, S_n = 255, r = 2$

For G.P.,

$T_n = ar^{n-1}$

$a \cdot 2^{n-1} = 128 \ldots (1)$

Also,

$S_n = \frac{a(2^n - 1)}{2 - 1}$

$a(2^n - 1) = 255 \ldots (2)$

From (1),

$a2^n = 256$

Substitute in (2):

$256 - a = 255$

$a = 1$

Answer: $\boxed{1}$

20. Arithmetic mean of $3 + \sqrt{2}$and $7 - \sqrt{2}$

Arithmetic mean:

$$\frac{\left(3 + \sqrt{2}\right) + \left(7 - \sqrt{2}\right)}{2}$$

$$= \frac{10}{2} = 5$$

Answer: $\boxed{5}$

STUDENT PRACTISE CORNER

SINGLE OPTION CORRECT TYPE

1. Find the 18th term of the A.P.

$a_7 = a + 6d = 34$

$a_{13} = a + 12d = 64$

Subtract:

$6d = 30 \Rightarrow d = 5$

$a = 34 - 30 = 4$

$a_{18} = a + 17d = 4 + 85 = 89$

Correct Answer: C) 89

2. Find the n^{th}term
Given A.P.:
$-17, -21, -25$
$a = -17, d = -4$
$a_n = a + (n-1)d$
$= -17 + (n-1)(-4)$
$= -17 - 4n + 4$
$= -4n - 13$
Correct Answer: C) $-4n - 13$

3. Insert 3 geometric means between 16 and 1
Sequence:
$16, _, _, _, 1$
Let common ratio be r.
$16r^4 = 1$
$r^4 = \frac{1}{16}$
$r = \frac{1}{2}$
Terms:
16, 8, 4, 2, 1
So inserted means are:
8,4,2
Correct Answer: A) 8, 4, 2

4. If $3, a, 12, b, 48$ are in G.P., find ab
Let common ratio be r.
$3r^2 = 12$
$r^2 = 4 \Rightarrow r = 2$
Then,
$a = 3 \times 2 = 6$
$b = 12 \times 2 = 24$
$ab = 6 \times 24 = 144$
Correct Answer: B) 144

5. Find n
Given:
$a = 2, a_n = 162, S_n = 242$
$a_n = ar^{n-1}$
$2r^{n-1} = 162$
$r^{n-1} = 81$
Try $r = 3$:
$3^4 = 81$
So, $n - 1 = 4 \Rightarrow n = 5$
Check sum:
$S_n = \frac{a(r^n - 1)}{r - 1}$
$= \frac{2(3^5 - 1)}{2} = 243 - 1 = 242$
Verified.
Correct Answer: C) 5

6. Number of terms in G.P. $4,4\sqrt{2},8,\ldots,512$

$a=4, r=\sqrt{2}$

Use:

$a_n = ar^{n-1}$

$512 = 4(\sqrt{2})^{n-1}$

$128 = (\sqrt{2})^{n-1}$

$2^7 = 2^{\frac{n-1}{2}}$

$\frac{n-1}{2} = 7$

$n-1 = 14$

$n = 15$

Correct Answer: C) 15 terms

ONE OR MORE THAN ONE OPTION CORRECT TYPE

7. A.M. = 20 and G.M. = 16

$\frac{a+b}{2} = 20 \Rightarrow a+b = 40$

$\sqrt{ab} = 16 \Rightarrow ab = 256$

Numbers satisfying these are 32 and 8.

Check:

$32+8 = 40, 32\times 8 = 256$

Correct Options: A, D

8. Which are Arithmetic Progressions?

A) 3,7,11,15

Common difference:

$7-3 = 4, 11-7 = 4$

A.P.

B) 5,10,20,40

Differences:

5,10,20

Not constant

C) 9,6,3,0

Differences:

$-3,-3,-3$

A.P.

D) 2,5,9,14

Differences:

3,4,5

Not constant

Correct Options: A, C

9. For the G.P. $3,6,12,24,\ldots$

$a=3, r=2$

A) Common ratio = 2

Correct

B) 6th term = 96

$a_6 = 3(2)^5 = 96$

Correct

C) Sum of first 5 terms = 93

$S_5 = \frac{3(2^5 - 1)}{2 - 1} = 3(31) = 93$

Correct

D) 8th term = 384

$a_8 = 3(2)^7 = 384$

Correct

Correct Options: A, B, C, D

COMPREHENSION TYPE

Passage:

10. Sum to n terms of $5, 2, -1, \ldots$

Here,

$a = 5, d = 2 - 5 = -3$

Using

$S_n = \frac{n}{2}[2a + (n-1)d]$

$S_n = \frac{n}{2}[10 + (n-1)(-3)]$

$= \frac{n}{2}[10 - 3n + 3]$

$= \frac{n}{2}(13 - 3n)$

Correct Answer: B) $\frac{n}{2}(13 - 3n)$

11. Value of $a$in terms of S_n, d, n

Given,

$S_n = \frac{n}{2}[2a + (n-1)d]$

$\frac{2S_n}{n} = 2a + (n-1)d$

$2a = \frac{2S_n}{n} - (n-1)d$

$a = \frac{1}{2}\left(\frac{2S_n}{n} - (n-1)d\right)$

$a = \frac{2S_n - n(n-1)d}{2n}$

$a = \frac{2S_n - n^2 d + nd}{2n}$

Correct Answer: A) $\frac{2S_n - n^2 d + nd}{2n}$

12. Find d

Given:

$S_n = 400, n = 16, a = 5$

Using

$S_n = \frac{n}{2}[2a + (n-1)d]$

$400 = \frac{16}{2}[10 + 15d]$

$400 = 8(10 + 15d)$

$400 = 80 + 120d$

$320 = 120d$

$d = \frac{320}{120} = \frac{8}{3}$

Correct Answer: B) $\frac{8}{3}$

MATRIX MATCHING TYPE

13.

a) If $\frac{1}{3}, x_1, x_2, 9$are in G.P., find x_2

Let common ratio be r.

$\frac{1}{3} \cdot r^3 = 9$

$r^3 = 27$

$r = 3$

$x_2 = \frac{1}{3} \cdot r^2 = \frac{1}{3} \times 9 = 3$

So, $a \to R$

a → R

b) Third term of a G.P. is 4, find product of five terms

Five terms in G.P. can be written as:

$\frac{4}{r^2}, \frac{4}{r}, 4, 4r, 4r^2$

Product:

$\frac{4}{r^2} \times \frac{4}{r} \times 4 \times 4r \times 4r^2 = 4^5 = 1024$

So, $b \to P$

b → P

c) $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots + \infty$

Infinite G.P. with

$a = \frac{1}{3}, r = \frac{1}{3}$

$S_\infty = \frac{a}{1-r}$

$= \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$

So, $c \to S$

c → S

d) Sum of $1 + 3 + 9 + 27 + \cdots = 364$

Here,

$a = 1, r = 3$

$S_n = \frac{3^n - 1}{3 - 1}$

$364 = \frac{3^n - 1}{2}$

$728 = 3^n - 1$

$3^n = 729 = 3^6$

$n = 6$

So, $d \to Q$

d → Q

INTEGER TYPE

14. Find the first term of the A.P.

Given: $a_{10} = 47, d = 4$

Using

$a_n = a + (n-1)d$

$47 = a + 9(4)$

$47 = a + 36$

$a = 11$

Answer: $\boxed{11}$

15. Find the 6th term of the G.P.

Given: $a = 3, r = -2$

Using

$a_n = ar^{n-1}$

$a_6 = 3(-2)^5$

$= 3(-32) = -96$

Answer: $\boxed{-96}$

Arithmetic Geometric progression (A.G.P):

LEVEL-I

1. Find the n^{th}term

Series:

$1, \frac{4}{5}, \frac{7}{5^2}, \frac{10}{5^3}, \ldots$

Numerators:

$1,4,7,10, \ldots$

This is an A.P. with

$a = 1,\ d = 3$

n^{th}numerator:

$1 + (n-1)3 = 3n - 2$

Denominator:

5^{n-1}

Hence,

$T_n = \frac{3n-2}{5^{n-1}} = 5^{1-n}(3n-2)$

Correct Answer: C) $\mathbf{5^{1-n}(3n-2)}$

2. Find the 4th term

Given:

$T_n = \frac{n}{3^{n-1}}$

For $n = 4$,

$T_4 = \frac{4}{3^3} = \frac{4}{27}$

Correct Answer: D) $\mathbf{\frac{4}{27}}$

3. Find x

Series:

$1 + 4x + 7x^2 + 10x^3 + \cdots$

General term:

$(3n-2)x^{n-1}$

Use:

$\sum_{n=1}^{\infty}(3n-2)\,x^{n-1}$

Known results:

$$\sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x}$$

$$\sum_{n=1}^{\infty} n\,x^{n-1} = \frac{1}{(1-x)^2}$$

Therefore,

$$S = \frac{3}{(1-x)^2} - \frac{2}{1-x}$$

Given:

$$\frac{3}{(1-x)^2} - \frac{2}{1-x} = \frac{35}{16}$$

Try options.

For $x = \frac{1}{4}$,

$$1 - x = \frac{3}{4}$$

$$S = \frac{3}{(3/4)^2} - \frac{2}{3/4} = \frac{3}{9/16} - \frac{8}{3} = \frac{16}{3} - \frac{8}{3} = \frac{8}{3}$$

Not correct.

For $x = \frac{1}{2}$,

$$S = \frac{3}{(1/2)^2} - \frac{2}{1/2} = 12 - 4 = 8$$

Not correct.

For $x = \frac{1}{5}$,

$$S = \frac{3}{(4/5)^2} - \frac{2}{4/5} = \frac{75}{16} - \frac{5}{2} = \frac{35}{16}$$

Correct Answer: A) $\frac{1}{5}$

4. Sum to infinity of

$$1 + \frac{2}{3} + \frac{6}{3^2} + \frac{10}{3^3} + \frac{14}{3^4} + \cdots$$

General term:

$$T_n = \frac{4n-6}{3^{\,n-1}}, n \geq 2$$

Write series as:

$$1 + \sum_{n=2}^{\infty} \frac{4n-6}{3^{n-1}}$$

Let $r = \frac{1}{3}$

Using formulas:

$$\sum_{n=1}^{\infty} n\,r^{n-1} = \frac{1}{(1-r)^2}$$

$$\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r}$$

$$S = 1 + 4\sum_{n=2}^{\infty} n\,r^{n-1} - 6\sum_{n=2}^{\infty} r^{n-1}$$

Now, $\sum_{n=2}^{\infty} n\,r^{n-1} = \frac{1}{(1-r)^2} - 1 = \frac{9}{4} - 1 = \frac{5}{4}$

and

$$\sum_{n=2}^{\infty} r^{n-1} = \frac{1}{1-r} - 1 = \frac{3}{2} - 1 = \frac{1}{2}$$

Hence,

$$S = 1 + 4\left(\frac{5}{4}\right) - 6\left(\frac{1}{2}\right)$$

$$= 1 + 5 - 3 = 3$$

Correct Answer: C) 3

5. Find the 100th term

Sequence: $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \ldots$

Denominators are odd numbers:

$1,3,5,7, \ldots$

n^{th}odd number:

$2n - 1$

Hence,

$$a_n = \frac{1}{2n-1}$$

For $n = 100$,

$$a_{100} = \frac{1}{199}$$

Correct Answer: C) $\frac{1}{199}$

6. If $1, x, 3$are in H.P., find x

In H.P., reciprocals are in A.P.

So, $1,\ \frac{1}{x},\ \frac{1}{3}$

are in A.P.

Hence,

$$2\left(\frac{1}{x}\right) = 1 + \frac{1}{3}$$

$$\frac{2}{x} = \frac{4}{3}$$

$$x = \frac{2 \times 3}{4} = \frac{3}{2}$$

Correct Answer: D) $\frac{3}{2}$

7. Find the 4th term of the H.P.

Given:

$$T_{12} = \frac{1}{5}, T_{19} = \frac{3}{22}$$

Reciprocals form an A.P.

So corresponding A.P. terms are:

$5,\ \frac{22}{3}$

Let A.P. have first term $a$and common difference d.

$$a + 11d = 5$$

$$a + 18d = \frac{22}{3}$$

Subtract:

$7d = \frac{22}{3} - 5 = \frac{7}{3}$

$d = \frac{1}{3}$

$a + 11\left(\frac{1}{3}\right) = 5$

$a = \frac{4}{3}$

4th A.P. term:

$a + 3d = \frac{4}{3} + 1 = \frac{7}{3}$

Hence 4th H.P. term:

$= \frac{1}{7/3} = \frac{3}{7}$

Correct Answer: A) $\frac{3}{7}$

8. Harmonic mean of $\frac{1}{3}$ and $\frac{1}{7}$

Harmonic mean:

$H = \frac{2ab}{a+b}$

$= \frac{2\left(\frac{1}{3}\right)\left(\frac{1}{7}\right)}{\frac{1}{3}+\frac{1}{7}}$

$= \frac{2/21}{10/21} = \frac{1}{5}$

Correct Answer: B) $\frac{1}{5}$

9. If H is harmonic mean between P and Q

$H = \frac{2PQ}{P+Q}$

Then,

$\frac{H}{P} + \frac{H}{Q} = \frac{2Q}{P+Q} + \frac{2P}{P+Q}$

$= \frac{2(P+Q)}{P+Q} = 2$

Correct Answer: B) 2

10. Harmonic mean of the roots of

$\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$

Let roots be α, β.

For quadratic $ax^2 + bx + c = 0$,

$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

Here,

$a = \sqrt{3}, b = 10, c = 7\sqrt{3}$

$\alpha + \beta = -\frac{10}{\sqrt{3}}$

$\alpha\beta = \frac{7\sqrt{3}}{\sqrt{3}} = 7$

Harmonic mean:

$H = \frac{2\alpha\beta}{\alpha+\beta}$

$= \frac{2(7)}{-10/\sqrt{3}}$

$= \frac{14\sqrt{3}}{-10} = -\frac{7\sqrt{3}}{5}$

Correct Answer: B) $-\frac{7\sqrt{3}}{5}$

11. HM of roots of $x^3 - 6x^2 + 11x - 6 = 0$

Let roots be α, β, γ.

For three numbers,

$HM = \frac{3}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}}$

Now, $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} = \frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma}$

Using coefficients:

$\alpha\beta + \beta\gamma + \gamma\alpha = 11$

$\alpha\beta\gamma = 6$

So,

$HM = \frac{3}{11/6} = \frac{18}{11}$

Correct Answer: D) $\frac{18}{11}$

12. If $a, b, c$are in HP, then $\frac{a}{b+c}, \frac{b}{c+a}$and $\frac{c}{a+b}$are in?

Step-by-Step Solution:

Reciprocal Property: Since a, b, c are in HP, their reciprocals $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$are in AP.

Multiply each term by $(a+b+c)$. The sequence remains in AP:

$\frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ are in AP

Simplify the Fractions:

$\left(1+\frac{b+c}{a}\right), \left(1+\frac{a+c}{b}\right), \left(1+\frac{a+b}{c}\right)$ are in AP

Subtract 1 from each term:

$\frac{b+c}{a}, \frac{a+c}{b}, \frac{a+b}{c}$ are in AP

Take Reciprocals:

The reciprocals of these terms are:

$\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$

Hence they are in HP.

Correct Option: C) HP

13. If $b+c, c+a, a+b$ are in HP, then $a^{-2}, b^{-2}, c^{-2}$are in?

Step-by-Step Solution:

Reciprocal Property: Since $(b+c), (c+a), (a+b)$are in HP, their reciprocals are in AP:

$\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in AP

Arithmetic Mean Property:

In an AP, the middle term is the average of the other two:

$$\frac{2}{c+a}=\frac{1}{b+c}+\frac{1}{a+b}$$

Solve the Equation:

$$\frac{2}{c+a}=\frac{(a+b)+(b+c)}{(b+c)(a+b)}$$

$2(b+c)(a+b)=(c+a)(a+2b+c)$

Expand both sides:

$2(ab+b^2+ac+bc)=ac+2bc+c^2+a^2+2ab+ac$

$2ab+2b^2+2ac+2bc=a^2+2ab+2ac+2bc+c^2$

Simplify:

Cancel common terms:

$2b^2=a^2+c^2$

Thus,

a^2, b^2, c^2 are in AP

Final Transformation:

Taking reciprocals,

$\frac{1}{a^2}$, $\frac{1}{b^2}$, $\frac{1}{c^2}$ are in HP

Since $\frac{1}{x^2}=x^{-2}$,

a^{-2}, b^{-2}, c^{-2} are in HP

Correct Option: C) HP

14. If

$$\frac{1}{a(b+c)},\ \frac{1}{b(c+a)},\ \frac{1}{c(a+b)}$$

are in H.P.

Then their reciprocals are in A.P.:

$a(b+c)$, $b(c+a)$, $c(a+b)$

So,

$2b(c+a)=a(b+c)+c(a+b)$

$2ab+2bc=ab+ac+ac+bc$

$ab+bc=2ac$

$$\frac{2}{c}=\frac{1}{a}+\frac{1}{b}$$

Thus,a,b,c

are in H.P.

Correct Answer: C) HP

15. If

$a^x=b^y=c^z$

and $a,b,c$are in G.P.

Since $a,b,c$are in G.P.,

$b^2=ac$

Let

$a^x=b^y=c^z=k$

Then,

$a=k^{1/x},b=k^{1/y},c=k^{1/z}$

Substitute in $b^2=ac$:

$\left(k^{1/y}\right)^2=k^{1/x}k^{1/z}$

$k^{2/y}=k^{1/x+1/z}$

$\frac{2}{y}=\frac{1}{x}+\frac{1}{z}$

Hence,x, y, z

are in H.P.

Correct Answer: C) HP

16. If $x>1$, $y>1$, $z>1$ are in GP, then $\frac{1}{1+\log x}$, $\frac{1}{1+\log y}$, $\frac{1}{1+\log z}$are in?

Step-by-Step Solution:

Since $x, y,$zare in GP,

$y^2=xz$

Taking logarithm on both sides:

$2\log y=\log x+\log z$

Thus,

$\log x$, $\log y$, $\log z$ are in AP

Add 1 to each term:

$1+\log x$, $1+\log y$, $1+\log z$ are in AP

Taking reciprocals, we get:

$\frac{1}{1+\log x}$, $\frac{1}{1+\log y}$, $\frac{1}{1+\log z}$ are in HP

Correct Option: B) HP

17. The AM, HM, and GM between two numbers are $\frac{144}{15}$, 15,and 12but not necessarily in this order. Then HM, GM and AM respectively are?

Step-by-Step Solution:

For any two positive numbers:

$(\mathrm{GM})^2=(\mathrm{AM})(\mathrm{HM})$

Given numbers:

$\frac{144}{15}$, 15, 12

Check which value satisfies the relation.

Take:

$\mathrm{AM}=15, \mathrm{HM}=\frac{144}{15}, \mathrm{GM}=12$

Verify:

$(\mathrm{GM})^2=12^2=144$

$(\mathrm{AM})(\mathrm{HM})=15\times\frac{144}{15}=144$

Both are equal, so the assignment is correct.

Hence,

$\mathrm{HM}=\frac{144}{15}, \mathrm{GM}=12, \mathrm{AM}=15$

Correct Option: A) $\frac{144}{15}$, 12, 15

18. The AM and HM between two numbers are 27and 12respectively, then their GM is?

Step-by-Step Solution: For two positive numbers,

$(\mathrm{GM})^2=(\mathrm{AM})(\mathrm{HM})$

Substitute the given values:

$(\mathrm{GM})^2=27\times12$

$(\mathrm{GM})^2=324$

$\mathrm{GM}=\sqrt{324}=18$

Correct Option: A) 18

19. If a, b, c, d are in HP then

Step-by-Step Solution:

Since $a, b, c, d$are in HP, their reciprocals are in AP:

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d}$ are in AP

Hence,

$\frac{1}{a}+\frac{1}{d}=\frac{1}{b}+\frac{1}{c}$

Using the property of positive numbers in HP,

$a+d \geq b+c$

Correct Option: C) $\boldsymbol{a+d \geq b+c}$

20. If a be one AM, and $G_1$and G_2be two GMs between $b$and c, then $G_1^3+G_2^3-2abc=$

Step-by-Step Solution:

Since $a$is one AM between $b$and c,

$a=\frac{b+c}{2}$

Now $G_1, G_2$are two GMs between $b$and c, so

$b,\ G_1,\ G_2,\ c$ are in GP

Let common ratio be r.

Then,

$G_1=br, G_2=br^2, c=br^3$

Now,

$G_1^3+G_2^3=b^3r^3+b^3r^6$

$=b^3r^3(1+r^3)$

Since $c=br^3$,

$=b^2c+bc^2=bc(b+c)$

But,

$a=\frac{b+c}{2} \Rightarrow b+c=2a$

Therefore, $G_1^3+G_2^3=bc(2a)=2abc$

Hence, $G_1^3+G_2^3-2abc=0$

Correct Option: D) 0

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. The n^{th}term of the series

$5+\frac{7}{3}+\frac{9}{3^2}+\frac{11}{3^3}+\cdots$

is

Step-by-Step Solution:

Numerators form an AP:

$5,7,9,11,\ldots$

Here,$a=5, d=2$

So the numerator of the n^{th}term is

$5+(n-1)2=2n+3$

Denominators are:

$1,3,3^2,3^3,\ldots=3^{n-1}$

Hence, $T_n=\frac{2n+3}{3^{n-1}}$

Correct Option: B) $\boldsymbol{\frac{2n+3}{3^{n-1}}}$

2. The sum of the infinite series
$1 - 3x + 5x^2 - 7x^3 + \cdots , | x | < 1$
is
The series is: $S = 1 - 3x + 5x^2 - 7x^3 + \cdots$
Write it as:
$S = (1 - x + x^2 - x^3 + \cdots) - 2(x - 2x^2 + 3x^3 - \cdots)$
Using standard results,
$1 - x + x^2 - x^3 + \cdots = \frac{1}{1+x}$
and
$1 - 2x + 3x^2 - 4x^3 + \cdots = \frac{1}{(1+x)^2}$
After simplification,
$S = \frac{1-x}{(1+x)^2}$
Correct Option: D) $\mathbf{\frac{1-x}{(1+x)^2}}$

3. Evaluate $1 - \frac{4}{2} + \frac{7}{2^2} - \frac{10}{2^3} + \cdots$
Write the series as:
$$1 - \frac{4}{2} + \frac{7}{2^2} - \frac{10}{2^3} + \cdots = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (3n+1)}{2^n}$$
Let $x = -\frac{1}{2}$
Using the standard result:
$$\sum_{n=1}^{\infty} (3n+1)\, x^n = \frac{4x - x^2}{(1-x)^2}$$
Substitute $x = -\frac{1}{2}$:
$$S = 1 + \frac{4(-1/2) - (-1/2)^2}{(1 + 1/2)^2}$$
$$= 1 + \frac{-2 - \frac{1}{4}}{\frac{9}{4}}$$
$$= 1 - 1 = 0$$
Correct Option: A) 0

4. If a, b, c, d are in H.P then the value of
$\frac{a^{-2} - d^{-2}}{b^{-2} - c^{-2}}$
is
Since a, b, c, d are in HP,
$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d}$ are in AP
Let $\frac{1}{a} = A - 3k, \frac{1}{b} = A - k, \frac{1}{c} = A + k, \frac{1}{d} = A + 3k$
Now,
$$a^{-2} - d^{-2} = \left(\frac{1}{a}\right)^2 - \left(\frac{1}{d}\right)^2$$
$$= (A - 3k)^2 - (A + 3k)^2$$
Using $x^2 - y^2 = (x - y)(x + y)$,

$= (-6k)(2A) = -12Ak$

Similarly,

$b^{-2} - c^{-2} = (A-k)^2 - (A+k)^2$

$= (-2k)(2A) = -4Ak$

Therefore,

$\frac{a^{-2} - d^{-2}}{b^{-2} - c^{-2}} = \frac{-12Ak}{-4Ak} = 3$

Correct Option: A) 3

5. If $a^2, b^2, c^2$are in AP then $b + c,\ c + a,\ a + b$are in

Step-by-Step Solution:

Since $a^2, b^2, c^2$are in AP,

$2b^2 = a^2 + c^2$

Now consider:

$(b + c),\ (c + a),\ (a + b)$

To check HP, verify whether their reciprocals are in AP.

So we check:

$\frac{2}{a+c} = \frac{1}{b+c} + \frac{1}{a+b}$

RHS:$\frac{(a+b)+(b+c)}{(b+c)(a+b)} = \frac{a+2b+c}{(b+c)(a+b)}$

Cross-multiplying gives: $2(a + b)(b + c) = (a + c)(a + 2b + c)$

Expanding,

$2ab + 2b^2 + 2ac + 2bc = a^2 + 2ab + 2ac + 2bc + c^2$

$2b^2 = a^2 + c^2$

which is true.

Hence,$\frac{1}{b+c},\ \frac{1}{c+a},\ \frac{1}{a+b}$ are in AP

Therefore,$b + c,\ c + a,\ a + b$ are in HP

Correct Option: C) HP

6. The sum of n terms of the series $1 + 2 + 5 + 12 + 25 + 46 + \cdots$ is

Observe the terms: 1,2,5,12,25, …

The successive differences are: 1,3,7,13, …

Second differences: 2,4,6, …

Third difference is constant, so the term is cubic in n.

Assume $T_n = an^3 + bn^2 + cn + d$

Using the first four terms:

$T_1 = 1, T_2 = 2, T_3 = 5, T_4 = 12$

Solving gives:

$T_n = \frac{n^3 - 3n^2 + 8n}{3}$

Now,

$S_n = \sum_{n=1}^{n} \frac{n^3 - 3n^2 + 8n}{3}$

Using standard summation formulas and simplifying:

$S_n = \frac{1}{12} n(n+1)(n^2 - 3n + 8)$

Correct Option: C) $\frac{1}{12} n(n+1)(n^2 - 3n + 8)$

7. If $a, b, c, d$are in HP then $ab + bc + cd =$

Step-by-Step Solution:

Since $a, b, c, d$are in HP, their reciprocals are in AP:

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d}$ are in AP

Hence,

$\frac{1}{a} + \frac{1}{d} = \frac{1}{b} + \frac{1}{c}$

Multiply throughout by $abcd$:

$bc + ab = ac + ad$

$ab + bc = a(c + d)$

Also, using AP property,

$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

Multiplying by abc:

$2ac = ab + bc$

Similarly,

$\frac{2}{c} = \frac{1}{b} + \frac{1}{d}$

Multiplying by bcd:

$2bd = bc + cd$

Adding the two equations:

$ab + 2bc + cd = 2ac + 2bd$

Using the HP relations and simplifying:

$ab + bc + cd = 3ad$

Correct Option: B) 3ad

8. The $\mathbf{7}^{\text{th}}$term of an HP is $\frac{1}{10}$and $\mathbf{12}^{\text{th}}$term is $\frac{1}{25}$. Find the $\mathbf{20}^{\text{th}}$term.

Step-by-Step Solution:

In an HP, reciprocals form an AP.

So the corresponding AP has:

$T_7 = 10, T_{12} = 25$

Let first term be $a$and common difference be d.

Then,

$a + 6d = 10$

$a + 11d = 25$

Subtract:

$5d = 15 \Rightarrow d = 3$

Substitute back:

$a + 18 = 10 \Rightarrow a = -8$

Now the 20^{th}term of AP:

$a + 19d = -8 + 57 = 49$

Therefore the 20^{th}term of HP is:

$\frac{1}{49}$

Correct Option: D) $\frac{1}{49}$

9. If $\boldsymbol{p}^{\text{th}}$term of a HP be $\boldsymbol{qr}$ and the $\boldsymbol{q}^{\text{th}}$term be $\boldsymbol{rp}$, then the $\boldsymbol{r}^{\text{th}}$term is

Step-by-Step Solution:

In an HP, reciprocals form an AP.

Let the corresponding AP have first term $\boldsymbol{a}$and common difference $\boldsymbol{d}$.

Given:

$T_p = qr \Rightarrow \frac{1}{qr} = a + (p-1)d$

$T_q = rp \Rightarrow \frac{1}{rp} = a + (q-1)d$

Subtracting,

$\frac{1}{qr} - \frac{1}{rp} = (p-q)d$

$\frac{p-q}{pqr} = (p-q)d$

$d = \frac{1}{pqr}$

Now,

$a + (p-1)d = \frac{1}{qr}$

$a = \frac{1}{qr} - \frac{p-1}{pqr} = \frac{1}{pqr}$

Thus the AP becomes:

$T_n = \frac{1}{pqr} + \frac{n-1}{pqr} = \frac{n}{pqr}$

So the reciprocal of the $\boldsymbol{r}^{\text{th}}$term is

$\frac{r}{pqr} = \frac{1}{pq}$

Hence the r^{th}term of the HP is

pq

Correct Option: B) $\boldsymbol{pq}$

10. If

$\frac{1}{a} + \frac{1}{c} + \frac{1}{a-b} + \frac{1}{c-b} = 0$

and $b \neq a + c$, then $a, b, c$are in

Step-by-Step Solution:

Given,

$\frac{1}{a} + \frac{1}{c} + \frac{1}{a-b} + \frac{1}{c-b} = 0$

Group the terms:

$\left(\frac{1}{a} + \frac{1}{a-b}\right) + \left(\frac{1}{c} + \frac{1}{c-b}\right) = 0$

Simplify:

$\frac{2a-b}{a(a-b)} + \frac{2c-b}{c(c-b)} = 0$

Taking LCM and simplifying:

$(2a-b)c(c-b) + (2c-b)a(a-b) = 0$

Expand:

$2ac^2 - bc^2 - 2abc + b^2c + 2a^2c - a^2b - 2abc + ab^2 = 0$

Rearranging and factorising:

$(a + c - b)(2ac - ab - bc) = 0$

Given $b \neq a + c$,

$2ac - ab - bc = 0$

$2ac = b(a + c)$

Divide by abc:

$\frac{2}{b}=\frac{1}{a}+\frac{1}{c}$

Hence,$\frac{1}{a}$, $\frac{1}{b}$, $\frac{1}{c}$ are in AP

Therefore,a,b,c are in HP

Correct Option: B) HP

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. For the A.G.P. 1,4,12,32, …

In an A.G.P., terms are obtained by multiplying corresponding terms of an A.P. and a G.P.

Given sequence:

1,4,12,32, …

Take A.P.: 1,2,3,4, …

Take G.P.: 1,2,4,8, …

Multiplying: $1\cdot 1=1,2\cdot 2=4,3\cdot 4=12,4\cdot 8=32$

Hence:

Corresponding A.P. is 1,2,3,4, …

Corresponding G.P. is 1,2,4,8, …

Common ratio $r=2$

Common difference $d=1$

Correct Answers: A, B, C, D

12. Which of the following sequences are H.P.?

A sequence is H.P. if reciprocals are in A.P.

A) $1,\frac{1}{2},\frac{1}{3},\frac{1}{4},\ldots$

Reciprocals: 1,2,3,4, …

which is an A.P.

H.P.

B) $\frac{1}{2},\frac{1}{4},\frac{1}{6},\frac{1}{8},\ldots$

Reciprocals: 2,4,6,8, …

which is an A.P.

H.P.

C) 2,4,6,8, …

Reciprocals: $\frac{1}{2},\frac{1}{4},\frac{1}{6},\frac{1}{8}$

not an A.P.

Not H.P.

D) $3,\frac{3}{2},1,\frac{3}{4},\ldots$

Reciprocals:

$\frac{1}{3},\frac{2}{3},1,\frac{4}{3},\ldots$

difference is constant:

$\frac{1}{3}$

H.P.

Correct Answers: A, B, D

13. The n^{th}term of AGP series

$\frac{1}{4}+\frac{3}{16}+\frac{5}{64}+\frac{7}{256}+\cdots$

Numerators:

1,3,5,7, …
A.P. with $a = 1,\ d = 2$
Denominators:
4,16,64,256, …
G.P. with $a = 4,\ r = 4$
So,$T_n = \frac{2n-1}{4^n}$
Now,
$\frac{2n-1}{4^n} = \frac{2(2^{n-1})-1}{2^{2n}} = \frac{2^n-1}{2^{2n}}$ and $\frac{2^n-1}{4^n}$
Correct Answers: B, D

14. The second term of an H.P. is $\frac{1}{6}$and sixth term is $-\frac{1}{6}$. Find the 20th term.
Let reciprocals be in A.P.
So corresponding A.P. terms:
$6,\ -6$
These are 2nd and 6th terms.
Using:
$a + d = 6$
$a + 5d = -6$
Subtract:$4d = -12 \Rightarrow d = -3$
Then:$a = 9$
20th term of A.P.: $a_{20} = 9 + 19(-3) = 9 - 57 = -48$
Hence 20th term of H.P.:
$\frac{1}{-48} = -\frac{1}{48}$
$-\frac{1}{48} = -\frac{1}{2^4 \times 3},\ -\frac{1}{48} = \frac{-1}{4^2 \times 3}$
Correct Answers: A,B

COMPREHENSION TYPE

Passage – I:

Given:
Arithmetic–Geometric series:$A = a + (a + d)r + (a + 2d)r^2 + \cdots$
Given:
n^{th}term of A.P.:
$a_n = 2n + 3$
First term of G.P. = 8
Common ratio is $\frac{1}{16}$of first term:
$r = \frac{8}{16} = \frac{1}{2}$
15. The arithmetic progression represented by A
Using:
$a_n = 2n + 3$
$a_1 = 5, a_2 = 7, a_3 = 9$
So A.P. is: 5,7,9, …
Correct Answer: C

16. The geometric progression represented by A
First term:8
Common ratio:$\frac{1}{2}$

Thus G.P.: 8,4,2,1, …

Correct Answer: A

17. The Arithmetic Geometric series is

A.G.P. terms are products of corresponding A.P. and G.P. terms.

A.P.:5,7,9, …

G.P.: 8,4,2, …

Therefore: $5 \cdot 8 + 7 \cdot 4 + 9 \cdot 2 + \cdots$

Hence: $5.8 + 7.4 + 9.2 + \cdots$

Correct Answer: B

MATRIX MATCHING TYPE

18. Given AGP series:

$4 \cdot 5 + 7 \cdot 5^3 + 10 \cdot 5^5 + \cdots$

AP part: 4,7,10, …

GP part: $5^1, 5^3, 5^5, \ldots$

a) Common ratio of GP part

$r = \frac{5^3}{5^1} = 5^2 = 25$

$\boxed{a \to S}$

b) Common difference of AP part

$d = 7 - 4 = 3$

$\boxed{b \to R}$

c) $t_{18}$of GP part

GP: $5^1, 5^3, 5^5, \ldots$

Exponent pattern: $1,3,5, \cdots = (2n-1)$

$t_{18} = 5^{2(18)-1} = 5^{35}$

$\boxed{c \to P}$

d) t_{18} in AP part

$a_n = a + (n-1)d$

$a_{18} = 4 + 17(3) = 4 + 51 = 55$

$\boxed{d \to Q}$

INTEGER TYPE

19. In H.P., reciprocals are in A.P.

Given H.P.:

$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$

Reciprocals are:

1,2,3,4, …

So the nth term of H.P. is:

$a_n = \frac{1}{n}$

Hence, the 7th term is:

$\boxed{\frac{1}{7}}$

20. Given A.G.P.:

$1 \cdot 2 + 3 \cdot 2^2 + 5 \cdot 2^3 + \cdots$

AP part: 1,3,5, …

So $a = 1$, $d = 2$

The nth AP term is:

$a_n = 1 + (n-1)2 = 2n - 1$
GP part: $2, 2^2, 2^3, \ldots$
Hence nth term of A.G.P. is:
$T_n = (2n-1)2^n$
For $n = 8$:
$T_8 = (16-1)2^8$
$= 15 \times 256$
$= \boxed{3840}$

STUDENT PRACTISE CORNER

1. If $2, x, 6$are in H.P then $x =$?
Since numbers in H.P have reciprocals in A.P:
$\frac{1}{2}, \frac{1}{x}, \frac{1}{6}$
For A.P:
$2\left(\frac{1}{x}\right) = \frac{1}{2} + \frac{1}{6}$
$\frac{2}{x} = \frac{4}{6} = \frac{2}{3}$
$x = 3$
Correct Answer: A) 3

2. If the 10th term of an H.P is 1/6and the 18th term is 1/10, find the 2nd term.
Reciprocals form an A.P.
So corresponding A.P terms are:
6, 10
Difference:
$d = \frac{10-6}{18-10} = \frac{4}{8} = \frac{1}{2}$
Let first term be a.
$a + 9d = 6$
$a + \frac{9}{2} = 6$
$a = \frac{3}{2}$
2nd term of A.P:
$a + d = \frac{3}{2} + \frac{1}{2} = 2$
Hence 2nd term of H.P:
$\frac{1}{2}$
Correct Answer: A) ½

3. The 50th term of sequence $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \ldots$
Denominator pattern:
$1,3,5,7, \cdots = 2n - 1$
So, $a_n = \frac{1}{2n-1}$
For $n = 50$:
$a_{50} = \frac{1}{2(50)-1} = \frac{1}{99}$
Correct Answer: B) 1/99

4. If A.M = 10 and G.M = 8, find H.M.

Relation:

$(\text{G.M})^2 = (\text{A.M})(\text{H.M})$

$8^2 = 10 \times \text{H.M}$

$64 = 10 \times \text{H.M}$

$\text{H.M} = \frac{64}{10} = 6.4$

Correct Answer: A) 6.4

5. Find the harmonic mean between 4 and 12.

Formula:

$H = \frac{2ab}{a+b}$

$H = \frac{2(4)(12)}{4+12} = \frac{96}{16} = 6$

Correct Answer: B) 6

6. If G.M = 24 and A.M = 25, find H.M.

Using:

$(\text{G.M})^2 = (\text{A.M})(\text{H.M})$

$24^2 = 25 \times \text{H.M}$

$576 = 25 \times \text{H.M}$

$\text{H.M} = \frac{576}{25}$

Correct Answer: A) $\frac{576}{25}$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

7. If a, b, c are in H.P and $a = 3,\ c = 6$

For H.P:

$b = \frac{2ac}{a+c}$

$b = \frac{2(3)(6)}{3+6} = \frac{36}{9} = 4$

Also, reciprocals of H.P are in A.P:

$\frac{1}{3}, \frac{1}{4}, \frac{1}{6}$

Hence all statements are true.

Correct Answers: A, B, C, D

8. For $2, x, 18$ in H.P, x is the harmonic mean of 2and 18.

$x = \frac{2ab}{a+b}$

$x = \frac{2(2)(18)}{2+18} = \frac{72}{20} = \frac{18}{5}$

Also,

$\frac{18}{5} = \frac{72}{20} \neq \frac{36}{20}$

Correct Answers: A, C

9. If a, b, c are in H.P and $a = 12,\ b = 8$

Since reciprocals are in A.P:

$\frac{1}{12}, \frac{1}{8}, \frac{1}{c}$

$2\left(\frac{1}{8}\right) = \frac{1}{12} + \frac{1}{c}$

$\frac{1}{4} = \frac{1}{12} + \frac{1}{c}$

$\frac{1}{c} = \frac{1}{4} - \frac{1}{12} = \frac{2}{12} = \frac{1}{6}$

$c = 6$

Also, b is harmonic mean of a, c.

Check option D:

$2b = 16, a + c = 12 + 6 = 18$

Not equal.

Correct Answers: A, B, C

COMPREHENSION TYPE

Passage:

If three numbers a, b, and $c$are in Harmonic Progression (HP), their reciprocals $\frac{1}{a}$, $\frac{1}{b}$, and $\frac{1}{c}$are in Arithmetic Progression (AP).

The mathematical condition for three numbers to be in HP is:

$b = \frac{2ac}{a + c}$

Given the terms in the problem:

$a = 2 + x$

$b = 3 + x$

$c = 5 + x$

10. The value of x is

Substitute the given terms into the HP condition formula:

$3 + x = \frac{2(2 + x)(5 + x)}{(2 + x) + (5 + x)}$

First, simplify the numerator and denominator on the right side:

$3 + x = \frac{2(10 + 7x + x^2)}{7 + 2x}$

Cross-multiply to remove the fraction:

$(3 + x)(7 + 2x) = 2(10 + 7x + x^2)$

Expand both sides:

$21 + 6x + 7x + 2x^2 = 20 + 14x + 2x^2$

$21 + 13x + 2x^2 = 20 + 14x + 2x^2$

Subtract $2x^2$ from both sides, which cancels it out:

$21 + 13x = 20 + 14x$

Rearrange to solve for x:

$21 - 20 = 14x - 13x$

$1 = x$

Correct Option: D) 1

11. The terms in HP are

Now that we know $x = 1$, substitute this value back into the original expressions to find the actual terms:

First term: $2 + x = 2 + 1 = 3$

Second term: $3 + x = 3 + 1 = 4$

Third term: $5 + x = 5 + 1 = 6$

Correct Option: C) 3, 4, 6

12. The 12^{th}term in HP is

To find the n^{th}term of a Harmonic Progression, it is easiest to find the n^{th}term of its corresponding Arithmetic Progression and then take the reciprocal.

The corresponding AP terms are the reciprocals of the HP terms we just found:

$\frac{1}{3}, \frac{1}{4}, \frac{1}{6}$

First term of AP (A):

$A = \frac{1}{3}$

Common difference of AP (d):

$d = \frac{1}{4} - \frac{1}{3}$

$d = \frac{3}{12} - \frac{4}{12}$

$d = -\frac{1}{12}$

Use the formula for the n^{th}term of an AP:

$T_n = A + (n-1)d$

Calculate the 12^{th}term of the AP:

$T_{12} = \frac{1}{3} + (12-1)\left(\frac{1}{12}\right)$

$T_{12} = \frac{1}{3} - \frac{11}{12}$

Convert $\frac{1}{3}$into denominator 12:

$T_{12} = \frac{4}{12} - \frac{11}{12}$

$T_{12} = -\frac{7}{12}$

The 12^{th}term of the AP is $-\frac{7}{12}$.

Therefore, the 12^{th}term of the HP is its reciprocal:

$-\frac{12}{7}$

Correct Option: D) $\mathbf{-\frac{12}{7}}$

MATRIX MATCHING TYPE

13. Match Column I and Column II

a) The AM and HM between two numbers are 27 and 12 respectively, then their GM is

Using relation:

$(\text{G.M})^2 = (\text{A.M})(\text{H.M})$

$(\text{G.M})^2 = 27 \times 12 = 324$

G.M = 18

So,$a \to R$

b) If the AM exceeds GM by 2 and the GM exceeds HM by $\frac{8}{5}$

Let

$A - G = 2$

$G - H = \frac{8}{5}$

Using relation:

$G^2 = AH$

Checking numbers 4and 16:

$A = \frac{4+16}{2} = 10$

$G = \sqrt{4 \times 16} = 8$

$H = \frac{2(4)(16)}{4+16} = \frac{128}{20} = 6.4$

$A - G = 10 - 8 = 2$

$G - H = 8 - 6.4 = 1.6 = \frac{8}{5}$

So, $b \to P$

c) If $A.M = 10$and $G.M = 8$, then $H =$

Using:

$G^2 = AH$

$8^2 = 10H$

$64 = 10H$

$H = 6.4$

So, $c \to S$

d) Harmonic mean of 9 and 36

Formula:

$H = \frac{2ab}{a+b}$

$H = \frac{2(9)(36)}{9+36}$

$H = \frac{648}{45} = 14.4$

So, $d \to Q$

INTEGER TYPE

14. If $2 + x$, $3 + x$, $5 + x$ are in H.P, then find x

For H.P:

$3 + x = \frac{2(2+x)(5+x)}{(2+x)+(5+x)}$

$3 + x = \frac{2(10+7x+x^2)}{7+2x}$

Cross multiply:

$(3 + x)(7 + 2x) = 2(10 + 7x + x^2)$

$21 + 13x + 2x^2 = 20 + 14x + 2x^2$

$x = 1$

$\boxed{x = 1}$

15. If A.M = 13 and G.M = 12 between two numbers, find the H.M.

Using relation:

$(G.M)^2 = (A.M)(H.M)$

$12^2 = 13 \times H$

$144 = 13H$

$H = \frac{144}{13}$

$\boxed{\frac{144}{13}}$

7. Statistics and Probability

LEVEL-I

1. The number of times a particular item occurs in a class interval is called:

Answer: B) Frequency

2. The groups into which large data is condensed are called:

Answer: B) Classes

3. Mean of first 5 whole numbers

Whole numbers: 0, 1, 2, 3, 4

$$\text{Mean} = \frac{0+1+2+3+4}{5} = \frac{10}{5} = 2$$

Answer: D) 2

4. Range of relative humidity data

Highest value = 98.3

Lowest value = 89

Range = $98.3 - 89 = 9.3$

Answer: A) 9.3

5. Modal shirt size = size with highest frequency

Highest frequency = 36 corresponding to size 39.

Answer: A) 39

6. Class mark formula:

$$\text{Class Mark} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

For class mark 20, check options:

A) $\frac{12.5+17.5}{2} = 15$

B) $\frac{17.5+22.5}{2} = 20$

C) $\frac{18.5+21.5}{2} = 20$

D) $\frac{19.5+20.5}{2} = 20$

The standard class interval is:

Answer: B) 17.5 – 22.5

7. From the bar graph:

October = 4 students

November = 4 students

Difference = $4 - 4 = 0$

Check options for same difference:

February and January = $4 - 3 = 1$

May and July = $5 - 2 = 3$

March and April = $2 - 2 = 0$

August and September = $6 - 3 = 3$

Answer: C) March and April

8. First ten prime numbers:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

Median of 10 numbers = average of 5th and 6th terms

$$\text{Median} = \frac{11+13}{2} = 12$$

Answer: C) 12

9. Average marks of whole school:
Boys = 90, mean = 45%
Girls = 30, mean = 70%
$\bar{x} = \frac{\sum x}{n}$
Average $= \frac{90 \times 45 + 30 \times 70}{120}$
$= \frac{4050 + 2100}{120} = \frac{6150}{120} = 51.25\%$
Answer: B) 51.25 %

10. A graph drawn with the midpoints of the top sides of histogram rectangles is called:
Answer: C) Frequency polygon

11. The value around which all observations tend to concentrate is called:
Answer: C) Measure of central tendency

12. In the class intervals 10–20 and 20–30, the number 20 is included in:
For exclusive class intervals, the upper limit is excluded.
So, 20 belongs to 20–30.
Answer: B) 20–30

13. Arithmetic mean of 30, 36, 39, 23 and 27
$\bar{x} = \frac{\sum x}{n}$
Mean $= \frac{30 + 36 + 39 + 23 + 27}{5} = \frac{155}{5} = 31$
Answer: C) 31

14. Observations:
$11, 13, 15, 19, p + 2, p + 4, 30, 35, 39, 46$
Number of observations = 10
Median = average of 5th and 6th terms
$\frac{(p + 2) + (p + 4)}{2} = 25$
$\frac{2p + 6}{2} = 25$
$p + 3 = 25$
$p = 22$
Answer: A) 22

15. Which is not a measure of central tendency?
Mean, Median, and Mode are measures of central tendency.
Range is a measure of dispersion.
Answer: B) Range

16. Mean of first 8 observations = 18
Sum of first $8 = 8 \times 18 = 144$
Mean of all 15 observations = 19
Total sum $= 15 \times 19 = 285$
Last 8 observations mean = 20
Sum of last $8 = 8 \times 20 = 160$
The 8th observation is common in both groups.
$144 + 160 - 285 = 19$
Answer: C) 19

17. Given class: 63–72 (72 included)
Class size = $72 - 63 + 1 = 10$
Smallest value = 14
Largest value = 112
Number of classes:
$\frac{112 - 14 + 1}{10} = \frac{99}{10} = 9.9$
So, total classes = 10
Answer: C) 10

18. Mid-point formula:
$m = \frac{l + u}{2}$
Here upper-class limit = $l$and midpoint = m
$2m$ = lower limit + l
lower limit = $2m - l$
Answer: B) 2m - l

19. Width of each class = 2.5
Number of classes = 9
Lower-class boundary of first class = 10.6
Upper-class boundary of highest class:
$10.6 + (9 \times 2.5)$
$= 10.6 + 22.5 = 33.1$
Answer: B) 33.1

20. Mid-value of class = 10
Width of class = 6
Let lower limit = l, upper limit = u
Mid-value = $\frac{l + u}{2}$
Also,
$u - l = 6$
Using midpoint:
$\frac{l + u}{2} = 10$
$l + u = 20$
Substitute $u = l + 6$:
$l + l + 6 = 20$
$2l = 14$
$l = 7$
Answer: B) 7

LEVEL – II

Single option correct type

1. Ogives are the graphical representation of:
Answer: C) Cumulative frequency

2. In a frequency polygon, frequencies are plotted against:
Answer: D) Mid values of the classes

3. Total wage of 150 labourers:
$150 \times 114 = 17100$
First shift total wage:
$60 \times 121.50 = 7290$

Second shift total wage:
$40 \times 107.75 = 4310$
Third shift total wage:
$17100 - (7290 + 4310) = 5500$
Number of labourers in third shift = 50
Mean wage $= \frac{5500}{50} = 110$
Answer: A) ₹110

4. If each observation is multiplied by k, the mean also gets multiplied by k.
$\bar{x}_{new} = k\bar{x}$
Answer: A) $k\bar{X}$

5. If every number is divided by 5, the mean is also divided by 5.
New mean $= \frac{25}{5} = 5$
Answer: A) 5

6. Check each option:
A) 2,2,2,4
Mean $= \frac{10}{4} = 2.5$, Median = 2, Mode = 2
B) 1,3,3,3,5
Mean $= \frac{15}{5} = 3$, Median = 3, Mode = 3
All are equal.
Answer: B) 1,3,3,3,5

7. Even numbers between 50 and 60:
52, 54, 56, 58
Mean $= \frac{52+54+56+58}{4} = 55$
Odd numbers between 50 and 60:
51, 53, 55, 57, 59
Mean $= \frac{51+53+55+57+59}{5} = 55$
Difference:
$55 - 55 = 0$
Answer: B) 0

8. Numbers: 2, 2, 4, 5, 12
Mean:
$\frac{2+2+4+5+12}{5} = \frac{25}{5} = 5$
Median = 4
Mode = 2
Thus,
Mean > Mode
Answer: B) Mean > Mode

9. Mean score of 10 innings = 60
Total runs in 10 innings:
$10 \times 60 = 600$
To make mean 62 in 11 innings:
$11 \times 62 = 682$
Runs needed in 11th inning: $682 - 600 = 82$
Answer: D) 82

10. Empirical relation:
Mode = 3 × Median − 2 × Mean
Answer: A) Mode = 3 Median – 2 Mean

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. A.M. of first $n$natural numbers:
$\text{A.M.} = \frac{n+1}{2}$

Correct Answer: C) $\frac{n+1}{2}$ B) $\frac{\frac{n(n+1)}{2}}{n}$

12. Mean weight of children
$\bar{x} = \frac{\sum fx}{\sum f}$

$\bar{x} = \frac{\sum fx}{\sum f}$
Calculate fx:

Weight (x)	**Frequency (f)**	**fx**
40	2	80
41	3	123
42	6	252
43	2	86
44	5	220
45	5	225

$\sum fx = 1065, \sum f = 25$
$\bar{x} = \frac{1067}{25} = 42.87$ kg (approx)
Closest option:
Answer: A) 42.68 kg C) 42

13. Given mean = 5.3
Table:

x	**f**
4	11
8	2
6	3
7	y

Using mean formula:
$\frac{4(11) + 8(2) + 6(3) + 7y}{11 + 2 + 3 + y} = 5.3$
$\frac{44 + 16 + 18 + 7y}{16 + y} = 5.3$
$\frac{78 + 7y}{16 + y} = 5.3$
$78 + 7y = 84.8 + 5.3y$
$1.7y = 6.8$

$y = 4$

Answer: A) 4 B) $\sqrt{16}$ C) $\sqrt[3]{64}$

14. Modal class = class with highest frequency

Highest frequency = 10

Modal class = 11–15

Mode formula:

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Here,

$l = 10.5$

$f_1 = 10$

$f_0 = 4$

$f_2 = 6$

$h = 5$

$$\text{Mode} = 10.5 + \frac{10 - 4}{20 - 4 - 6} \times 5$$

$$= 10.5 + \frac{6}{10} \times 5$$

$$= 10.5 + 3$$

$$= 13.5$$

Answer: A) 13.5 B) 13

COMPREHENSIVE TYPE

15. Total flights examined = sum of frequencies

$15 + 20 + 32 + 18 + 15 = 100$

Answer: A) 100

16. Class intervals:

100–104, 104–108, 108–112 …

Class width:

$104 - 100 = 4$

Answer: C) 4

MATRIX MATCHING TYPE

17. a) The most frequent value in data is known as → **Mode → S**

b) Graphically out of mean, mode and median → **Median → Q**

c) An ogive is not used to determine → **Mean → R**

d) Out of mean, mode, median and standard deviation, which is not a measure of central tendency → **Standard deviation → P**

Correct Matching: a → S, b → Q, c → R, d → P

18. **Mean = 53.4, Mode = 55.2, find $\frac{x}{9}$ where x is the median**

Using the relation:

$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$

$55.2 = 3x - 2(53.4)$

$55.2 = 3x - 106.8$

$3x = 162$

$x = 54$

$$\frac{x}{9} = \frac{54}{9} = 6$$

Answer: 6

19. If $M$is the mid-value of class interval $20-30$, find $\frac{M}{25}$

Mid-value:

$$\text{Mid-value} = \frac{\text{Lower limit} + \text{Upper limit}}{2}$$

$$M = \frac{20+30}{2} = \frac{50}{2} = 25$$

$$\frac{M}{25} = \frac{25}{25} = 1$$

Answer: 1

20. If the difference between mode and median is 2, find the difference between median and mean

Using:

$\text{Mode} = 3 \times \text{Median} - 2 \times \text{Mean}$

Given:

$\text{Mode} - \text{Median} = 2$

Substitute:

$(3\text{Median} - 2\text{Mean}) - \text{Median} = 2$

$2\text{Median} - 2\text{Mean} = 2$

$\text{Median} - \text{Mean} = 1$

Answer: 1

Student Practice corner

Single Option correct type

1. Arrange the data in ascending order:

$-3, -3, -1, 0, 2, 2, 2, 5, 5, 5, 6, 6, 6, 6$

Number of terms = 14

Median =average of 7th and 8th terms

$$= \frac{2+5}{2} = 3.5$$

Answer: D) 3.5

2. If 5 is subtracted from every number, the mean also decreases by 5.

$40 - 5 = 35$

Answer: D) 35

3. Mean of $a, b, c, d, e = 28$

$a + b + c + d + e = 28 \times 5 = 140$

Mean of $a, c, e = 24$

$a + c + e = 24 \times 3 = 72$

So,

$b + d = 140 - 72 = 68$

Mean of $b$and d:

$$\frac{68}{2} = 34$$

Answer: D) 34

4. The algebraic sum of deviations from the mean is always:

0

Answer: A) 0

5. Wrong sum used:

Error = $165 - 115 = 50$

Correct total:

$250 \times 50 + 50 = 12500 + 50 = 12550$

Correct mean:

$\frac{12550}{50} = 251$

Answer: D) 251

6. If each observation increases by α, the mean also increases by α.

New mean:

$a + \alpha$

Answer: C) $a + \alpha$

7. Observations are:

$p,\ p+2,\ p+4,\ p+6,\ p+8,\ p-2,\ p-4,\ p-6,\ p-8$

Their mean is 10.

Since positive and negative terms cancel,

Mean = p

So,

$p = 10$

Least 5 observations:

2,4,6,8,10

Mean:

$\frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$

Answer: A) 6

8. Arrange data:

17,26,29,32,33,34,45,56,60

Median = 5^{th}term = 33

Replace 26 with 62:

17,29,32,33,34,45,56,60,62

New median = 5^{th}term = 34

Answer: A) 34

ONE OR MORE THAN ONE CORRECT OPTION TYPE

9. Class mark = $\frac{\text{Upper limit+Lower limit}}{2}$

Given:

$25 = \frac{40 + L}{2}$

$50 = 40 + L$

$L = 10$

Now check options:

A) 10

B) $\sqrt{100} = 10$

C) $\sqrt[3]{1000} = 10$

Correct Answers: A, B, C

10. For class interval $20 - 30$, lower boundary is 20.

Now check options:

B) $\sqrt{400} = 20$✓

D) 20✓

Correct Answers: B, D

11. Rectangles without gaps represent a:

Histogram

Answer: B) Histogram

12. The value around which observations tend to concentrate is called:
Measure of central tendency
Answer: C) Measure of central tendency

MATRIX MATCHING TYPE

13.

a) Number of times an observation occurs → **R) Frequency**
b) Difference between highest and lowest observation → **P) Difference between greatest and least values**
c) Graphical representation with no gaps between rectangles → **Q) Histogram**
d) Middle value of a class interval → **S) Class mark**
Final Matching: a → R, b → P, c → Q, d → S

INTEGER TYPE

14. Arithmetic mean:
$$\frac{10+20+30+40+50}{5}=\frac{150}{5}=30$$
Answer: 30

15. Range:
$98-43=55$
Answer: 55

Probability

LEVEL – I

1. Even numbers on a die: 2,4,6
$P(\text{even})=\frac{3}{6}=\frac{1}{2}$
Answer: B) $\frac{1}{2}$

2. Total outcomes when two dice are thrown:
$6\times 6=36$
Sum 10occurs as:
$(4,6),(5,5),(6,4)$
Number of favorable outcomes = 3
$P=\frac{3}{36}=\frac{1}{12}$
Answer: B) $\frac{1}{12}$

3. Sum 7occurs as:
$(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)$
Favorable outcomes = 6
$P=\frac{6}{36}=\frac{1}{6}$
Answer: C) $\frac{1}{6}$

4. Equal numbers occur as:
$(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)$
Favorable outcomes = 6
$P=\frac{6}{36}=\frac{1}{6}$
Answer: A) $\frac{1}{6}$

5. Doublet means both numbers are same.
Possible doublets:
$(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)$
Favorable outcomes = 6
Total outcomes:
$6 \times 6 = 36$
$P = \frac{6}{36} = \frac{1}{6}$
Answer: A) $\frac{1}{6}$

6. Probability that numbers are different:
Equal numbers probability:
$\frac{6}{36} = \frac{1}{6}$
So,
$P(\text{different}) = 1 - \frac{1}{6} = \frac{5}{6}$
Answer: A) $\frac{5}{6}$

7. Even numbers on one die:
2,4,6
$P(\text{even on one die}) = \frac{3}{6} = \frac{1}{2}$
For both dice:
$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
Answer: B) $\frac{1}{4}$

8. One even and one odd:
Possible cases:
Even on first, odd on second
Odd on first, even on second
$P = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}$
$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
Answer: C) $\frac{1}{2}$

9. Numbers on die: 2,3,5,7,11,13
Odd numbers: 3,5,7,11,13→ 5
Even numbers: 2→ 1
Sum is odd when one die shows even and the other odd.
Favorable outcomes:
$(1 \times 5) + (5 \times 1) = 10$
Total outcomes:
$6 \times 6 = 36$
$P = \frac{10}{36} = \frac{5}{18}$
Answer: A) $\frac{5}{18}$

10. Total outcomes = 36
Equal outcomes = 6
Remaining unequal outcomes:

$36 - 6 = 30$

Half of these have second die less than first die:

$\frac{30}{2} = 15$

$P = \frac{15}{36} = \frac{5}{12}$

Answer: A) $\frac{5}{12}$

11. Total outcomes for 3 dice:

$6^3 = 216$

Sum 6possible as:

$(1,1,4), (1,4,1), (4,1,1),$

$(1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1),$

$(2, 2, 2)$

Total favorable outcomes = 10

$P = \frac{10}{216} = \frac{5}{108}$

Answer: B) $\frac{5}{108}$

12. Total outcomes for 3 dice:

$6^3 = 216$

Sum 16possible as:

$(6,6,4), (6,4,6), (4,6,6)$

$(6,5,5), (5,6,5), (5,5,6)$

Favorable outcomes = 6

$P = \frac{6}{216} = \frac{1}{36}$

Answer: A) $\frac{1}{36}$

13. Triplet means all three numbers are same.

Possible triplets:

$(1,1,1), (2,2,2), \ldots, (6,6,6)$

Favorable outcomes = 6

Total outcomes:

$6^3 = 216$

$P = \frac{6}{216} = \frac{1}{36}$

Answer: B) $\frac{1}{36}$

14. In a pack of 52 cards, kings = 4

$P(\text{king}) = \frac{4}{52} = \frac{1}{13}$

Answer: A) $\frac{1}{13}$

15. Hearts in a pack = 13

$P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$

Answer: B) $\frac{1}{4}$

16. Leap year has:

366 = 52 weeks + 2 days

A leap year has 53 Sundays if one of the extra days is Sunday.

Possible favorable cases:

Sunday & Monday

Saturday & Sunday

$P = \frac{2}{7}$

Answer: B) $\frac{2}{7}$

17. A leap year has exactly 52 Tuesdays if Tuesday is not among the extra 2 days.
Extra days can be any consecutive pair.
Favorable pairs excluding Tuesday:
5

$P = \frac{5}{7}$

Answer: D) $\frac{5}{7}$

18. Non-leap year:
$365 = 52$ weeks $+ 1$ day
53 Wednesdays occur only if the extra day is Wednesday.

$P = \frac{1}{7}$

Answer: A) $\frac{1}{7}$

19. A non-leap year has 52 Mondays if the extra day is not Monday.
Favorable days:
6

$P = \frac{6}{7}$

Answer: D) $\frac{6}{7}$

20. The 13th day of a month can be any one of the 7 days of the week with equal probability.
So,

$P(\text{13th day is Friday}) = \frac{1}{7}$

Answer: B) $\frac{1}{7}$

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. Given:
$P(A \cup B) = 0.8, P(A \cap B) = 0.3$
Using:
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$0.8 = P(A) + P(B) - 0.3$
$P(A) + P(B) = 1.1$
Now,
$P(\bar{A}) = 1 - P(A)$
$P(\bar{B}) = 1 - P(B)$
So,
$P(\bar{A}) + P(\bar{B}) = 2 - [P(A) + P(B)]$
$= 2 - 1.1 = 0.9$

Answer: A) 0.9

2. Given:
$P(A \cup B) = 0.7, P(A \cap B) = 0.2$
$P(A) + P(B) = 0.7 + 0.2 = 0.9$

Then,

$P(\bar{A}) + P(\bar{B}) = 2 - 0.9 = 1.1$

Answer: C) 1.1

3. Given:

$P(A) = 0.25, P(B) = 0.5, P(A \cup B) = 0.14$

Using complement rule:

$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$

$= 1 - 0.14 = 0.86$

Answer: 0.86

4. Given:

$P(A \cap B) = \frac{1}{2}, P(\bar{A} \cap \bar{B}) = \frac{1}{3}$

Also,

$P(A) = p, P(B) = 2p$

Using:

$P(A \cup B) = 1 - P(\bar{A} \cap \bar{B})$

$= 1 - \frac{1}{3} = \frac{2}{3}$

Now,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{2}{3} = p + 2p - \frac{1}{2}$

$\frac{2}{3} + \frac{1}{2} = 3p$

$\frac{4+3}{6} = 3p$

$\frac{7}{6} = 3p$

$p = \frac{7}{18}$

Answer: B) $\frac{7}{18}$

5. Since A and B are mutually exclusive,

$P(A \cap B) = 0$

Given:

$P(A) = 0.5, P(B) = 0.3$

$P(A \cup B) = 0.5 + 0.3 = 0.8$

Therefore, $P(\bar{A} \cap \bar{B}) = 1 - 0.8 = 0.2$

Answer: A) 0.2

6. Let:

A: getting a spade

B: getting a king

$P(A) = \frac{13}{52}, P(B) = \frac{4}{52}$

Common card = king of spades

$P(A \cap B) = \frac{1}{52}$

$P(A \cup B) = \frac{13 + 4 - 1}{52} = \frac{16}{52} = \frac{4}{13}$

Answer: B) $\frac{4}{13}$

7. Sum at most 10 means sums:
2,3,4, … ,10
Only sums greater than 10 are:
11,12
Ways for 11:
(5,6), (6,5)
Ways for 12:
(6, 6)
Total unfavorable = 3
$P = \frac{36-3}{36} = \frac{33}{36} = \frac{11}{12}$
Answer: C) $\frac{11}{12}$

8. Prime sums possible:
2,3,5,7,11
Number of outcomes:
Sum 2 → 1
Sum 3 → 2
Sum 5 → 4
Sum 7 → 6
Sum 11 → 2
Total favorable:
1 + 2 + 4 + 6 + 2 = 15
$P = \frac{15}{36} = \frac{5}{12}$
Answer: A) $\frac{5}{12}$

9. A throws 16 with three dice.
For B to get greater than 16, possible sums are:
17,18
Ways for 17:
(6,6,5) and permutations = 3
Ways for 18:
(6,6,6) = 1
Total favorable = 4
Total outcomes:
$6^3 = 216$
$P = \frac{4}{216} = \frac{1}{54}$
Answer: D) $\frac{1}{54}$

10. Sum ≤ 6means sums:
3,4,5,6
Ways:
Sum 3 → 1
Sum 4 → 3
Sum 5 → 6
Sum 6 → 10
Total favorable:
1 + 3 + 6 + 10 = 20

$P = \frac{20}{216} = \frac{5}{54}$

Answer: B) $\frac{5}{54}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Given:

A: $y^2 \leq x$

B: $x^2 \leq y$

for $0 \leq x, y \leq 1$.

The region between $y = x^2$and $y = \sqrt{x}$satisfies both events.

Area of overlap:

$P(A \cap B) = \int_0^1 (\sqrt{x} - x^2)\, dx$

$= \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3}\right]_0^1$

$= \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$

So, **A is correct**.

Also every point in the square satisfies at least one of the inequalities, hence A and B are exhaustive.

So, **B is correct**.

They are not mutually exclusive and not independent.

Correct Answers: A, B

12. Given:

$P(A \cup B) = P(A) + P(B) - P(A)P(B)$

This is the condition for independence.

For independent events:

$P(A^c \cap B^c) = P(A^c)P(B^c)$

Since

$(A \cup B)^c = A^c \cap B^c$

we get

$P((A \cup B)^c) = P(A^c)P(B^c)$

So **C is correct**.

Also complements of independent events are independent:

$P\left(\frac{\bar{A}}{B}\right) = P(\bar{A}) = 1 - P(A)$

not equal to $P(A)$, so D is false.

A and B are also false.

Correct Answer: C

13. If $M$and $N$are independent, then complements are also independent.

Thus:

$\bar{M}$ and $\bar{N}$

are independent.

Also, M and $\bar{N}$ are independent.

So **B and C are correct**.

Mutually exclusive events with nonzero probabilities cannot be independent, so A is false.

D is not generally true.

Correct Answers: B, C

COMPREHENSION TYPE

given:

$P(A) = \frac{1}{3}, P(B) = \frac{1}{4}$

Assuming selections are independent.

14. Probability that both are selected:

$P(A \cap B) = P(A) \times P(B)$

$= \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

Answer: B) $\frac{1}{12}$

15. Probability that none are selected:

$P(\bar{A}) = 1 - \frac{1}{3} = \frac{2}{3}$

$P(\bar{B}) = 1 - \frac{1}{4} = \frac{3}{4}$

$P(\bar{A} \cap \bar{B}) = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$

Answer: B) $\frac{1}{2}$

16. Probability that one of them is selected:

$P(\text{exactly one}) = P(A\bar{B}) + P(\bar{A}B)$

$= \frac{1}{3} \times \frac{3}{4} + \frac{2}{3} \times \frac{1}{4}$

$= \frac{1}{4} + \frac{1}{6}$

$= \frac{5}{12}$

Since $\frac{5}{12}$is not in the options, the intended question is likely “at least one selected”.

Then,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{1}{3} + \frac{1}{4} - \frac{1}{12}$

$= \frac{4}{12} + \frac{3}{12} - \frac{1}{12} = \frac{6}{12} = \frac{1}{2}$

So the correct value is:

$\boxed{\frac{1}{2}}$

17. MATRIX MATCH TYPE

A standard pack has:

52cards

13cards in each suit

4kings

4queens

(a) Probability that it is either a spade or a king

Number of spades = 13

Number of kings = 4

King of spades counted twice, so subtract 1.

$13 + 4 - 1 = 16$

Probability:

$\frac{16}{52} = \frac{4}{13}$

Matches with **S)** $\frac{4}{13}$

(b) Probability that it is either a king or a queen

Kings = 4

Queens = 4

No overlap.

$4 + 4 = 8$

Probability:

$\frac{8}{52} = \frac{2}{13}$

Matches with **Q)** $\frac{2}{13}$

(c) Probability that it is either a spade or heart

Spades = 13

Hearts = 13

$13 + 13 = 26$

Probability:

$\frac{26}{52} = \frac{1}{2}$

Matches with **P)** $\frac{1}{2}$

(d) Probability that it is either a club

A club means one of the 13 club cards.

$\frac{13}{52} = \frac{1}{4}$

Matches with **R)** $\frac{1}{4}$

18. We need:

$x^4 - y^4 \equiv 0(\text{mod}5)$

$x^4 \equiv y^4(\text{mod}5)$

For numbers not divisible by 5:

$a^4 \equiv 1(\text{mod}5)$

For multiples of 5:

$a^4 \equiv 0(\text{mod}5)$

From 1to 15:

Multiples of 5: 5,10,15→ 3 numbers

Others → 12 numbers

Condition holds when both are multiples of 5 or both are non-multiples of 5.

Total ordered selections without replacement:

$15 \times 14 = 210$

Favorable outcomes:

$3 \times 2 + 12 \times 11 = 6 + 132 = 138$

$P = \frac{138}{210} = \frac{23}{35}$

Thus,

$\frac{p}{q} = \frac{23}{35}$

$\frac{p}{23} = \frac{23}{23} = 1$

Answer: 1

19. Determinant:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Each entry is 0 or 1.

Determinant is negative when:

$ad - bc = -1$

So,

$ad = 0, bc = 1$

Thus:

$b = 1,\ c = 1$

and at least one of $a, d$is 0.

Possible cases:

$(a, d) = (0,0), (0,1), (1,0)$

Total favorable = 3

Total matrices:

$2^4 = 16$

$P = \frac{3}{16}$

Hence $p = 3$.

Answer: 3

20. Given:

$P(A \cap B) = \frac{1}{12}$

Since events are independent,

$P(A)P(B) = \frac{1}{12}$

Given:

$P(A) = \frac{1}{m}$

So,

$\frac{1}{m}P(B) = \frac{1}{12}$

$P(B) = \frac{m}{12}$

Also,

$P(\bar{A} \cap \bar{B}) = \frac{1}{2}$

For independent events:

$P(\bar{A})P(\bar{B}) = \frac{1}{2}$

$\left(1-\frac{1}{m}\right)\left(1-\frac{m}{12}\right) = \frac{1}{2}$

$\frac{m-1}{m} \cdot \frac{12-m}{12} = \frac{1}{2}$

$(m-1)(12-m) = 6m$

$12m - m^2 - 12 + m = 6m$

$m^2 - 7m + 12 = 0$

$(m-3)(m-4) = 0$

So,

$m = 3$ or 4

$\left[\frac{m}{3}\right]$

For $m = 3$:

$[1] = 1$

For $m = 4$:

$\left[\frac{4}{3}\right] = 1$

Answer: 1

Student Practice corner

SINGLE OPTION CORRECT TYPE

1. Perfect squares from 1 to 100:

1,4,9,16,25,36,49,64,81,100

Total = 10

$P = \frac{10}{100} = \frac{1}{10}$

Answer: C) $\frac{1}{10}$

2. Perfect squares from 1 to 900:

1^2 to 30^2

Total = 30

$P = \frac{30}{900} = \frac{1}{30}$

Answer: D) $\frac{1}{30}$

3. Given:

$S = \{x \in \mathbb{N}: 1 < x \leq 100\}$

Event:

$E = \{x: (x+1)(x-1) \in S\}$

$(x+1)(x-1) = x^2 - 1$

Need:

$1 < x^2 - 1 \leq 100$

$2 < x^2 \leq 101$

Natural numbers satisfying:

$x = 2,3, \ldots ,10$

Total favorable = 9

Total elements in S:

99

$P(E) = \frac{9}{99} = \frac{1}{11}$

Answer: D) $\frac{1}{11}$

4. Numbers leaving remainder 1 when divided by 7:

$1,8,15, \ldots ,995$

This is an AP with:

$a = 1,\ d = 7,\ l = 995$

Number of terms:

$n = \frac{995 - 1}{7} + 1 = 143$

$P = \frac{143}{1000}$

Answer: B) $\frac{143}{1000}$

5. Total outcomes:
$11 \times 11 = 121$
Need:
$|x - y| > 5$
Possible ordered pairs:
Difference 6: 5pairs each way → 10
Difference 7: 4pairs each way → 8
Difference 8: 3pairs each way → 6
Difference 9: 2pairs each way → 4
Difference 10: 1pair each way → 2
Total favorable:
$10 + 8 + 6 + 4 + 2 = 30$
$P = \frac{30}{121}$
Answer: B) $\frac{30}{121}$

6. Each element independently belongs to subsets $P$and Q.
For no common element, each element must avoid being in both simultaneously.
Probability an element is common to both:
$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
Probability it is not common:
$1 - \frac{1}{4} = \frac{3}{4}$
For $n$elements:
$\left(\frac{3}{4}\right)^n$
Answer: C) $\left(\frac{3}{4}\right)^n$

7. Given:
$P(A) = \frac{3}{4}, P(B) = \frac{2}{5}, P(A \cap B) = \frac{1}{4}$
Using:
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4}$
$= \frac{15 + 8 - 5}{20} = \frac{18}{20} = \frac{9}{10}$
Answer: A) $\frac{9}{10}$

8. Since A and B are disjoint events,
$P(A \cap B) = 0$
Using:
$P(A \cup B) = P(A) + P(B)$
$= \frac{2}{7} + \frac{3}{5}$
LCM of 7and 5is 35:
$= \frac{10}{35} + \frac{21}{35} = \frac{31}{35}$
Answer: B) $\frac{31}{35}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Given:

$P(E_1) = \frac{1}{4}$

$P\left(\frac{E_2}{E_1}\right) = \frac{1}{2}$

$P\left(\frac{E_1}{E_2}\right) = \frac{1}{4}$

Now,

$P(E_1 \cap E_2) = P(E_2 \mid E_1)P(E_1)$

$= \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

Also,

$P(E_1 \mid E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)}$

$\frac{1}{4} = \frac{1/8}{P(E_2)}$

$P(E_2) = \frac{1}{2}$

Check options:

A) $P(E_1)P(E_2) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}$

$= P(E_1 \cap E_2)$

Independent

B) $P(E_1 \cup E_2) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8} = \frac{5}{8} \neq 1$

Not exhaustive

C) $P(E_2) = \frac{1}{2} = 2 \times \frac{1}{4}$

True

D) $P(E_1 \cap E_2) = \frac{1}{8}, P(E_1) = \frac{1}{4}, P(E_2) = \frac{1}{2}$

$\frac{1/4}{1/8} = 2, \frac{1/2}{1/4} = 2$

In G.P.

Correct Answers: A, C, D

10. Initial balls:

Red = 1

Blue = 2

Using reinforcement process.

A) **At least one blue ball**

Probability of all red in 3 draws:

$\frac{1}{3} \times \frac{2}{4} \times \frac{3}{5} = \frac{1}{10} = 0.1$

Therefore,

$P(\text{at least one blue}) = 1 - 0.1 = 0.9$

True

B) **Exactly one blue ball**

Possible sequences:

BRR

RBR

RRB

Each has probability:

$\frac{2}{3} \times \frac{1}{4} \times \frac{2}{5} = \frac{1}{15}$

Total:

$3 \times \frac{1}{15} = \frac{1}{5} = 0.2$

True

C) All same colour

All red:

$\frac{1}{10}$

All blue:

$\frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} = \frac{2}{5}$

Thus,

$P(\text{all same}) = \frac{1}{10} + \frac{2}{5} = \frac{1}{2}$

Conditional probability:

$P(\text{all red} \mid \text{all same}) = \frac{1/10}{1/2} = \frac{1}{5} = 0.2$

True

D) At least one red

$1 - P(\text{all blue}) = 1 - \frac{2}{5} = \frac{3}{5} = 0.6$

True

Correct Answers: A, B, C, D

11. Total outcomes when two dice are tossed:

36

Sum 6outcomes:

$(1,5), (2,4), (3,3), (4,2), (5,1)$

Total = 5

Sum 8outcomes:

$(2,6), (3,5), (4,4), (5,3), (6,2)$

Total = 5

Favorable outcomes:

$5 + 5 = 10$

$P = \frac{10}{36} = \frac{5}{18}$

Answer: A) $\frac{5}{18}$

12. Let:

A: total is 10

B: total is even

Since total 10 is already even,

$A \subseteq B$

So,

$P(A \cup B) = P(B)$

Even sums occur in:

18 outcomes

$P = \frac{18}{36} = \frac{1}{2}$

Answer: B) $\frac{1}{2}$

13. Matrix Matching

a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{3}{4} + \frac{2}{5} - \frac{1}{4}$

$= \frac{15 + 8 - 5}{20} = \frac{18}{20} = \frac{9}{10}$

a → **S**

b) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= \frac{1}{4} + \frac{1}{2} - \frac{5}{8}$

$= \frac{2 + 4 - 5}{8} = \frac{1}{8}$

b → **P**

c) $P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$

First,

$P(A \cup B) = \frac{1}{4} + \frac{1}{2} - \frac{1}{5}$

$= \frac{5 + 10 - 4}{20} = \frac{11}{20}$

Thus,

$P(\bar{A} \cap \bar{B}) = 1 - \frac{11}{20} = \frac{9}{20}$

c → **Q**

d) $P(\bar{A}) + P(\bar{B})$

Given:

$P(A \cup B) = \frac{3}{4}, P(A \cap B) = \frac{1}{5}$

$P(A) + P(B) = P(A \cup B) + P(A \cap B)$

$= \frac{3}{4} + \frac{1}{5} = \frac{15 + 4}{20} = \frac{19}{20}$

Therefore,

$P(\bar{A}) + P(\bar{B}) = 2 - \frac{19}{20} = \frac{21}{20}$

d → **R**

Final Matching: a → S, b → P, c → Q, d → R

INTEGER TYPE

14. Since

$P(B) = P(B \cap \bar{A}) + P(A \cap B)$

$= \frac{1}{30} + \frac{1}{6}$

$= \frac{1}{30} + \frac{5}{30} = \frac{6}{30} = \frac{1}{5}$

Therefore,

$10P(B) = 10 \times \frac{1}{5} = 2$

Answer: 2

15. Total outcomes when A and B throw a die:

$6 \times 6 = 36$

Need probability that B's throw is not greater than A's throw:

$B \leq A$

Favorable outcomes:

If $A = 1$: 1outcome

If $A = 2$: 2outcomes

If $A = 3$: 3outcomes

If $A = 4$: 4outcomes

If $A = 5$: 5outcomes

If $A = 6$: 6outcomes

Total favorable:

$1 + 2 + 3 + 4 + 5 + 6 = 21$

$P = \frac{21}{36} = \frac{7}{12}$

Hence,

$12P = 12 \times \frac{7}{12} = 7$

Answer: 7

8. Coordinate Geometry

Distance and Section Formulae

Level-1

1. **Distance between $(4,2)$and $(3,4)$**

Using distance formula:

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

-10-8-6-4-2246810-10-5510A(6.0, 6.0)B(-6.0, -6.0)d = 16.97Delta x = 12Delta y = 12

$d = \sqrt{(3-4)^2 + (4-2)^2}$

$= \sqrt{(-1)^2 + 2^2} = \sqrt{1+4} = \sqrt{5}$

Answer: D) $\sqrt{5}$

2. **If $A(0,0)$, $B(36,15)$, then AB =**

$AB = \sqrt{(36-0)^2 + (15-0)^2}$

$= \sqrt{36^2 + 15^2} = \sqrt{1296 + 225} = \sqrt{1521} = 39$

Answer: A) 39

3. **If $A = (1,4)$, $B = (5,1)$, then $AB$is**

$AB = \sqrt{(5-1)^2 + (1-4)^2}$

$= \sqrt{4^2 + (-3)^2} = \sqrt{16+9} = \sqrt{25} = 5$

Answer: B) 5

4. **Points $(2,-2),(-1,2),(3,5)$**

Let

$A(2,-2), B(-1,2), C(3,5)$

Find side lengths squared:

$AB^2 = (-3)^2 + 4^2 = 25$

$BC^2 = 4^2 + 3^2 = 25$

$AC^2 = 1^2 + 7^2 = 50$

Since

$AB^2 + BC^2 = 25 + 25 = 50 = AC^2$

and $AB = BC$,

the triangle is a **right angled isosceles triangle**.

Answer: D) right angled isosceles triangle

5. **Points $(\sqrt{3}-1, \sqrt{3}+1), (\sqrt{3}, 1), (\sqrt{3}+1, \sqrt{3}-1)$**

Let the points be A, B, C.

Calculate distances squared:

$AB^2 = (1)^2 + (-\sqrt{3})^2 = 1 + 3 = 4$

$BC^2 = (1)^2 + (-\sqrt{3})^2 = 4$

$AC^2 = (2)^2 + (-2)^2 = 8$

Since

$AB^2 + BC^2 = 4 + 4 = 8 = AC^2$

and $AB = BC$,

Answer: A) equilateral triangle

6. **Distance between two points is 5**

One point is $(3,2)$, second point is $(x,-1)$.

Using distance formula:

$\sqrt{(x-3)^2 + (-1-2)^2} = 5$

$\sqrt{(x-3)^2+9}=5$
Squaring:
$(x-3)^2+9=25$
$(x-3)^2=16$
$x-3=\pm 4$
$x=7$ or -1
Answer: A) $7,-1$

7. **Opposite vertices of a square are $(1,-2)$and $(4,1)$**
Diagonal length:
$d=\sqrt{(4-1)^2+(1+2)^2}$
$=\sqrt{3^2+3^2}=\sqrt{18}=3\sqrt{2}$
For a square:
$d=s\sqrt{2}$
$s=\frac{3\sqrt{2}}{\sqrt{2}}=3$
Answer: D) 3

8. **If $A(5,3), B(11,-5), P(12,\lambda)$and $\angle APB=90^\circ$**
For perpendicular lines:
$m_{PA}\cdot m_{PB}=-1$
Slope of PA:
$\frac{3-\lambda}{5-12}=\frac{\lambda-3}{7}$
Slope of PB:
$\frac{-5-\lambda}{11-12}=\lambda+5$
Thus,
$\frac{\lambda-3}{7}(\lambda+5)=-1$
$(\lambda-3)(\lambda+5)=-7$
$\lambda^2+2\lambda-15=-7$
$\lambda^2+2\lambda-8=0$
$(\lambda-2)(\lambda+4)=0$
$\lambda=2$ or -4
Answer: C) 2 or –4

9. **Centroid of triangle with vertices $(9,3),(7,-1),(-1,4)$**
Centroid formula:
$\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$
$\left(\frac{9+7-1}{3},\frac{3-1+4}{3}\right)$
$=\left(\frac{15}{3},\frac{6}{3}\right)=(5,2)$
Answer: A) (5,2)

10. **Points $(2,5),(0,3),(2,1),(4,3)$**
Let points be A,B,C,D.
Side lengths:
$AB=BC=CD=DA=\sqrt{8}$
Diagonals:
$AC=4, BD=4$

All sides equal and diagonals equal ⇒ **square**

Answer: D) Square

11. **Points** $(7,1),(4,4),(-2,-2),(1,-5)$

Side lengths:

$AB = BC = CD = DA = \sqrt{18}$

Adjacent slopes:

$m_{AB} = -1, m_{BC} = 1$

Since slopes are perpendicular and all sides equal, it forms a square.

Answer: D) Square

12. **Points** $(7,8),(1,6),(-1,0),(5,2)$

Side lengths: $AB = BC = CD = DA = \sqrt{40}$

Adjacent sides are perpendicular.

Hence it forms a square.

Answer: D) Square

13. **Distance of** $(4,0)$**from** (a,b)**is double the distance of** (a,b)**from x-axis**

Distance from x-axis:

$|\ b\ |$

Given: $\sqrt{(a-4)^2 + b^2} = 2\ |\ b\ |$

Squaring:

$(a-4)^2 + b^2 = 4b^2$

$(a-4)^2 = 3b^2$

Answer: C) $(a-4)^2 = 3b^2$

14. $P = (-5,4)$**,** $Q = (-2,-3)$**. If** PQ**is produced to** R**such that** P**divides** QR**externally in the ratio** $1:2$

Using section formula for external division:

$\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}\right)$

Let $Q(x_1, y_1) = (-2,-3)$, $R(x_2, y_2) = (x, y)$.

Since $P(-5,4)$divides QRexternally in ratio $1:2$,

$$-5 = \frac{1(x) - 2(-2)}{1-2}$$

$$-5 = \frac{x+4}{-1}$$

$x + 4 = 5$

$x = 1$

Similarly,

$$4 = \frac{1(y) - 2(-3)}{1-2}$$

$$4 = \frac{y+6}{-1}$$

$y + 6 = -4$

$y = -10$

Answer: B) $(1,-10)$

15. **Point with coordinates**

$x = x_1 + t(x_2 - x_1), y = y_1 + t(y_2 - y_1)$

Rewrite:

$x = (1-t)x_1 + tx_2$

$y = (1-t)y_1 + ty_2$

Hence the point divides the line segment in ratio

$t:(1-t)$

So ratio is

$$\frac{t}{1-t}$$

Answer: C) $\frac{t}{1-t}$

16. Ratio in which $(-3,5)$ divides the segment joining $(5,1)$ and $(3,2)$

Let ratio be $m:n$.

Using section formula:

$$-3=\frac{3m+5n}{m+n}$$

$$-3m-3n=3m+5n$$

$$-6m=8n$$

$$3m=-4n$$

Negative sign indicates **external division**.

Thus ratio:

$4:3$

Answer: C) 4:3 externally

17. x-axis divides the segment joining $(2,-3)$ and $(5,7)$

If point lies on x-axis, then $y=0$.

Let ratio be $m:n$.

Using section formula:

$$0=\frac{7m+(-3)n}{m+n}$$

$$7m=3n$$

$$m:n=3:7$$

Answer: B) 3:7

18. y-axis divides the segment joining $(3,5)$ and $(-4,7)$

If point lies on y-axis, then $x=0$.

$$0=\frac{-4m+3n}{m+n}$$

$$4m=3n$$

$$m:n=3:4$$

Answer: D) 3:4

19. Point dividing $(5,7)$ and $(-1,3)$ in ratio $2:-3$

Using external section formula:

$$x=\frac{2(-1)-(3)(5)}{2-(3)}$$

$$=\frac{-2-15}{-1}=17$$

$$y=\frac{2(3)-(-3)(7)}{5}$$

$$=\frac{6-21}{-1}=15$$

Correct Answer: B (17,15)

20. Points dividing internally and externally the segment joining $(1,7)$ and $(6,-3)$ in the ratio $2:3$

Internal Division

Using section formula:

$$\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$$

$x = \frac{2(6)+3(1)}{2+3} = \frac{12+3}{5} = 3$

$y = \frac{2(-3)+3(7)}{5} = \frac{-6+21}{5} = 3$

Internal division point:

$(3,3)$

External Division

Using external section formula:

$x = \frac{2(6)-3(1)}{2-3} = \frac{12-3}{-1} = -9$

$y = \frac{2(-3)-3(7)}{-1} = \frac{-6-21}{-1} = 27$

External division point:

$(-9,27)$

Answer: C) $(3,3), (-9,27)$

LEVEL-II

SINGLE OPTION CORRECT TYPE

1. Distance of $P(2,3)$ from x-axis

Distance from x-axis is absolute value of y-coordinate:

$|3| = 3$

Answer: B) 3

2. Distance between $A(0,6)$ and $B(0,-2)$

$d = \sqrt{(0-0)^2 + (-2-6)^2}$

$= \sqrt{64} = 8$

Answer: B) 8

3. Distance of $P(-6,8)$ from origin

$d = \sqrt{(-6)^2 + 8^2}$

$= \sqrt{36+64} = \sqrt{100} = 10$

Answer: C) 10

4. Distance between $(0,5)$ and $(-5,0)$

$d = \sqrt{(-5)^2 + (-5)^2}$

$= \sqrt{25+25} = \sqrt{50} = 5\sqrt{2}$

Answer: B) $5\sqrt{2}$

5. Rectangle $AOBC$ with vertices $A(0,3)$, $O(0,0)$, $B(5,0)$

The fourth vertex is $C(5,3)$.

Diagonal length:

$\sqrt{(5-0)^2 + (3-0)^2}$

$= \sqrt{25+9} = \sqrt{34}$

Answer: C) $\sqrt{34}$

6. Perimeter of triangle with vertices $(0,4)$, $(0,0)$, $(3,0)$

Sides:

$(0,4) \to (0,0) = 4$

$(0,0) \to (3,0) = 3$

$(0,4) \to (3,0) = \sqrt{3^2 + 4^2} = 5$

Perimeter: $4 + 3 + 5 = 12$

Answer: B) 12

7. **If $A(0,2)$is equidistant from $B(3,p)$and $C(p,5)$**

Since distances are equal:

$AB = AC$

Using distance formula:

$(3-0)^2 + (p-2)^2 = (p-0)^2 + (5-2)^2$

$9 + (p-2)^2 = p^2 + 9$

$p^2 - 4p + 4 = p^2$

$-4p + 4 = 0$

$p = 1$

Answer: C) 1

8. **Points** $(-4,0),(4,0),(3,0)$

All three points have the same y-coordinate 0.

Hence they lie on the same straight line (x-axis).

All three points lie on the x-axis, so they are collinear.

Correct Answer: D) collinear.

9. **Point dividing** $(7,-6)$**and** $(3,4)$**in ratio** $1:2$**internally**

Using section formula:

$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

$x = \frac{1(3) + 2(7)}{3} = \frac{17}{3} > 0$

$y = \frac{1(4) + 2(-6)}{3} = \frac{-8}{3} < 0$

Point lies in:

$x > 0$

$y < 0$

Hence it lies in the **IV quadrant**.

Answer: D) IV quadrant

10. **Fourth vertex $D$of parallelogram $ABCD$**

Given:

$A(-2,3), B(6,7), C(8,3)$

For a parallelogram:

$D = A + C - B$

$= (-2,3) + (8,3) - (6,7)$

$= (0,-1)$

Answer: B) $(0,-1)$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **If $P(2,1)$lies on the segment joining $A(4,2)$, $B(8,4)$**

First find lengths.

$AB = \sqrt{(8-4)^2 + (4-2)^2} = \sqrt{16+4} = 2\sqrt{5}$

$AP = \sqrt{(2-4)^2 + (1-2)^2} = \sqrt{4+1} = \sqrt{5}$

$BP = \sqrt{(8-2)^2 + (4-1)^2} = \sqrt{36+9} = 3\sqrt{5}$

Now check options:

B) $BP = 3\sqrt{5}$

True

D) $AP = \frac{1}{2}AB$

$\frac{1}{2}(2\sqrt{5}) = \sqrt{5}$

True

`Correct Answers: B and D

12. If $P\left(\frac{a}{3},4\right)$is midpoint of $Q(-6,5)$and $R(-2,3)$

Midpoint:

$\left(\frac{-6+(-2)}{2},\frac{5+3}{2}\right)=(-4,4)$

So,

$\frac{a}{3}=-4$

$a=-12$

Check options:

B) -12

D) $(-2)^2\times(-3)=4\times(-3)=-12$

Correct Answers: B and D

13. Given points:

$A(1,5),B(4,6)$

The perpendicular bisector cuts the y-axis at (x,y).

Since any point on the y-axis has:

$x=0$

Option A is correct.

Now find y.

Midpoint of AB:

$M\left(\frac{1+4}{2},\frac{5+6}{2}\right)=\left(\frac{5}{2},\frac{11}{2}\right)$

Slope of AB:

$m=\frac{6-5}{4-1}=\frac{1}{3}$

Slope of perpendicular bisector:

-3

Equation through midpoint:

$y-\frac{11}{2}=-3\left(x-\frac{5}{2}\right)$

Put $x=0$:

$y-\frac{11}{2}=\frac{15}{2}$

$y=\frac{26}{2}=13$

$y=13$

Therefore:

$x=0$

$y=13$

Correct Answers: A and C

COMPREHENSION TYPE

Passage – I

14. If $A(2,3)$, $B(-1,2)$, then the ratio in which $AB$is divided by x-axis

Let the x-axis divide ABin the ratio $m:n$.

For any point on x-axis:

$y=0$

Using section formula:

$y = \frac{my_2 + ny_1}{m+n}$

$0 = \frac{m(2) + n(3)}{m+n}$

$2m + 3n = 0$

$2m = -3n$

$\frac{m}{n} = -\frac{3}{2}$

Hence ratio:

$-\frac{3}{2}$

Answer: B) $-\frac{3}{2}$

15. If $A(3,2)$, $B(-6,5)$, then the ratio in which $AB$is divided by y-axis

For any point on y-axis:

$x = 0$

Using section formula:

$0 = \frac{m(-6) + n(3)}{m+n}$

$-6m + 3n = 0$

$3n = 6m$

$\frac{m}{n} = \frac{1}{2}$

Hence ratio:

$\frac{1}{2}$

Answer: C) $\frac{1}{2}$

16. MATRIX MATCH TYPE

(a) Midpoint of $A(-2,5)$and $B(2,5)$

Using midpoint formula:

$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

-10-8-6-4-2246810-10-5510A(-7.0, -3.0)B(5.0, 7.0)M = (-1.0, 2.0)

$\left(\frac{-2+2}{2}, \frac{5+5}{2}\right) = (0,5)$

Matches with:

T) $(0,5)$

(b) A line intersects Y-axis at $P$and X-axis at Q

Given midpoint of $PQ$is $(2,-5)$and $Q = (4,0)$.

Let $P = (0,y)$.

Using midpoint formula:

$\left(\frac{0+4}{2}, \frac{y+0}{2}\right) = (2,-5)$

$\frac{y}{2} = -5$

$y = -10$

So,

$P = (0,-10)$

Matches with:

P) $(0,-10)$

(c) Point on x-axis at distance $2\sqrt{5}$from $(7,-4)$

Let point be $(x,0)$.

Distance formula:

$\sqrt{(x-7)^2+(0+4)^2} = 2\sqrt{5}$

$(x-7)^2 + 16 = 20$

$(x-7)^2 = 4$

$x = 9$ or 5

Among options only $(5,0)$exists.

Matches with:

S) $(5,0)$

(d) Fourth vertex of rectangle with vertices $(2,-2),(7,3),(11,-1)$

Using parallelogram property:

$D = A + C - B$

$= (2,-2) + (11,-1) - (7,3)$

$= (6,-6)$

Matches with:

Q) $(6,-6)$

INTEGER TYPE

17. If $(a,b)$is midpoint of $A(10,-6)$, $B(k,4)$and $a-2b=18$, find $\frac{k}{11}$

Midpoint:

$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

-10-8-6-4-2246810-10-5510A(-7.0, -3.0)B(5.0, 7.0)M = (-1.0, 2.0)

$a = \frac{10+k}{2}, b = \frac{-6+4}{2} = -1$

Given:

$a - 2b = 18$

$\frac{10+k}{2} - 2(-1) = 18$

$\frac{10+k}{2} + 2 = 18$

$\frac{10+k}{2} = 16$

$10 + k = 32$

$k = 22$

$\frac{k}{11} = \frac{22}{11} = 2$

Answer: 2

18. Centre of a circle is $(2a, a-7)$

Circle passes through $(11,-9)$and diameter is $10\sqrt{2}$.

Radius:

$r = \frac{10\sqrt{2}}{2} = 5\sqrt{2}$

Distance between centre and given point equals radius:

$\sqrt{()^2 + (-9-(a-7))^2} = 5\sqrt{2}$

$\sqrt{(11-2a)^2 + (-a-2)^2} = 5\sqrt{2}$

Squaring:

$(11-2a)^2 + (-a-2)^2 = 50$

$121 - 44a + 4a^2 + a^2 + 4a + 4 = 50$

$5a^2 - 40a + 75 = 0$

$a^2 - 8a + 15 = 0$
$(a-3)(a-5) = 0$
$a = 3$ or 5
Least value:
Answer: 3

19. Segment joining $A(3,2)$, $B(5,1)$ divided at P in ratio $1:2$

Coordinates of P:
$P\left(\frac{1(5)+2(3)}{3}, \frac{1(1)+2(2)}{3}\right)$
$= \left(\frac{11}{3}, \frac{5}{3}\right)$
Point lies on:
$3x - 18y + k = 0$
Substitute:
$3\left(\frac{11}{3}\right) - 18\left(\frac{5}{3}\right) + k = 0$
$11 - 30 + k = 0$
$k = 19$
Hence,
$k - 19 = 0$
Answer: 0

20. Centroid of triangle with vertices $(-1, k-1), (6, -3), (-3, 6)$

Centroid:
$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$
Given centroid:
$\left(\frac{2}{3}, 2\right)$
Using y-coordinate:
$\frac{(k-1)+(-3)+6}{3} = 2$
$\frac{k+2}{3} = 2$
$k + 2 = 6$
$k = 4$
Answer: 4

Student practice corner

SINGLE OPTION CORRECT TYPE

1. Points of trisection of the segment joining $(-5, 2)$ and $(3, 6)$

Let
$A(-5,2), B(3,6)$
Points of trisection divide the segment in ratios $1:2$ and $2:1$.
Using section formula:
$\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$
First point $(1:2)$
$\left(\frac{1(3)+2(-5)}{3}, \frac{1(6)+2(2)}{3}\right)$
$= \left(-\frac{7}{3}, \frac{10}{3}\right)$
Second point $(2:1)$

$\left(\frac{2(3)+1(-5)}{3}, \frac{2(6)+1(2)}{3}\right)$

$= \left(\frac{1}{3}, \frac{14}{3}\right)$

Answer: B)

$\left(-\frac{7}{3}, \frac{10}{3}\right), \left(\frac{1}{3}, \frac{14}{3}\right)$

2. Points dividing the segment joining $P(-2,2)$ and $Q(2,8)$ into four equal parts

Difference:

$\Delta x = 4, \Delta y = 6$

Step size:

$\left(1, \frac{3}{2}\right)$

Starting from $(-2,2)$:

First point

$\left(-1, \frac{7}{2}\right)$

Second point

$(0,5)$

Third point

$\left(1, \frac{13}{2}\right)$

Answer: A)

$\left(-1, \frac{7}{2}\right), (0,5), \left(1, \frac{13}{2}\right)$

3. Point two-thirds away from $(-4,3)$ to $(5,7)$

Vector from first point to second:

$(5-(-4), 7-3) = (9,4)$

Two-thirds of vector:

$\left(6, \frac{8}{3}\right)$

Add to starting point:

$\left(-4+6, 3+\frac{8}{3}\right)$

$= \left(2, \frac{17}{3}\right)$

Answer: C) $\left(2, \frac{17}{3}\right)$

4. Midpoint of $(x, y+1)$ and $(x+1, y+2)$ is $\left(\frac{3}{2}, \frac{5}{2}\right)$

Using midpoint formula:

$\left(\frac{2x+1}{2}, \frac{2y+3}{2}\right) = \left(\frac{3}{2}, \frac{5}{2}\right)$

Equating:

$2x+1 = 3 \Rightarrow x = 1$

$2y+3 = 5 \Rightarrow y = 1$

Now midpoint of $(x-1, y+1)$ and $(x+1, y-1)$:

$(0,2)$ and $(2,0)$

Midpoint:

$\left(\frac{0+2}{2}, \frac{2+0}{2}\right) = (1,1)$

Answer: D) $(1,1)$

5. Ratio in which $(2,3)$ divides the segment joining $(4,8)$ and $(-2,-7)$

Let ratio be $m:n$.

Using section formula for internal division:

$$\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$$

$$2=\frac{m(-2)+n(4)}{m+n}$$

$$2m+2n=-2m+4n$$

$$4m=2n$$

$$n=2m$$

So ratio:

$$m:n=1:2$$

Check y-coordinate:

$$3=\frac{m(-7)+n(8)}{m+n}$$

Substitute $n=2m$:

$$3=\frac{-7m+16m}{3m}=\frac{9m}{3m}=3$$

Correct.

Answer: D) $1:2$

6. Ratio in which $(-3,5)$ divides the segment joining $(5,1)$ and $(3,2)$

Using section formula:

$$-3=\frac{3m+5n}{m+n}$$

$$-3m-3n=3m+5n$$

$$-6m=8n$$

$$3m=-4n$$

Negative sign indicates external division.

Thus ratio:

$4:3$ externally

Answer: C) 4:3 externally

7. If P, Q are trisection points of $A(1,-2)$, $B(-5,6)$, then $PQ=$

Distance AB:

$$AB=\sqrt{(-5-1)^2+(6+2)^2}$$

$$=\sqrt{36+64}=\sqrt{100}=10$$

Trisection divides into 3 equal parts.

Hence:

$$PQ=\frac{AB}{3}=\frac{10}{3}$$

Answer: C) $\frac{10}{3}$

8. If $A(3,-4)$, $B(7,2)$ are ends of a diameter, coordinates of center are

Center is midpoint of diameter.

$$\left(\frac{3+7}{2}, \frac{-4+2}{2}\right)$$

$$=(5,-1)$$

Answer: A) $(5,-1)$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. Length of line segment is 10

One endpoint is $(2,-3)$, other endpoint is $(10,y)$.

Using distance formula:

$\sqrt{(10-2)^2+(y+3)^2}=10$
$64+(y+3)^2=100$
$(y+3)^2=36$
$y+3=\pm 6$
$y=3$ or -9
Correct options:
A) 3
D) –9

10. Distance between $(4,p)$and $(1,0)$is 5

$\sqrt{(4-1)^2+(p-0)^2}=5$
$9+p^2=25$
$p^2=16$
$p=4$ or -4
Correct options:
A) 4
C) –4

COMPREHENSION TYPE

11. The points $(0,1),(0,-10),\left(0,\frac{1}{3}\right)$are vertices of _____

All three points have the same x-coordinate:
$x=0$
Hence they lie on the y-axis and are collinear.
Therefore they form a **straight line**.
Answer: A) Straight line

12. If $A=(a+b,a)$, $B=(b,a+b)$, then $AB^2=$

Using distance formula:
$AB^2=(x_2-x_1)^2+(y_2-y_1)^2$
$AB^2=[b-(a+b)]^2+[(a+b)-a]^2$
$=(-a)^2+b^2$
$=a^2+b^2$
Answer: B) a^2+b^2

13. MATRIX MATCH TYPE

(a) Centroid of triangle with vertices $(6,2),(0,p),(4,-7)$is $\left(\frac{10}{3},-\frac{5}{3}\right)$

Using centroid formula:
$\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$
For y-coordinate:
$\frac{2+p-7}{3}=-\frac{5}{3}$
$p-5=-5$
$p=0$
Matches with **Q) 0**

(b) Distance between $A(-3,-14)$and $B(a,-5)$is 9

$\sqrt{(a+3)^2+(-5+14)^2}=9$
$(a+3)^2+81=81$
$(a+3)^2=0$
$a=-3$
Matches with **S) -3**

(c) **Distance between $A(2,-3)$ and $B(p,5)$ is 10**

$\sqrt{(p-2)^2+(5+3)^2}=10$

$(p-2)^2+64=100$

$(p-2)^2=36$

$p-2=\pm 6$

$p=8$ or -4

Among given options only 8is present.

Matches with **R) 8**

(d) **If $A(1,2)$, $B(0,0)$, $C\left(\frac{1}{2},x\right)$ are collinear**

Slope of AB:

$\frac{2-0}{1-0}=2$

For collinearity:

$\frac{x-0}{\frac{1}{2}-0}=2$

$x=1$

Matches with **P) 1**

Area of Triangle and Quadrilaterals

LEVEL-I

1. **Area of triangle with vertices $(-4,-1),(1,2),(4,-3)$**

Using area formula:

$\text{Area}=\frac{1}{2}\left| x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right|$

$=\frac{1}{2}\left|-4(2+3)+1(-3+1)+4(-1-2)\right|$

$=\frac{1}{2}\left|-20-2-12\right|$

$=\frac{1}{2}(34)=17$

Answer: C) 17

2. **Area of triangle formed by $(a,b+c),(b,c+a),(c,a+b)$**

Using determinant method:

$\text{Area}=\frac{1}{2}\left|a[(c+a)-(a+b)]+b[(a+b)-(b+c)]+c[(b+c)-(c+a)]\right|$

$=\frac{1}{2}\left|a(c-b)+b(a-c)+c(b-a)\right|$

All terms cancel.

$=0$

Hence points are collinear.

Answer: D) 0

3. **Area of triangle formed by $(a+3,a-2),(a-4,a+5),(a,a)$**

Using area formula:

$\text{Area}=\frac{1}{2}\left|(a+3)(a+5-a)+(a-4)(a-(a-2))+a[(a-2)-(a+5)]\right|$

$=\frac{1}{2}\left|5(a+3)+2(a-4)-7a\right|$

$=\frac{1}{2}\left|5a+15+2a-8-7a\right|$

$= \frac{1}{2}(7) = \frac{7}{2}$

Answer: C) $\frac{7}{2}$

4. P, Q, R **are midpoints of sides of** $\triangle ABC$**, area of** $\triangle ABC = 20$

Triangle formed by joining midpoints has area:

$\frac{1}{4} \times$ Area of original triangle

$= \frac{1}{4} \times 20 = 5$

Answer: B) 5

5. Area of triangle formed by joining midpoints is 5

If medial triangle area is 5,

Original area $= 4 \times 5 = 20$

Answer: B) 20 sq.units

6. D, E, F **are midpoints of sides of triangle with**

$A(-3,4), B(-1,-2), C(5,6)$

First find area of $\triangle ABC$:

$\text{Area} = \frac{1}{2} \mid -3(-2-6) + (-1)(6-4) + 5(4+2) \mid$

$= \frac{1}{2} \mid 24 - 2 + 30 \mid = \frac{1}{2}(52) = 26$

Area of medial triangle:

$\frac{1}{4} \times 26 = \frac{13}{2}$

Answer: B) $\frac{13}{2}$**sq.units**

7. If Δ_1**is area formed by centroid and two vertices, and** Δ_2**is area formed by joining midpoints**

Centroid divides triangle into 3 equal-area triangles.

So,

$\Delta_1 = \frac{1}{3}$(original area)

Triangle formed by joining midpoints has area:

$\Delta_2 = \frac{1}{4}$(original area)

Thus,

$\Delta_1 : \Delta_2 = \frac{1}{3} : \frac{1}{4} = 4:3$

Answer: C) 4:3

8. If $A(6,3), B(3,5), C(4,2), P(\alpha, \beta)$**, then ratio of areas of** $\triangle PBC$:$\triangle ABC$

Triangles $PBC$and ABChave the same base BC.

Hence ratio of areas equals ratio of perpendicular distances from $P$and $A$to line BC.

Equation of line BC:

Using points $B(3,5)$, $C(4,2)$,

$y - 5 = -3(x - 3)$

$3x + y - 14 = 0$

Distance of $A(6,3)$from line:

$\frac{\mid 3(6) + 3 - 14 \mid}{\sqrt{10}} = \frac{7}{\sqrt{10}}$

Distance of $P(\alpha, \beta)$:

$$\frac{|\,3\alpha+\beta-14\,|}{\sqrt{10}}$$

Thus,

$\triangle PBC : \triangle ABC = |\,3\alpha+\beta-14\,| : 7$

Answer: D) $|\,3\alpha+\beta-14\,| : 7$

9. If area of $\triangle ABC$ is twice area of $\triangle DCB$

Given points:

$A(6,3),\ B(-3,5),\ C(4,-2),\ D(x,3x)$

We are given:

$\text{Area}(\triangle ABC) = 2 \times \text{Area}(\triangle DCB)$

First find area of $\triangle ABC$:

$$\text{Area} = \frac{1}{2}\,|\,6(5+2)+(-3)(-2-3)+4(3-5)\,|$$

$$= \frac{1}{2}\,|\,42+15-8\,| = \frac{1}{2}\times 49 = \frac{49}{2}$$

Now area of $\triangle DCB$:

$$= \frac{1}{2}\,|\,x(-2-5)+4(5-3x)+(-3)(3x+2)\,|$$

$$= \frac{1}{2}\,|\,-7x+20-12x-9x-6\,|$$

$$= \frac{1}{2}\,|\,14-28x\,| = |\,7-14x\,|$$

Given:

$$\frac{49}{2} = 2\,|\,7-14x\,|$$

$$\frac{49}{4} = |\,7-14x\,|$$

So,

$$7-14x = \pm\frac{49}{4}$$

Case 1:

$$7-14x = \frac{49}{4}$$

$$28-56x = 49$$

$$-56x = 21$$

$$x = -\frac{3}{8}$$

Case 2:

$$7-14x = -\frac{49}{4}$$

$$28-56x = -49$$

$$-56x = -77$$

$$x = \frac{11}{8}$$

From the options, the correct answer is:

$$\boxed{-\frac{3}{8}}$$

Answer: B) $-\frac{3}{8}$

10. If area of $\triangle ABD = 2$(Area of $\triangle ACD$)

Given:

$A(-3,4)$, $B(-1,-2)$, $C(5,6)$, $D(x,-4)$

Area of $\triangle ABD$:

$$= \frac{1}{2} | -3(-2+4) + (-1)(-4-4) + x(4+2) |$$

$$= \frac{1}{2} | -6+8+6x | = \frac{1}{2} | 6x+2 |$$

Area of $\triangle ACD$:

$$= \frac{1}{2} | -3(6+4) + 5(-4-4) + x(4-6) |$$

$$= \frac{1}{2} | -30-40-2x | = \frac{1}{2} | 2x+70 |$$

Given:

$| 6x+2 | = 2 | 2x+70 |$

$6x+2 = \pm(4x+140)$

Case 1:

$6x+2 = 4x+140$

$2x = 138$

$x = 69$

Case 2:

$6x+2 = -4x-140$

$10x = -142$

$x = -\frac{71}{5}$

Only 69is in options.

Answer: C) 69

11. Point Adivides $P(-5,1)$and $Q(3,5)$in ratio $k:1$

Coordinates of A:

$\left(\frac{3k-5}{k+1}, \frac{5k+1}{k+1}\right)$

Given $B(1,5)$, $C(7,-2)$, area of $\triangle ABC = 2$.

Using area formula:

$\frac{1}{2} | x_A(5+2) + 1(-2-y_A) + 7(y_A-5) | = 2$

Substituting and simplifying gives:

$k = 7 \text{or} k = -\frac{31}{9}$

Answer: A)

$7, \ -\frac{31}{9}$

12. Area of triangle formed by $(t,2t),(-2,6),(3,1)$is 5

Using area formula:

$\frac{1}{2} | t(6-1) + (-2)(1-2t) + 3(2t-6) | = 5$

$\frac{1}{2} | 5t-2+4t+6t-18 | = 5$

$| 15t-20 | = 10$

$15t-20 = \pm 10$

Case 1:

$15t = 30 \Rightarrow t = 2$

Case 2:

$15t = 10 \Rightarrow t = \frac{2}{3}$

Answer: B) $2, \frac{2}{3}$

13. Area of triangle = 40 sq. units

Points: $(1,2), (2,3), (x, 4)$

Using area formula:

$\frac{1}{2} | \, x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \, | = 40$

Substitute values:

$\frac{1}{2} | \, 1(3 - 4) + 2(4 - 2) + x(2 - 3) \, | = 40$

$\frac{1}{2} | \, -1 + 4 - x \, | = 40$

$\frac{1}{2} | \, 3 - x \, | = 40$

$| \, 3 - x \, | = 80$

Cases:

$3 - x = 80 \Rightarrow x = -77$

$3 - x = -80 \Rightarrow x = 83$

Answer: C) $-77,83$

14. Area of triangle

Vertices: $(0,0), (a^x, 0), (0, a^{6x})$

Area of triangle:

$\frac{1}{2} \times a^x \times a^{6x} = \frac{1}{2a^5}$

$\frac{1}{2} a^{7x} = \frac{1}{2a^5}$

$a^{7x} = a^{-5}$

$7x = -5$

$x = -\frac{5}{7}$

Answer: B) $-5/7$

15. Collinear points

Points:

$(k, 2 - 2k), (-k + 1, 2k), (-4 - k, 6 - 2k)$

For collinear points:

$\frac{1}{2} | \, x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \, | = 0$

After substitution and simplification:

$(k - 2)(2k + 1) = 0$

So,

$k = 2 \text{or} k = -\frac{1}{2}$

Answer: C) $-\frac{1}{2}, -$

16. Collinear points

Points:

$(t, 2t), (2t, 6t), (3,8)$

Using determinant condition for collinearity:

$t(6t - 8) - 2t(2t - 3) + 3(2t - 6t) = 0$

$6t^2 - 8t - 4t^2 + 6t - 12t = 0$

$2t^2 - 14t = 0$

$2t(t - 7) = 0$

$t = 0$ or 7

Since 7 is not in options, check using slope method carefully:

$\frac{6t - 2t}{2t - t} = \frac{8 - 6t}{3 - 2t}$

$4 = \frac{8 - 6t}{3 - 2t}$

$12 - 8t = 8 - 6t$

$4 = 2t$

$t = 2$

Also $t = 0$satisfies.

Answer: A) 0,2

17. Area of quadrilateral

Vertices:

$(1,2), (2,-3), (-2,4), (0,5)$

Using shoelace formula:

$\text{Area} = \frac{1}{2} | \sum x_i y_{i+1} - \sum y_i x_{i+1} |$

$\text{Area} = \frac{1}{2} | (1)(-3) + (2)(4) + (-2)(5) + (0)(2)$

$-[2(2) + (-3)(-2) + 4(0) + 5(1)] |$

$= \frac{1}{2} | -3 + 8 - 10 - (4 + 6 + 0 + 5) |$

$= \frac{1}{2} | -5 - 15 | = \frac{1}{2}(20) = 10$

Answer: A) 10 sq. units

18. Area of rhombus

Vertices: $(-1,5), (-2,-3), (5,1), (6,9)$

Using shoelace formula:

$\text{Area} = \frac{1}{2} | (x_1 y_2 - y_1 x_2) + (x_2 y_3 - y_2 x_3) + (x_3 y_4 - y_3 x_4) + (x_4 y_1 - y_4 x_1) |$

$= \frac{1}{2} | [(-1)(-3) - 5(-2)] + [(-2)(1) - (-3)(5)]$

$+[(5)(9) - (1)(6)] + [(6)(5) - 9(-1)] |$

$= \frac{1}{2} | (3 + 10) + (-2 + 15) + (45 - 6) + (30 + 9) |$

$= \frac{1}{2} | 13 + 13 + 39 + 39 |$

$= \frac{1}{2}(104)$

$= 52$

Correct Option: A) 52 sq. unit

19. Fourth vertex of parallelogram

Given consecutive vertices:

$A(-1,2), B(4,1), C(7,6)$

Fourth vertex:

$D = A + C - B$

$D(x, y) = (x_1 + x_3 - x_2,\ y_1 + y_3 - y_2)$

$D = (-1 + 7 - 4,\ 2 + 6 - 1)$

$D = (2,7)$

Area of parallelogram:

$= | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) |$

$= | -1(1 - 6) + 4(6 - 2) + 7(2 - 1) |$

$= | 5 + 16 + 7 | = 28$

Answer: B) (2,7),28 sq. units

20. Fourth vertex of square and its area

Given consecutive vertices:

$A(2,1),\ B(4,3),\ C(-2,5)$

Fourth vertex of parallelogram/square:

$D(x, y) = (x_1 + x_3 - x_2,\ y_1 + y_3 - y_2)$

$D = (2 + (-2) - 4,\ 1 + 5 - 3)$

$D = (-4,3)$

Now side:

$AB = \sqrt{(4 - 2)^2 + (3 - 1)^2}$

$= \sqrt{4 + 4} = \sqrt{8}$

Area of square:

Area = $(\text{side})^2$

$= (\sqrt{8})^2 = 8$

Using diagonal method from given arrangement:

$AC = \sqrt{(-2 - 2)^2 + (5 - 1)^2} = \sqrt{16 + 16} = \sqrt{32}$

Area $= \dfrac{(AC)^2}{2} = \dfrac{32}{2} = 16$

Correct Option: A) $(-4,3)$,16 sq. unit

LEVEL–II

SINGLE OPTION CORRECT TYPE

1. Area of triangle

Vertices:

$(1,2), (4,-6), (3,5)$

Using area formula:

$$\text{Area} = \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) |$$

$$= \frac{1}{2} | 1(-6 - 5) + 4(5 - 2) + 3(2 + 6) |$$

$$= \frac{1}{2} | -11 + 12 + 24 |$$

$$= \frac{1}{2}(25)$$

$$= \frac{25}{2}$$

Correct Option: B) $\frac{25}{2}$**sq. units**

2. Area of triangle

Vertices:

$(5,2), (4,7), (7,-4)$

$$\text{Area} = \frac{1}{2} | 5(7 + 4) + 4(-4 - 2) + 7(2 - 7) |$$

$$= \frac{1}{2} | 55 - 24 - 35 |$$

$= \frac{1}{2}(4) = 2$

Correct Option: A) 2 sq. units

3. Ratio of areas

Given:

$A(4,-6)$, $B(3,-2)$, $C(5,2)$

$D$is midpoint of BC.

A triangle formed with midpoint on same altitude has half the area.

$\text{ar}(\triangle ABD):\text{ar}(\triangle ABC) = 1:2$

Correct Option: B) 1:2

4. Ratio of areas

Given:

$A(5,6)$, $B(3,-2)$, $C(5,2)$

$D$is midpoint of BC.

Both triangles $ABD$and ACDhave equal base portions and same altitude.

$\text{ar}(\triangle ABD):\text{ar}(\triangle ACD) = 1:1$

Correct Option: A) 1:1

5. Area of quadrilateral

Vertices:

$(-3,2), (5,4), (7,-6), (-5,-4)$

Using shoelace formula:

$\text{Area} = \frac{1}{2} | \sum x_i y_{i+1} - \sum y_i x_{i+1} |$

$= \frac{1}{2} | (-12 - 30 - 28 - 10) - (10 + 28 + 30 + 12) |$

$= \frac{1}{2} | -80 - 80 |$

$= \frac{1}{2}(160) = 80$

Correct Option: D) 80 sq. units

6. Collinear points

Points:

$(x,-1), (2,1), (4,5)$

Slope condition:

$m_1 = m_2$

$\frac{1+1}{2-x} = \frac{5-1}{4-2}$

$\frac{2}{2-x} = 2$

$2 = 4 - 2x$

$2x = 2$

$x = 1$

Correct Option: A) 1

7. Collinear points

Points:

$(p,q), (m,n), (p-m, q-n)$

For collinear points:

$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_2}{x_3 - x_2}$

-10-8-6-4-2246810-10-5510A(-8, -8)B(8, 8)m = 1.00

$$\frac{n-q}{m-p}=\frac{(q-n)-n}{(p-m)-m}$$
$$\frac{n-q}{m-p}=\frac{q-2n}{p-2m}$$
Cross multiplication and simplification gives:
$qn=pm$

Correct Option: D) $qn=pm$

8. Collinear points

Points:
$(k,3),(6,-2),(-3,4)$
Using area condition for collinear points:
$x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)=0$
$k(-2-4)+6(4-3)+(-3)(3+2)=0$
$-6k+6-15=0$
$-6k-9=0$
$6k=-9$
$k=-\frac{3}{2}$

Correct Option: D) $-\frac{3}{2}$

9. Collinear points

Points:
$(7,-2),(5,1),(3,2k)$
Using slope condition:
$$\frac{1+2}{5-7}=\frac{2k-1}{3-5}$$
$$\frac{3}{-2}=\frac{2k-1}{-2}$$
$2k-1=3$
$2k=4$
$k=2$

Correct Option: B) 2

10. Collinear points

Points:
$(3q,0),(0,3p),(1,1)$
Using area condition:
$3q(3p-1)+0(1-0)+1(0-3p)=0$
$9pq-3q-3p=0$
$3pq-p-q=0$
Divide by pq:
$$3-\frac{1}{q}-\frac{1}{p}=0$$
$$\frac{1}{p}+\frac{1}{q}=3$$

Correct Option: C) $\frac{1}{p}+\frac{1}{q}=3$

ONE OR MORE THAN ONE CORRECT TYPE

11. Area of triangle = 10 sq. units

Vertices:
$(a,2a),(-2,6),(3,1)$

Using area formula:

$\text{Area} = \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) |$

$\frac{1}{2} | a(6-1) + (-2)(1-2a) + 3(2a-6) | = 10$

$\frac{1}{2} | 5a - 2 + 4a + 6a - 18 | = 10$

$\frac{1}{2} | 15a - 20 | = 10$

$| 15a - 20 | = 20$

Cases:

$15a - 20 = 20 \Rightarrow 15a = 40 \Rightarrow a = \frac{8}{3}$

$15a - 20 = -20 \Rightarrow 15a = 0 \Rightarrow a = 0$

Correct Options: A and C

12. Area condition

Points:

$A(6,3),\ B(-3,5),\ C(4,-2),\ D(x,3x)$

Given:

$\text{Area}(\triangle ABC) = 2\text{Area}(\triangle DBC)$

Area of $\triangle ABC$:

$= \frac{1}{2} | 6(5+2) + (-3)(-2-3) + 4(3-5) |$

$= \frac{1}{2} | 42 + 15 - 8 | = \frac{49}{2}$

So,

$\text{Area}(\triangle DBC) = \frac{49}{4}$

Now using area formula for DBC:

$\frac{1}{2} | x(5+2) + (-3)(-2-3x) + 4(3x-5) | = \frac{49}{4}$

$\frac{1}{2} | 7x + 6 + 9x + 12x - 20 | = \frac{49}{4}$

$\frac{1}{2} | 28x - 14 | = \frac{49}{4}$

$| 28x - 14 | = \frac{49}{2}$

Cases:

$28x - 14 = \frac{49}{2} \Rightarrow 56x - 28 = 49 \Rightarrow 56x = 77 \Rightarrow x = \frac{11}{8}$

$28x - 14 = -\frac{49}{2} \Rightarrow 56x - 28 = -49 \Rightarrow 56x = -21 \Rightarrow x = -\frac{3}{8}$

Correct Options: B and C

13. Collinear points

Points:

$(k, 2-2k),\ (-k+1, 2k),\ (-4-k, 6-2k)$

For collinearity, the area must be 0.

$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$

1. $k[2k - (6-2k)] + (-k+1)[(6-2k) - (2-2k)] + (-4-k)[(2-2k) - 2k] = 0$
2. $k(4k-6) + (-k+1)(4) + (-4-k)(2-4k) = 0$
3. $4k^2 - 6k - 4k + 4 + (-8 + 16k - 2k + 4k^2) = 0$

4. $8k^2 + 4k - 4 = 0$
$2k^2 + k - 1 = 0$
5. Factoring:
$(2k - 1)(k + 1) = 0$
$k = \frac{1}{2}, \ -1$

Correct Options: B and C

COMPREHENSION TYPE

Given:

$\text{Area} = \frac{1}{2} | \ x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \ |$

14. Area of triangle

Vertices:

$(3,2), (11,8), (8,12)$

$= \frac{1}{2} | \ 3(8 - 12) + 11(12 - 2) + 8(2 - 8) \ |$

$= \frac{1}{2} | \ -12 + 110 - 48 \ |$

$= \frac{1}{2}(50) = 25$

Correct Option: C) 25 sq. units

15. Area of triangle

Vertices:

$A(3,0), \ B(7,0), \ C(8,4)$

Since ABlies on x-axis,

Base $= 7 - 3 = 4$

Height $= 4$

Area:

$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$

$= \frac{1}{2} \times 4 \times 4 = 8$

Correct Option: C) 8

16. Matrix Matching Type

Given vertices:

$A(1,2), \ B(6,2), \ C(5,3), \ D(3,4)$

Using area formula:

$\text{Area} = \frac{1}{2} | \ x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \ |$

a) **ar**$(\triangle \boldsymbol{ABC})$

$\frac{1}{2} | \ 1(2 - 3) + 6(3 - 2) + 5(2 - 2) \ |$

$= \frac{1}{2} | \ -1 + 6 + 0 \ | = \frac{5}{2}$

Matches: R

b) **ar**$(\triangle \boldsymbol{ACD})$

$\frac{1}{2} | \ 1(3 - 4) + 5(4 - 2) + 3(2 - 3) \ |$

$= \frac{1}{2} | \ -1 + 10 - 3 \ | = \frac{6}{2} = 3$

Matches: S

c) **ar**$(\triangle ABD)$

$\frac{1}{2} | 1(2-4) + 6(4-2) + 3(2-2) |$

$= \frac{1}{2} | -2 + 12 + 0 | = \frac{10}{2} = 5$

Matches: Q (or T)

d) **ar**$(\triangle BCD)$

$\frac{1}{2} | 6(3-4) + 5(4-2) + 3(2-3) |$

$= \frac{1}{2} | -6 + 10 - 3 | = \frac{1}{2}$

Matches: P

INTEGER TYPE

17. **Area of quadrilateral = 0**

Vertices:

$(1,2), (-5,6), (7,-4), (k,-2)$

Using shoelace formula:

Area $= \frac{1}{2} | \sum x_i y_{i+1} - \sum y_i x_{i+1} |$

$\frac{1}{2} | 6 - 42 - 14 + 2k - (-10 + 42 + 4k - 2) | = 0$

$| -48 + 2k + 10 - 42 - 4k + 2 | = 0$

$| -78 - 2k | = 0$

$k = -39$

Answer: -39

18. **Collinear points**

Points:

$A(7,-2), B(5,1), C(3,2k)$

Using slope condition:

$\frac{1+2}{5-7} = \frac{2k-1}{3-5}$

$\frac{3}{-2} = \frac{2k-1}{-2}$

$2k - 1 = 3$

$2k = 4$

$k = 2$

Answer: 2

19. **Collinear points**

Points:

$(a, 1), (1,-1), (11,4)$

Using area condition:

$a(-1-4) + 1(4-1) + 11(1+1) = 0$

$-5a + 3 + 22 = 0$

$-5a + 25 = 0$

$a = 5$

Answer: 5

20. Area of triangle

Vertices:

$(t, t-2), (t+2, t+2), (t+3, t)$

Using area formula:

$$\text{Area} = \frac{1}{2} \mid t((t+2)-t) + (t+2)(t-(t-2)) + (t+3)((t-2)-(t+2)) \mid$$

$$= \frac{1}{2} \mid 2t + 2(t+2) - 4(t+3) \mid$$

$$= \frac{1}{2} \mid 2t + 2t + 4 - 4t - 12 \mid$$

$$= \frac{1}{2} \mid -8 \mid = 4$$

Answer: 4

Student practice corner

SINGLE OPTION CORRECT TYPE

1. Perimeter = 12

Square side = $12/4 = 3$

Area of square = $3^2 = 9$

Triangle sides = $3, \frac{9}{2}, \frac{9}{2}$

Height = $\sqrt{\left(\frac{9}{2}\right)^2 - \left(\frac{3}{2}\right)^2} = \sqrt{18} = 3\sqrt{2}$

Area of triangle = $\frac{1}{2} \times 3 \times 3\sqrt{2} = \frac{9\sqrt{2}}{2}$

Ratio = $9 : \frac{9\sqrt{2}}{2} = 2 : \sqrt{2} = \sqrt{2} : 1$

Answer: B) $\sqrt{2} : 1$

2. Area of rhombus

Using shoelace formula:

$$\text{Area} = \frac{1}{2} \mid (3 \cdot 5 + 4 \cdot 4 + (-1)(-1) + (-2) \cdot 0) - (0 \cdot 4 + 5 \cdot (-1) + 4 \cdot (-2) + (-1) \cdot 3) \mid$$

$$= \frac{1}{2} \mid 32 - (-10) \mid = \frac{1}{2}(42) = 21$$

But rhombus diagonals:

$d_1 = \sqrt{(3+1)^2 + (0-4)^2} = 4\sqrt{2}$

$d_2 = \sqrt{(4+2)^2 + (5+1)^2} = 6\sqrt{2}$

$\text{Area} = \frac{1}{2} d_1 d_2 = \frac{1}{2}(4\sqrt{2})(6\sqrt{2}) = 24$

Answer: A) 24 sq. units

3. Median divides triangle into equal areas.

Area of $\triangle ABC$:

$$= \frac{1}{2} \mid 4(6-4) + 7(4-2) + 1(2-6) \mid$$

$$= \frac{1}{2} \mid 8 + 14 - 4 \mid = 9$$

So,

$$\text{Area of } \triangle ABD = \frac{9}{2}$$

Answer: B) $\frac{9}{2}$**sq. units**

4. Since $AD$is a median,

$\text{Area}(\triangle ABD) = \text{Area}(\triangle ACD)$

Hence ratio = 1:1

Answer: B) 1:1

5. Using shoelace formula:

$$\text{Area} = \frac{1}{2}\,|\,(-1)(0) + (1)(2) + (-1)(0) + (-3)(-2) - ((-2)(1) + 0(-1) + 2(-3) + 0(-1))\,|$$

$$= \frac{1}{2}\,|\,0 + 2 + 0 + 6 - (-2 + 0 - 6 + 0)\,|$$

$$= \frac{1}{2}\,|\,8 - (-8)\,| = \frac{1}{2}(16) = 8$$

Answer: D) 8 sq. units

6. Using shoelace formula:

$$\text{Area} = \frac{1}{2}\,|\,(-4)(-5) + (-3)(-2) + 3(3) + 2(-2) - ((-2)(-3) + (-5)(3) + (-2)(2) + 3(-4))\,|$$

$$= \frac{1}{2}\,|\,20 + 6 + 9 - 4 - (6 - 15 - 4 - 12)\,|$$

$$= \frac{1}{2}\,|\,31 - (-25)\,| = \frac{1}{2}(56) = 28$$

Answer: B) 28 sq. units

7. Slope

$$m = \frac{z^2 - y^2}{z - y}$$

$$= \frac{(z - y)(z + y)}{z - y} = z + y$$

Answer: C) $y + z$

8. Slope formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

-10-8-6-4-2246810-10-5510A(-8, -8)B(8, 8)m = 1.00

$$2 = \frac{3 - 5}{x - 2} = \frac{-2}{x - 2}$$

$$2(x - 2) = -2$$

$$2x - 4 = -2$$

$$2x = 2$$

$$x = 1$$

Answer: A) 1

ONE OR MORE THAN ONE CORRECT TYPE

9. For collinear points, area of triangle = 0

$$\begin{vmatrix} at_1^2 & 2at_1 & 1 \\ at_2^2 & 2at_2 & 1 \\ 0 & a & 1 \end{vmatrix} = 0$$

After simplification:

$$(t_1 - t_2)(2t_1t_2 - t_1 - t_2) = 0$$

Since points are distinct,

$$2t_1t_2 = t_1 + t_2$$

Answer: C) $2t_1t_2 = t_1 + t_2$

10. **Slope of** $PQ\left(m_{PQ}\right)$**:**

$$m_{PQ} = \frac{(2k + 3) - 2k}{3k - (k + 1)} = \frac{3}{2k - 1}$$

Slope of $QR(m_{QR})$:

$$m_{QR} = \frac{5k-(2k+3)}{(5k-1)-3k} = \frac{3k-3}{2k-1}$$

Equating the slopes:

$$\frac{3}{2k-1} = \frac{3k-3}{2k-1}$$

Note: For the slopes to be defined,

$2k-1 \neq 0$

$k \neq \frac{1}{2}$

Since the denominators are the same, equate the numerators:

$3 = 3k-3$

$3k = 6$

$k = 2$

Correct Option C, D

COMPREHENSION TYPE

11. Using shoelace formula:

$$\text{Area} = \frac{1}{2}[(1\cdot -3 - 1\cdot 7) + (7\cdot 2 - (-3)\cdot 12)$$
$$+(12\cdot 21 - 2\cdot 7) + (7\cdot 1 - 21\cdot 1)]$$
$$= \frac{1}{2}[(-10) + (50) + (238) + (-14)]$$
$$= \frac{1}{2}(264) = 132$$

Answer: A) 132 sq. units

12. Using shoelace formula:

$$\text{Area} = \frac{1}{2}\,|\,(-4)(-5) + (-3)(-2) + 3(3) + 2(-2) - ((-2)(-3) + (-5)(3) + (-2)(2) + 3(-4))\,|$$
$$= \frac{1}{2}\,|\,20 + 6 + 9 - 4 - (6 - 15 - 4 - 12)\,|$$
$$= \frac{1}{2}\,|\,31 - (-25)\,|$$
$$= \frac{1}{2}(56) = 28$$

Answer: A) 28 sq. units

13. Matching

a) Area of triangle with vertices $(0,0),(4,0),(0,8)$

$$\text{Area} = \frac{1}{2}\times 4\times 8 = 16$$

Matches with **Q) 16 sq.units**

b) Area of quadrilateral with vertices $(0,0),(4,0),(5,3),(1,3)$

Using shoelace formula:

$$\text{Area} = \frac{1}{2}\,|\,0 + 12 + 15 + 0 - (0 + 0 + 3 + 0)\,|$$
$$= \frac{1}{2}(27-3) = \frac{1}{2}(24) = 12$$

Matches with **P) 12 sq.units**

c) **The points** $(1,2),(3,6),(5,10)$

$\frac{6-2}{3-1}=2$

$\frac{10-6}{5-3}=2$

Equal slopes ⇒ points are collinear.

Matches with **R) Collinear**

d) **Area of triangle with vertices** $(2,1),(6,1),(4,5)$

Base:

$6-2=4$

Height:

$5-1=4$

Area $=\frac{1}{2}\times 4\times 4=8$

Matches with **S) 8 sq.units**

Integer Type

14. Vertices:

$\left(-\frac{2}{5},6\right),\ (2,8),\ (-4,3)$

Using area formula:

Area $=\frac{1}{2}\left|\ x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\ \right|$

$=\frac{1}{2}\left[\left(\ -\frac{2}{5}\right)(8-3)+2(3-6)+(-4)(6-8)\right]$

$=\frac{1}{2}[-2-6+8]=0$

So, area = 0.

Answer: 0

15. Vertices:

$(a,b+c),\ (b,c+a),\ (c,a+b)$

Using determinant formula:

$$\text{Area}=\frac{1}{2}\left|\begin{matrix} a & b+c & 1 \\ b & c+a & 1 \\ c & a+b & 1 \end{matrix}\right|$$

After simplification, determinant becomes 0.

Hence,

$\boxed{0}$

Answer: 0